

The Surreal Numbers as a Real Vector Space

Conway–Hahn Coordinates, Hamel Bases, Strong Duality,
Canonical Operators, and the Euler Quotient

1 October 2026

Built from three independent manuscripts (batch 73X, manuscripts 51, 52 and 53)

Abstract

The surreal numbers form a proper-class vector space over their canonical copy of $\mathbb{R}$. Three manuscripts written independently on the same day answer the question “what is its natural basis?” in the same way and prove the answer in different ways; this report prints them as one text. The monomials ω^γ are a canonical *Hahn* basis (unique set-supported, reverse-well-ordered strong expansions), the basis of every one-dimensional layer of the associated graded algebra $\text{gr } \mathbf{No} \cong \mathbb{R}[\mathbf{No}]$, and a Hamel basis only of the finite-support subspace. No set spans $\mathbf{No}$; in NBG with global choice a class Hamel basis exists, may contain every monomial, and must contain a proper class of infinite-support vectors, with supports of unbounded cardinality. Every infinite support window A has Hamel dimension $|\mathbb{R}|^{|A|}$, and for every nonzero set-sized exponent group Γ the finite-monomial part of $\mathbb{R}((t^\Gamma))$ has codimension $|\mathbb{R}((t^\Gamma))|$.

The linear structure that *is* canonical is described in detail: support projections, convex subspaces (classical), the commutant of the coordinate projections, positive diagonal operators, finite real linear systems, valuation bases on finite-dimensional subspaces and their global failure, strong linear maps as Hahn-compatible column families, the strong real dual (summation-finite weights), the vanishing of the order-bounded dual, and the rigidity of set-indexed convergence that keeps normal forms from being Schauder expansions. For real exponent groups the Euler operator $\mathfrak{D}(t^\gamma) = \gamma t^\gamma$ makes $\mathbb{R}((t^\Gamma))$ modulo its finite-monomial part a continuum-dimensional $\mathbb{R}(X)$ -vector space; every real-linear map commuting with $\mathfrak{D}$ is a strong coefficient multiplier, and the finite-monomial part has no $\mathfrak{D}$ -invariant complement. Results that re-prove material of other reports of the collection (completion of Hahn vector spaces, monomial automorphisms, Euler derivations) are printed as pointers with the sources’ proofs as second routes.

Status. AI-assisted, unrefereed research report. Every theorem has a written conventional proof; source 52’s program checks finite exact-arithmetic instances only. **Nothing in this report is formalized**; the Lean declarations of the repository that formalize special cases are named in [section 20.5](#). The three manuscripts were written at repository snapshot [f7e7d86](#). Text marked [write] was added when the manuscripts were merged into this report.

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1 The question and the answer

The surreal numbers **No** contain the real numbers as a canonical ordered subfield. Consequently their additive class is naturally a vector class over $\mathbb{R}$. This observation is elementary; the word *basis* is not. In ordinary linear algebra a basis means a Hamel basis, so every vector is a *finite* linear combination. The defining coordinate theorem for surreals, however, is Conway normal form, and its support may have any set-sized ordinal order type. Treating those two notions as interchangeable obscures the main structure.

The answer developed in this article is:

The canonical basis of No over $\mathbb{R}$ is the monomial class $\{\omega^\gamma : \gamma \in \text{No}\}$ in the Hahn, strong-summation, and associated-graded senses. It is not a Hamel basis. Hamel bases exist only after a class-choice principle is specified, are necessarily proper classes, and have no comparably canonical known description.

The distinction can be summarized as follows.

Notion of basis	Allowed expansion	Verdict for No over $\mathbb{R}$
Hamel basis	Finite linear combinations	Exists in NBG with global choice; necessarily a proper class and noncanonical.
Conway–Hahn basis	Set-indexed, reverse-well-ordered, locally finite strong sums	Canonically $\{\omega^\gamma\}$; every surreal has one unique expansion.
Associated-graded basis	One coordinate in each dominance grade	Canonically the same monomials.
Schauder basis	Limits of partial sums in a topology	Not in the full fine order or natural valuation topology: every set-indexed convergent net is eventually constant.
Transcendence basis	Algebraic field generation	Choice-dependent; the monomials nevertheless contain explicit proper-class algebraically independent families.

1.1 Why the repository makes this question timely

The ProveIt repository already separates several uses of the word “vector.” Its report on vector and tensor fields studies finite tuples with surreal coefficients. Its reports on Hahn duality and Hahn spectral theory study set-sized coefficient spaces $V((t^\Gamma))$, strong sums, completion, and several dual notions. A separate report and the work of Kaplan, Krapp, and Serra study automorphisms through Conway normal forms. The present article asks a more foundational question: view the scalar class **No** itself as an $\mathbb{R}$ -vector object and determine which linear-algebraic structures are canonical.

An informal MathOverflow discussion in 2013 already noticed both sides of the issue: global choice should produce a class-sized real basis, while the “homogeneous” pieces ought to be real multiples of powers of ω [18]. Conway normal form makes the latter intuition exact, but only as a Hahn decomposition rather than an algebraic direct sum.

1.2 Main results at a glance

Besides the central basis theorem, the article proves the following.

- A. No set of surreals spans **No** over $\mathbb{R}$ (theorem 5.1).
- B. Under global choice a Hamel class basis exists (theorem 5.3), but every one has a proper class of vectors outside the finite-support skeleton (corollary 5.8).
- C. Every infinite reverse-well-ordered support window A has Hamel dimension $|\mathbb{R}|^{|A|}$ (theorem 5.15).
- D. Normal-form truncation gives canonical complementary projections for every exponent subclass and a one-dimensional graded piece at every surreal exponent (theorems 6.1 and 6.7).
- E. Strongly linear maps are exactly Hahn-compatible column matrices (theorem 10.3), and the strong real dual is classified by summation-finite coefficient weights (theorem 10.7).
- F. The order-bounded real dual is zero (theorem 11.2), although the strong dual is nonzero and even contains functionals with proper-class active support.
- G. The fine natural valuation topology has only eventually constant set-indexed convergent nets (theorem 11.8).
- H. Monomial-preserving strong ordered-field automorphisms are controlled by ordered exponent automorphisms and positive real characters (theorem 18.1), while additive real-valued characters of the exponent group yield diagonal strong derivations (theorem 18.5).
- I. Rational independence of exponents is exactly algebraic independence of the corresponding monomials (theorem 18.7).

1.3 Claim status and novelty discipline

The existence and uniqueness of Conway normal form, the omega map, Hahn support arithmetic, and the Erdős–Kaplansky theorem are classical. Several results below are immediate or short consequences once the correct basis notion is fixed; others package familiar Hahn-space ideas in the proper-class surreal setting. The exact combination—especially the quotient witness, strong real-dual classification, order-dual collapse, and the anti-Schauder observation—was not located as a single published theorem package in the targeted search. This is not a certification that each statement is new.

[write] Source 51’s title page adds the following. **Repository context.** The report was prepared after inspecting VladimirReshetnikov/ProveIt, especially its surreal-number reports on vector and tensor fields, Hahn vector-space duality, infinite-dimensional Hahn spectral theory, and automorphisms. Repository snapshot: f7e7d8607d3b1c49c93b844cae20a77ab3689fe7 (f7e7d86). **Status.** This is an AI-assisted mathematical research draft with complete proofs of the results stated as new deductions in this text. It has not been refereed and the new statements have not been formalized in Lean. Literature searches cannot certify novelty; the article therefore separates classical inputs, elementary consequences, and proposed research questions explicitly.

1.4 Source 52’s formulation and its summary theorems

[write] Source 52 opens with the same question. Its text is printed here in its own convention $t^\gamma = \omega^{-\gamma}$ (section 2.2); its E_Γ and C_Γ are written Fin_Γ and Lf_Γ , its Euler operator D is written $\mathfrak{D}$, and its $\mathfrak{c}$ is written $\mathfrak{c}$. Its section references now point to the sections of this report that print the material.

There are two very different answers to the question “What is a natural basis of the surreal numbers over $\mathbb{R}$?” If a basis means that each vector is a *finite* linear combination of basis vectors, the question

concerns a Hamel basis. If it means unique coordinates in an admissible generalized series, Conway normal form already supplies the answer:

$$x = \sum_{\alpha < \lambda}^H r_\alpha \omega^{a_\alpha}, \quad r_\alpha \in \mathbb{R} \setminus \{0\}, \quad a_\alpha > a_\beta \quad (\alpha < \beta), \quad (1.1)$$

where λ is an ordinal. Thus

$$\mathfrak{M} = \{\omega^a : a \in \mathbf{No}\}$$

is the canonical *Hahn coordinate family*. The support is a set, not a proper class, and the sum is a Hahn sum, not an assertion of convergence in the full surreal order topology. Conway normal form is classical; see [6, 11, 17], especially the statement of Theorem 2.10 in [17].

The interesting research is not to relabel this theorem as a Hamel-basis theorem. It is to understand what fails when infinite coordinates are replaced by finite coordinates, how much of the vector space remains invisible, and which natural operations organize that invisible part.

1.4.1 A guide to the results

The article has three levels. Sections 3–10 settle the basic basis question, explain size and topology, and establish useful structural results. Sections 13–15 develop a more substantial operator-theoretic picture. The final sections provide examples, verification boundaries, and a research program.

For a set-sized ordered abelian group Γ , write

$$H_\Gamma = \mathbb{R}((t^\Gamma)), \quad \text{Fin}_\Gamma = \bigoplus_{\gamma \in \Gamma} \mathbb{R}t^\gamma, \quad \text{Lf}_\Gamma = \overline{\text{Fin}_\Gamma}^v.$$

Here t^γ is a formal monomial, realized inside $\mathbf{No}$ as $\omega^{-\gamma}$ whenever Γ is an ordered subgroup of $\mathbf{No}$. The main conclusions are as follows.

Theorem 1.1 (Summary of the basis and dimension results). *For every nonzero set-sized ordered abelian group Γ :*

1. *The monomials form a Hamel basis of Fin_Γ , but not of H_Γ , and*

$$\dim_{\mathbb{R}}(H_\Gamma/\text{Fin}_\Gamma) = \dim_{\mathbb{R}} H_\Gamma = |H_\Gamma|.$$

2. *Every finite-dimensional real subspace of H_Γ has a unique reduced leading-coordinate basis. The whole H_Γ has no valuation-independent Hamel basis.*
3. *Lf_Γ consists exactly of series whose support below every valuation cutoff is finite. Moreover, $\text{Lf}_\Gamma = H_\Gamma$ exactly when Γ is ordered isomorphic to $\mathbb{Z}$.*

Proofs appear in Sections 5, 9, and 12. The term “valuation-independent” is defined there; ordinary Hamel bases still exist for set-sized spaces under choice.

Theorem 1.2 (Summary of the Euler results). *Let $0 \neq \Gamma \subseteq \mathbb{R}$ be an additive subgroup, and set*

$$\mathfrak{D} \left(\sum_{\gamma}^H x_{\gamma} t^{\gamma} \right) = \sum_{\gamma}^H \gamma x_{\gamma} t^{\gamma}.$$

Then:

1. *$Q_\Gamma = H_\Gamma/\text{Fin}_\Gamma$ is canonically an $\mathbb{R}(X)$ -vector space, with X acting by $\mathfrak{D}$, and $\dim_{\mathbb{R}(X)} Q_\Gamma = \mathfrak{c} = 2^{\aleph_0}$.*

2. Every real-linear $T : H_\Gamma \rightarrow H_\Gamma$ satisfying $T\mathfrak{D} = \mathfrak{D}T$ has the form

$$T(x) = \sum_{\gamma}^H a_{\gamma} x_{\gamma} t^{\gamma} \quad \text{for a unique } (a_{\gamma}) \in \mathbb{R}^{\Gamma}.$$

In particular, T automatically preserves Hahn summation.

3. The exact sequence $0 \rightarrow \text{Fin}_{\Gamma} \rightarrow H_{\Gamma} \rightarrow Q_{\Gamma} \rightarrow 0$ does not split as a sequence of $\mathbb{R}[X]$ -modules. In ordinary language, no algebraic complement of the finite-monomial span is invariant under $\mathfrak{D}$.

These statements are proved in Sections 13 and 14. They sharpen the mere observation that monomials do not span: the missing directions have a canonical scalar calculus, and no basis extension can separate them while preserving the natural Euler operator.

1.4.2 Relation to ProveIt

The repository’s *Three Duals at Surreal Scales* studies strong Hahn operators, scalar-extension completion, and continuous functionals invisible on a distinguished coefficient space [23]. Its coefficient-space problem is not identical to the present problem, where the ground scalars remain $\mathbb{R}$ and the distinguished subspace is the finite *monomial* span. Nevertheless, the completion-versus-strong-summation distinction is an explicit precedent, not a novelty claimed here.

The report on tail-spans and differential transcendence uses Hahn supports, cofinite linear data, and Euler-type derivations [24]. Its differential-algebraic independence results are different from the present quotient as a module over $\mathbb{R}[X]$. We do not infer algebraic or differential-algebraic independence from our $\mathbb{R}(X)$ -linear independence. The project guide explicitly separates manuscript proofs, finite checks, and formalization [22]; we preserve that separation.

[write] Source 52’s title page adds: **Status.** This is an AI-assisted mathematical research draft, not an independently refereed or Lean-verified paper. Classical foundations and repository precedents are identified separately from the deductions developed here. No named open conjecture or certified priority claim is asserted. The Euler-quotient and commutant package is a proposed research contribution whose priority requires a broader literature audit.

1.5 Source 53’s answer in one page

[write] Source 53 gives the same answer in four levels, and recommends terminology that this report adopts.

The additive group of **No** is canonically an $\mathbb{R}$ -vector space: a real scalar r acts on a surreal x by multiplication inside the surreal field. The hard part of the question is not whether scalar multiplication exists, but what “basis” should mean for a proper class whose canonical expansions may have arbitrary set-sized ordinal length.

The classical Conway normal-form theorem gives a unique expansion

$$x = \sum_{i < \lambda} r_i \omega^{a_i}, \quad a_0 > a_1 > \cdots, \quad r_i \in \mathbb{R} \setminus \{0\}, \quad (1.2)$$

where λ is an ordinal. This looks like a coordinate expansion, and it is one, but it is not a Hamel expansion unless λ is finite. The correct answer therefore has four levels.

1. **Hamel basis.** A Hamel basis allows only finite sums. The monomials ω^a span exactly the finite-normal-form subspace **No**_{fin}, not all of **No**. Any Hamel basis for all surreals is a proper class and is highly noncanonical. Its existence is a class-set-theoretic statement, not merely an application of the set-sized basis theorem.

2. **Hahn or summation basis.** If admissible sums are allowed to have set-sized reverse-well-ordered support, then the monomials are a canonical basis: normal form gives existence and uniqueness. This is the most faithful answer to “what is the natural basis?”
3. **Graded basis.** At each leading exponent a , the quotient of surreals of leading exponent at most a by those of leading exponent below a is one-dimensional, generated by the class of ω^a . Thus the monomials are an ordinary basis of the associated graded object.
4. **Topological basis.** For the full natural valuation topology, the finite monomial span is closed. Hence the monomials do not become a Schauder basis by taking valuation limits. Coefficientwise or Hahn summation, not natural-valuation completion, is the relevant infinitary operation.

Recommended terminology. Call $\{\omega^a : a \in \mathbf{No}\}$ the *canonical monomial Hahn basis* of $\mathbf{No}$ over $\mathbb{R}$. Reserve *Hamel basis* for an algebraic class basis with finite expansions. The two notions solve different problems and should not be conflated.

The report develops this answer into exact theorems. Especially useful are the support-slice dimension formula

$$\dim_{\mathbb{R}} \mathbf{H}(A) = |\mathbb{R}|^{|A|},$$

the coordinate-projection calculus, the classification of diagonal positive operators, the classification of convex $\mathbb{R}$ -subspaces by cuts in the exponent class, and the closure theorem for $\mathbf{No}_{\text{fin}}$.

[write] Source 53 states its main conclusion on its title page:

Main conclusion. The monomials

$$\mathfrak{M} = \{\omega^a : a \in \mathbf{No}\}$$

are the unique natural coordinate directions supplied by Conway normal form. They are $\mathbb{R}$ -linearly independent, but they are *not* a Hamel basis of $\mathbf{No}$: finite linear combinations produce only finite normal forms. They are, however, a canonical *Hahn basis* (a summation basis): every surreal has a unique set-supported, reverse-well-ordered expansion

$$x = \sum_{a \in \text{supp}(x)}^{\mathbf{H}} c_a(x) \omega^a.$$

They are also an ordinary basis of every one-dimensional graded layer of the natural valuation filtration, and their classes form the canonical basis of $\text{gr } \mathbf{No} \cong \mathbb{R}[\mathbf{No}]$.

A true Hamel basis exists only after a class-level choice principle is specified; it is necessarily a proper class, necessarily contains vectors with arbitrarily large supports, and has no comparable canonicity. In fact, the finite monomial span is closed in the full natural valuation topology, so the normal-form expansion is not an ordinary topological completion of finite sums.

[write] *Reading of “unique natural coordinate directions”.* The monomials are canonical relative to the omega map, that is, to the simplicity structure of $\mathbf{No}$; they are not determined by the ordered real vector space alone. Source 52’s shear (section 9.4) is a strong, order-preserving real-linear automorphism fixing $\mathbb{R}$ that sends t^γ to $t^\gamma + t^\delta$. Read the box’s “unique natural” as “canonical, given the omega map”.

Source 53’s source and priority discipline

The normal-form theorem, the Hahn-field interpretation, the natural valuation, and the simplicity-hierarchical viewpoint are classical; standard sources include Conway, Gonshor, Alling, Ehrlich, van den Dries–Ehrlich, and the survey of Mantova and Matusinski [6, 11, 1, 8, 9, 25, 17]. The exact dimension of a full product K^I is the Erdős–Kaplansky theorem; we use a modern proof by Kuba [15].

The PROVEIT repository already contains formalized normal-form extraction and evaluation, strong summation, real-linear coefficient maps on a purely infinite ideal, and exponent-reindexing automorphisms. Precise paths are recorded in [section 20.4](#). The projection, diagonal-operator, positivity, convex-subspace, support-complexity, and natural-valuation closure theorems are proved directly below. They are elementary consequences of the cited normal-form machinery. No claim is made that these deductions have not appeared elsewhere under Hahn-space terminology.

[write] Source 53’s title page adds: Independent research report prepared for Vladimir Reshetnikov. Classical inputs are cited; deductions proved in this report are not asserted to be new in the priority sense. The manuscript has not undergone independent refereeing. Selected foundational ingredients have existing Lean implementations in the repository, but the new synthesis and theorems below have not been machine-checked as a unit.

2 How this report is assembled

[write] This section, [section 20.5](#), [section C](#) and every paragraph marked [write] were written when the three manuscripts were merged. Nothing else in the report is new text.

2.1 Three manuscripts on one question

The three manuscripts were written independently on 1 October 2026 at the same repository snapshot `f7e7d86`, and they do not know each other. They share no paragraph (the longest common word run of any pair is a bibliography entry), and they answer the same question with the same answer and different proofs.

Source	Manuscript (batch 73X)	Contribution and where it is printed
51 (base)	<i>The Surreal Numbers as a Real Vector Class: Conway–Hahn Coordinates, Hamel Bases, Strong Duality, and Canonical Operators</i> (32 pp.)	The spine of every Part: foundations, the four basis notions, Hamel bases and windows, projections and the associated graded, strong maps and the strong dual, order duality and topology, monomial symmetries and derivations (sections 3 to 6, 10, 11 and 18), 20 research questions.
52	<i>Surreal Numbers as a Real Vector Space: Canonical Hahn Coordinates, Hamel Obstructions, and the Euler Quotient</i> (29 pp.)	Full codimension of the finite-monomial span, valuation bases, the symmetry obstruction and the shear, coordinate subspaces, completion (as a second route to the three-duals report), the strong dual on a slice, positivity in any non-Archimedean field, and the whole of Part IV (the Euler quotient); its finite checks.
53	<i>Surreal Numbers as a Vector Space over $\mathbb{R}$: Hamel Bases, Hahn Bases, Canonical Coordinates, Valuation Layers, and Linear Operators</i> (28 pp.)	The coordinate model, unbounded support of Hamel basis vectors, slice quotients, convex subspaces, commutant and positive diagonal operators, covariance and rigidity, finite real linear systems, closedness of the finite span and the trichotomy, the recommended terminology (section 19), the map to the Lean development, 31 research questions.

Merge rules. Every result, proof, example, remark, warning, limitation and research question of every source is printed. A result proved by more than one source is printed once, in the base’s wording unless another source’s statement is stronger; a [write] note names the other sources, and a genuinely different proof is printed as “Second proof (source NN)”. Where another source’s statement adds a clause, the clause is printed as a complement next to the shared statement. Source 51 is the base because it works on the whole class **No** under the weakest foundations (NBG with global choice, tied with source 52) and covers the most of the common spine. Source 52’s Euler-quotient material has no counterpart and is printed whole as [Part IV](#). Questions asked by several sources are printed once, with every source’s wording ([section 21](#)).

Results of other reports. Three groups of statements re-prove results already in the collection, without (51) or with only partial (52) credit. They are printed as pointers, with the sources’ proofs as second routes and no novelty claim:

- source 52’s completion section is the diagonal case of the report *Three Duals at Surreal Scales* ([surcomplex/three-duals-of-hahn-vector-spaces](#)), Theorems 4.2, 4.5, 4.6 and 10.1, Lemma 4.4 and Proposition 5.1; its invisible functionals parallel that report’s Theorem 6.4 ([section 12](#));
- source 51’s monomial automorphisms are the canonical Hahn lifts of *Growth-Scale Rigidity for Surreal and Surcomplex Automorphisms* ([surreal/exponential-automorphism-rigidity](#)), Proposition 8.3, and the monomial maps of *Omnific-Preserving Automorphisms* ([surreal/omnific-preserving-automorphisms](#)), (5.1) and Theorem 5.1;
- source 51’s diagonal derivations and source 52’s Euler operator are the Euler derivations of *Omnific Integers and Omnific-Diophantine Geometry* ([surreal/omnific-diophantine-geometry](#)), (7.1), Lemma 7.1, (16.1) and Lemma 16.2, and of the tail-span report ([surreal/tail-spans-and-differential-transcendence](#)), (7.2) ([section 18](#)).

Numbers of other reports are those of their committed PDFs on 1 October 2026.

Labels. Every label carries the prefix `rvs:`. Source 51’s labels are `rvs:<original>`, source 52’s are `rvs:eu:<original>` (named after its Euler-quotient package), source 53’s are `rvs:cv:<original>` (named after its convex-subspace and positivity material), and labels of text added here are `rvs:w:....`. A label of a statement printed once under another source’s wording is attached to the printed statement.

2.2 Conventions

Sources 51 and 53 use Conway’s decreasing convention: a normal form is $\sum r_\beta \omega^{\gamma_\beta}$ with reverse-well-ordered support, and $v(x) = -\text{lead}(x)$. Source 52 writes a Hahn variable

$$t^\gamma = \omega^{-\gamma},$$

so that its supports are *well* ordered and $v(x) = \min \text{supp}_t x$. The two valuations are the same function. This report uses the ω -convention throughout, except in the material of source 52 (parts of [sections 3](#) to [5](#), [sections 9](#) and [12](#), its theorems in [sections 10](#) and [11](#), and all of [Part IV](#)), which keeps its t -convention. There its support is written supp_t , so that $\text{supp}_t x = -\text{supp } x$.

2.3 Notions and symbols

Notion	This report	In the sources
Monomial	ω^γ ; in 52's text $t^\gamma = \omega^{-\gamma}$	51: ω^γ ; 53: ω^a ; 52: t^γ , and ω^a in its introduction.
Monomial class	$\mathfrak{M}$	51, 52: $\mathfrak{M}$; 53: $\mathcal{M}$ (renamed).
Coefficient	$[\omega^\gamma]x$ (51), $c_\gamma(x)$ (53)	52: x_γ is the coefficient of t^γ , that is $[\omega^{-\gamma}]x$.
Support	$\text{supp } x$, reverse well ordered	52: $\text{supp } x$, well ordered in t -exponents; printed $\text{supp}_t x$ (renamed).
Leading exponent	$\text{lead } x = \max \text{supp } x$	53 printed le ; its macro now prints lead .
Valuation	$v(x) = -\text{lead } x = \min \text{supp}_t x$	Same function in all three.
Strong (Hahn) sum	$\sum^H$	51: $\sum^H$; 52: $\sum^H$; 53: $\sum^H$; the same symbol.
Finite-support class	$\mathbf{No}_{\text{fin}} = \text{span}_{\mathbb{R}} \mathfrak{M}$	52 wrote $E_{\mathbf{No}}$ (renamed $\mathbf{No}_{\text{fin}}$).
Hahn workspace	$H_\Gamma = \mathbb{R}((t^\Gamma))$, Γ a set	52's notation. The collection's <code>NOTATION.md</code> recommends F_Γ ; here F is the dominance flag $F_{\leq \gamma}$ of 51 and 53.
Finite-monomial part of H_Γ	Fin_Γ	52: E_Γ (renamed). Not the $E_\Gamma(V)$ of the three-duals report: for $V = \mathbb{R}$ that is all of H_Γ .
Valuation closure of Fin_Γ	Lf_Γ (left-finite supports)	52: C_Γ (renamed). Not the three-duals $C_\Gamma(V)$; with $V = \mathbb{R}$ that is H_Γ .
Euler quotient	$Q_\Gamma = H_\Gamma / \text{Fin}_\Gamma$	52.
Support window, slice	$H(A)$, $H(A)^{\text{fin}}$	51: $H(A)$; 53: $\mathcal{H}_A$, $\mathcal{H}_A^{\text{fin}}$ (renamed); 52's $H[A] \subseteq H_\Gamma$ is the same notion in t -exponents.
Support projection	P_A	51: P_A ; 53: P_S ; 52: P_A on H_Γ .
Diagonal operator	D_a (51), D_λ (53)	The same family; $a, \lambda : \mathbf{No} \rightarrow \mathbb{R}$ arbitrary.
Diagonal derivation	D_d , d additive	51.
Euler operator	$\mathfrak{D}(t^\gamma) = \gamma t^\gamma$ on H_Γ , $\Gamma \subseteq \mathbb{R}$	52: D (renamed). In the ω -convention $\mathfrak{D}\omega^\beta = -\beta\omega^\beta$: $\mathfrak{D}$ is D_d with $d(\beta) = -\beta$, not D_{id} .
Continuum	$\mathfrak{c} = 2^{\aleph_0}$	52: $\mathfrak{c}$, typed <code>\ct</code> (renamed); 51's ct is the constant term $[\omega^0]$.
Real Vandermonde parameter	$s \in \mathbb{R}$	51 and 53 wrote t (renamed: t is 52's Hahn variable).
Staged spaces in class-basis proofs	$V_\alpha = \text{span}_{\mathbb{R}} \mathbf{No}_{<\alpha}$	53's staged Hahn fields $\mathbb{R}((\omega^{\Gamma_\alpha}))$ are W_α (renamed).
Shifts and shears	$S_b x = \omega^b x$ (53)	52's shear is S and its reweightings are $T_{\gamma,a}$; 51's monomial automorphisms are $T_{\chi,\phi}$.
Group-algebra basis	e_γ (51), T^a (53)	Both name the basis of $\mathbb{R}[\mathbf{No}]$.
Zero neighbourhoods	$B_\gamma(0) = \{v > \gamma\}$ (51), $U_\gamma = \{\text{lead} < \gamma\}$ (53)	$U_\gamma = B_{-\gamma}(0)$.

Tempting false readings.

- Fin_Γ and Lf_Γ are not the $E_\Gamma(V)$ and $C_\Gamma(V)$ of the three-duals report with $V = \mathbb{R}$: there $E_\Gamma(\mathbb{R}) = C_\Gamma(\mathbb{R}) = H_\Gamma$, because $\mathbb{R}$ is one-dimensional. The correct dictionary is the diagonal embedding of [section 12](#).

- $\mathfrak{D}$ is not the Berarducci–Mantova derivation, not a global surreal derivative, and in the ω -convention it is D_d with $d = -\text{id}$ on real exponents. It is the differential-equations report’s ∂_τ on real exponents, the tail-span report’s (7.2), and the omnific–Diophantine report’s ∂_λ with $\lambda = -\text{id}$.
- ct is a functional (constant term); $\mathfrak{c}$ is a cardinal.
- A “Hahn basis” is not a weakened Hamel basis and not a Schauder basis (section 4); a “valuation basis” is a Hamel basis with a non-cancellation property (section 9).
- In 51’s and 53’s Vandermonde families the parameter s is a real number, not a monomial.

Part I

Hahn coordinates and Hamel bases

3 Foundations and Conway–Hahn coordinates

3.1 Class-theoretic convention

We work in von Neumann–Bernays–Gödel set theory with global choice (NBG+GC), as is common in recent work on class-sized automorphism groups of **No** [14]. A *class* may be a set or a proper class. A class vector space over the set field $\mathbb{R}$ has the usual operations, with quantification over its vectors understood class-theoretically. A *Hamel class basis* is a linearly independent class such that every vector is a finite linear combination of basis vectors.

The use of NBG+GC has three purposes.

1. It permits direct statements about **No** as one algebraic object.
2. It permits a class-length recursion choosing compatible bases of its set-sized birthday stages.
3. It keeps every actual infinite sum set-indexed. No proper-class sum is ever formed in this article.

Readers preferring ZFC may interpret every construction inside a fixed set-sized surreal subfield or birthday cutoff; all local arguments are then ordinary set mathematics. Only the global basis-existence statement and the proper-class conclusions require the class framework.

3.2 Set-sized workspaces and class vector spaces (sources 53 and 52)

[write] Sources 53 and 52 state their class-theoretic conventions as follows. Source 53 also allows a universe-relative reading; source 52 fixes ZFC for sets and Gödel–Bernays class theory with global choice for classes. All three readings agree with section 3.1.

In ordinary ZFC, a vector space is a set. The totality **No** of all surreal numbers is a proper class. Accordingly, a statement such as “**No** has a Hamel basis” requires one of two conventions.

- Work in a class theory such as NBG, GBC, or MK and speak of a *class vector space*. This is the convention used for global statements in this report.
- Fix a Grothendieck universe or a Lean universe level and work with the set-sized surreal field available at that level. Ordinary basis theorems then apply inside the ambient metatheory. The PROVEIT formalization follows this disciplined universe-relative approach.

The elementary linear identities below are insensitive to this distinction. The distinction becomes essential only for existence and cardinality statements about a Hamel basis of the entire class.

Definition 3.1 (Class real vector space). A *class real vector space* is a possibly proper class V with class operations $+$ and $\mathbb{R} \times V \rightarrow V$ satisfying the usual vector-space axioms. A class family $B \subseteq V$ is *Hamel independent* if every finite linear relation among distinct members of B is trivial. It is a *class Hamel basis* if, in addition, every $v \in V$ is a finite linear combination of members of B .

Since $\mathbb{R}$ is canonically embedded in the surreal field, $r \cdot x$ is defined by surreal multiplication. Thus $\mathbf{No}$ is a class vector space over $\mathbb{R}$, and its order is compatible with positive real scalar multiplication. The entire surreal field $\mathbf{No}$ is a proper class. Accordingly, “ $\mathbf{No}$ is a real vector space” means a *class vector space*: its addition and real scalar multiplication satisfy the vector-space identities, but its collection of vectors is not a set. We use ordinary ZFC for all set-sized Hahn spaces. Statements explicitly involving a class Hamel basis use Gödel–Bernays class theory with global choice. No cardinal-valued dimension is assigned to the whole proper class.

A set-sized ordered subgroup $\Gamma \subseteq (\mathbf{No}, +)$ defines the concrete Hahn subfield

$$H_\Gamma = \left\{ \sum_{\gamma \in S}^H x_\gamma \omega^{-\gamma} : S \subseteq \Gamma \text{ well ordered, } x_\gamma \in \mathbb{R} \right\}. \quad (3.1)$$

Normal form identifies this with the abstract Hahn field $\mathbb{R}((t^\Gamma))$. No divisibility assumption on Γ is needed for its field operations. It need not be real closed.

Warning 3.2 (The notation t^γ). For arbitrary surreal γ , $t^\gamma = \omega^{-\gamma}$ denotes the Conway monomial map with the sign reversed. It is not an unproved identification with $\exp(\gamma \log t)$ for general surreal exponents. We use only the monomial identity $t^{\gamma+\delta} = t^\gamma t^\delta$. The Euler construction later is restricted to real exponents.

3.3 Reverse well ordering and supports

Definition 3.3 (Reverse well ordered). A linearly ordered set S is *reverse well ordered* if every nonempty subset of S has a largest element. Equivalently, S admits a unique enumeration without repetition

$$S = \{\gamma_\beta : \beta < \lambda\}, \quad \gamma_0 > \gamma_1 > \cdots,$$

for some ordinal λ .

Every subset of a reverse-well-ordered set is reverse well ordered. This simple hereditary property is responsible for the existence of arbitrary support truncations later in the article.

Theorem 3.4 (Conway normal form). *Every surreal number x has a unique expansion*

$$x = \sum_{\beta < \lambda} r_\beta \omega^{\gamma_\beta}, \quad (3.2)$$

where λ is an ordinal, the exponents $\gamma_\beta \in \mathbf{No}$ are strictly decreasing, and $r_\beta \in \mathbb{R} \setminus \{0\}$. Conversely, every expression of this form represents a surreal number.

[write] Source 53 states the same theorem with the arithmetic included (addition coefficientwise, multiplication by Hahn convolution with $\omega^a \omega^b = \omega^{a+b}$, so that $\mathbf{No} = \mathbb{R}((\omega^{\mathbf{No}}))$ with the usual proper-class qualification) and comments:

This is the decisive structural theorem. It is stated in this form in modern surveys and in the analysis literature [17, 7]; Ehrlich derives the same “Conway names” from the structure of $\mathbf{No}$ as a

complete simplicity-hierarchical ordered vector space over $\mathbb{R}$ [9]. This is the classical Conway–Gonshor normal-form theorem [6, 11, 25, 17]. The map $\gamma \mapsto \omega^\gamma$ here is Conway’s *omega map*. It sends the additive ordered class **No** isomorphically onto the multiplicative ordered class of monomials and chooses the simplest positive representative of each Archimedean equivalence class. It should not be confused merely with formal use of ordinary real exponentiation.

For x in the form (3.2), define

$$\text{supp}(x) = \{\gamma_\beta : \beta < \lambda\}, \quad [\omega^\gamma]x = \begin{cases} r_\beta, & \gamma = \gamma_\beta, \\ 0, & \gamma \notin \text{supp}(x). \end{cases}$$

The first notation is the *normal-form support*; the second is the coefficient of ω^γ . The support is always a set, even though the exponent class is proper.

Definition 3.5 (The Hahn product over the surreal exponent class). Let

$$H_{\mathbb{R}}(\mathbf{No}) = \{f : \mathbf{No} \rightarrow \mathbb{R} : \text{supp}(f) = \{\gamma : f(\gamma) \neq 0\} \text{ is a reverse-well-ordered set}\}. \quad (3.3)$$

The functions are class functions, but each one has set support. Addition and real scalar multiplication are coefficientwise.

The normal-form theorem gives a canonical $\mathbb{R}$ -linear class isomorphism

$$\mathbf{No} \cong H_{\mathbb{R}}(\mathbf{No}), \quad x \longmapsto (\gamma \mapsto [\omega^\gamma]x). \quad (3.4)$$

Hahn–Neumann support arithmetic further identifies the ordered field structure with the generalized power-series field

$$\mathbf{No} = \mathbb{R}((\omega^{\mathbf{No}})). \quad (3.5)$$

In particular,

$$\omega^\gamma \omega^\delta = \omega^{\gamma+\delta},$$

and multiplication of two normal forms is the convolution of their coefficients. The support theorem guarantees that each output coefficient is a finite sum and that the resulting support is reverse well ordered [12, 19, 11].

[write] Source 53 states the identification (3.4) as a theorem with a converse, and draws two consequences.

Theorem 3.6 (Restricted class-product model). *The coefficient map*

$$C : \mathbf{No} \longrightarrow \mathbb{R}^{\mathbf{No}}, \quad C(x) = (c_a(x))_{a \in \mathbf{No}},$$

is an injective $\mathbb{R}$ -linear class map. Its image is exactly the class of functions $c : \mathbf{No} \rightarrow \mathbb{R}$ whose support is a set and is reverse well ordered. Under this identification, addition and scalar multiplication are pointwise.

Proof. Normal form gives the support condition, existence, and uniqueness. Coefficientwise addition and multiplication by a real constant give linearity. Conversely, if c has set-sized reverse-well-ordered support, enumerate that support in decreasing order and apply the converse clause of theorem 3.4. $\square$

Corollary 3.7 (Coefficients separate points). *For $x, y \in \mathbf{No}$,*

$$x = y \iff c_a(x) = c_a(y) \text{ for every } a \in \mathbf{No}.$$

Each $c_a : \mathbf{No} \rightarrow \mathbb{R}$ is an $\mathbb{R}$ -linear functional.

3.4 Three canonical pieces

Normal form immediately gives the familiar decomposition

$$x = x_{>0} + c_0(x) + x_{<0}, \quad (3.6)$$

where $x_{>0}$ is purely infinite and $x_{<0}$ is infinitesimal. In vector-space language,

$$\mathbf{No} = \mathbf{No}_{>0} \oplus \mathbb{R} \oplus \mathbf{No}_{<0}. \quad (3.7)$$

This is only the first instance of the support-cut decompositions developed in [section 6](#).

3.5 Strong Hahn summation

Normal form is not an ordinary finite sum and, as [section 11](#) will show, it is not generally a topological limit of finite partial sums. The appropriate operation is strong Hahn summation.

Definition 3.8 (Hahn-summable family). Let I be a set and $(x_i)_{i \in I}$ a family in $\mathbf{No}$. The family is *Hahn summable*, or *strongly summable*, if

1. $\bigcup_{i \in I} \text{supp}(x_i)$ is reverse well ordered, and
2. for every exponent γ , only finitely many i have $[\omega^\gamma]x_i \neq 0$.

Its strong sum is

$$\sum_{i \in I}^H x_i = \sum_{\gamma} \left(\sum_{i \in I} [\omega^\gamma]x_i \right) \omega^\gamma, \quad (3.8)$$

where the inner sum is finite.

This definition is purely algebraic and support-controlled. It is invariant under reindexing of the family and is associative under the usual flattening hypotheses. In particular, every normal form is the strong sum of its monomial terms:

$$x = \sum_{\gamma \in \text{supp}(x)}^H [\omega^\gamma]x \omega^\gamma. \quad (3.9)$$

[write] Source 53 gives the same definition and adds:

This is an algebraic summability notion, not convergence in the natural valuation topology. It is the notion implemented for actual surreal sign sequences in the PROVEIt strong-summation development; see [section 20.4](#).

The same rule in the t -convention (source 52)

[write] Source 52 states [definition 3.8](#) for Hahn series in t -exponents, where joint supports are well ordered, and defines strong maps ([definition 10.1](#)) in the same breath.

Definition 3.9 (Strong Hahn summability). A set-indexed family $(x_i)_{i \in I}$ of Hahn series is summable if

$$S = \bigcup_{i \in I} \text{supp}_t(x_i)$$

is well ordered and, for every γ , only finitely many i satisfy $(x_i)_\gamma \neq 0$. Its sum is the series with coefficient $\sum_i (x_i)_\gamma$ at γ . A real-linear map is *strong* if it takes every summable family to a summable family and commutes with its sum.

This is the standard generalized-series convention underlying strong additivity in surreal differential algebra [4]. A subfamily is still summable, and a regrouping is legitimate whenever the same joint-support and finite-occurrence conditions hold.

Every vector has the summable monomial decomposition

$$x = \sum_{\gamma \in \text{supp}_t x}^H x_\gamma t^\gamma.$$

This statement includes the support restriction. It does not license arbitrary real coefficient functions on the entire exponent group. For example, $\{1/n : n \geq 1\}$ is not well ordered in its usual order, so $\sum_{n \geq 1} t^{1/n}$ is not a Hahn series. In contrast, $\{-1/n : n \geq 1\}$ is well ordered and $\sum_{n \geq 1} t^{-1/n}$ is a valid Hahn series.

Example 3.10 (A geometric normal form). The reverse-well-ordered set $\{0, -1, -2, \dots\}$ supports the identity

$$\frac{1}{1 - \omega^{-1}} = 1 + \omega^{-1} + \omega^{-2} + \dots = \sum_{n < \omega}^H \omega^{-n}. \quad (3.10)$$

The right side is a legitimate surreal normal form. It is not a finite linear combination of monomials.

Example 3.11 (What is not allowed). The class-indexed expression $\sum_{\gamma \in \mathbf{No}} \omega^\gamma$ is not a surreal number: its support is a proper class. Likewise, a set-supported formal expression whose support is not reverse well ordered is not a Conway normal form. The word “transfinite” therefore does not mean that arbitrary class sums are permitted.

3.6 Leading exponent, valuation, and Archimedean scale

For $x \neq 0$, let

$$\text{lead}(x) = \max \text{supp}(x), \quad \text{lc}(x) = [\omega^{\text{lead}(x)}]x.$$

Then

$$x = \text{lc}(x) \omega^{\text{lead}(x)} (1 + \varepsilon), \quad |\varepsilon| < 1/n \quad (n \in \mathbb{N}_{>0}), \quad (3.11)$$

so ε is infinitesimal. The sign of x is the sign of its leading real coefficient, and the monomial $\omega^{\text{lead}(x)}$ is the simplest positive representative of its Archimedean magnitude.

We use the additive natural valuation

$$v(x) = -\text{lead}(x) \quad (x \neq 0), \quad v(0) = +\infty. \quad (3.12)$$

Thus larger values mean smaller surreal magnitude. This sign convention agrees with writing the formal Hahn variable as $t^\gamma = \omega^{-\gamma}$.

[write] Source 53 adds: its residue field is $\mathbb{R}$ and its value group is the additive ordered class $\mathbf{No}$.

4 The natural basis: Hahn, not Hamel

4.1 Four basis notions

For a set-sized vector space, “basis” normally means Hamel basis. In non-Archimedean analysis and transseries, one also encounters topological, orthogonal, and strong-summation bases. The distinctions are essential here.

Definition 4.1 (Hamel and Hahn bases). Let V be a class vector space over $\mathbb{R}$.

1. A *Hamel class basis* B is linearly independent and every $x \in V$ is a unique finite sum $\sum_{b \in F} r_b b$, with $F \subset B$ finite.
2. A *Hahn basis* indexed by a linearly ordered class Γ is a family $(e_\gamma)_{\gamma \in \Gamma}$ together with a specified strong summation rule such that every vector has a unique expansion

$$x = \sum_{\gamma \in S}^H r_\gamma e_\gamma$$

with S a set in the admissible support class.

A Hahn basis is not a weakened Hamel basis. It is a basis for a richer algebraic object: a vector space equipped with distinguished infinite sums. Likewise, a Schauder basis requires a topology and convergence of partial sums, a condition independent of formal Hahn summability.

[write] Source 53 gives the same definition of a Hahn basis (as “local terminology”) and remarks:

The term is descriptive; closely related notions in the literature include Hahn groups, Hahn spaces, generalized series, and summable families.

[write] Source 52 separates three meanings:

A *Hamel basis* gives unique finite combinations. A *Hahn basis*, as the term is used here, gives unique admissible Hahn expansions. A *valuation basis* is a Hamel basis for which no finite linear combination suffers leading-value cancellation. These requirements are not interchangeable. Neither is a Hahn basis automatically a Schauder basis: that concept requires a specified topology and a convergence statement.

Conway’s monomials are natural because each positive Archimedean class has a simplest representative. The simplicity relation is extra structure beyond the ordered real-vector-space reduct. Section 9 makes this qualification precise.

Theorem 4.2 (The canonical Conway–Hahn basis). *The monomial class*

$$\mathfrak{M} = \{\omega^\gamma : \gamma \in \mathbf{No}\} \tag{4.1}$$

is a canonical Hahn basis of $\mathbf{No}$ over $\mathbb{R}$. More precisely, the map

$$\mathbf{H}_{\mathbb{R}}(\mathbf{No}) \longrightarrow \mathbf{No}, \quad f \longmapsto \sum_{\gamma \in \text{supp}(f)}^H f(\gamma) \omega^\gamma$$

is an $\mathbb{R}$ -linear class isomorphism, and its coordinate expansion is unique.

Proof. Existence and uniqueness are exactly [theorem 3.4](#). Linearity follows from coefficientwise addition and scalar multiplication. Every support involved is a reverse-well-ordered set, and no class-indexed sum is taken. $\square$

[write] Source 53 proves the same theorem, phrased as: for every x the monomial family $(c_a(x)\omega^a)_{a \in \text{supp}(x)}$ is strongly summable with Hahn sum x , and this representation is unique. It adds:

Remark 4.3. This theorem is the mathematically natural positive answer to the question in the title. It says that $\mathbf{No}$ is not merely analogous to a coordinate space: it has canonical coordinate functionals c_a and canonical coordinate vectors ω^a . What changes from elementary linear algebra is the admissible summation law.

The theorem answers the user’s basis question in the sense best adapted to surreal arithmetic. It also explains why the answer is useful rather than merely formal: the basis vectors multiply by addition of their indices, order is read from the first nonzero coordinate, and every truncation of an expansion is again a surreal.

4.2 The finite-support skeleton

Define

$$\mathbf{No}_{\text{fin}} = \{x \in \mathbf{No} : \text{supp}(x) \text{ is finite}\}. \quad (4.2)$$

Then

$$\mathbf{No}_{\text{fin}} = \bigoplus_{\gamma \in \mathbf{No}} \mathbb{R}\omega^\gamma \subsetneq \mathbf{No} = \prod_{\gamma \in \mathbf{No}}^{\text{H}} \mathbb{R}\omega^\gamma. \quad (4.3)$$

The symbol on the right denotes the Hahn product: set-supported, reverse-well-ordered coordinate families, not the unrestricted direct product.

[write] Source 53 introduces the same subspace as a span:

Definition 4.4 (Finite monomial span). Let

$$\mathfrak{M} = \{\omega^a : a \in \mathbf{No}\}, \quad \mathbf{No}_{\text{fin}} = \text{span}_{\mathbb{R}}(\mathfrak{M}).$$

Thus $\mathbf{No}_{\text{fin}}$ consists exactly of surreal numbers with finite support.

As a class algebra,

$$\mathbf{No}_{\text{fin}} \cong \mathbb{R}[\mathbf{No}], \quad (4.4)$$

the group algebra of the additive exponent class, interpreted as finite formal sums. It is a subring but not a field.

Proposition 4.5 (Where the monomials are a Hamel basis). *The class $\mathfrak{M}$ is a Hamel class basis of $\mathbf{No}_{\text{fin}}$, and it is real-linearly independent in $\mathbf{No}$. It does not Hamel-span $\mathbf{No}$.*

Proof. A finite-support normal form is, by definition, a finite real linear combination of distinct monomials, and uniqueness of normal form proves linear independence. Strictness follows from [example 3.10](#), whose support is infinite. $\square$

[write] Source 53 proves the same proposition with the same argument, and adds:

Corollary 4.6. *No Hamel basis of $\mathbf{No}$ can consist entirely of monomials, or even entirely of finite-support surreals.*

Proof. A finite combination of finite-support vectors again has finite support. $\square$

[write] Source 52 proves the version for every set-sized Hahn workspace H_Γ :

Proposition 4.7 (Uniqueness and finite-span obstruction). *The monomials are linearly independent over $\mathbb{R}$, and their finite linear span is exactly the finite-support subspace. When $\Gamma \neq 0$, this subspace is proper in H_Γ .*

Proof. In a finite relation between distinct monomials, each coefficient is a distinct normal-form coefficient and must vanish. A finite combination has finite support, and conversely any finite-support series is such a combination. Choose $\delta > 0$ in Γ . The support $\{n\delta : n \geq 0\}$ is well ordered, and

$$\sum_{n \geq 0}^{\text{H}} t^{n\delta} = \frac{1}{1 - t^\delta}$$

has infinitely many nonzero coefficients. The equality follows by multiplying by $1 - t^\delta$ and canceling coefficients; the required product and sum are Hahn-admissible. $\square$

In particular, with $t = \omega^{-1}$,

$$\frac{1}{1 - \omega^{-1}} = 1 + \omega^{-1} + \omega^{-2} + \dots$$

is already a counterexample to the claim that the Conway monomials are a Hamel basis. Notice how elementary the vector is: rationality as a *field expression* does not imply a finite real-linear monomial expansion. Thus the most natural algebraic direct sum is not the whole surreal field. Passing from $\mathbf{No}_{\text{fin}}$ to $\mathbf{No}$ is analogous to passing from polynomials to a Hahn-series field, not merely adjoining more basis vectors.

Remark 4.8 (Why “all Archimedean lines” is not a direct-sum answer). Each canonical line $\mathbb{R}\omega^\gamma$ is Archimedean as an ordered additive group, but $\mathbf{No}$ is not their algebraic direct sum. Moreover, there are many nonhomogeneous one-dimensional real subspaces such as $\mathbb{R}(\omega + 1)$. The omega map is what canonically selects one pure line per Archimedean scale.

4.3 Why the monomial basis is natural

The word “natural” can conceal taste. Here it is supported by five precise properties.

Principle 4.9 (Simplicity normalization). *For every Archimedean class of positive surreals, ω^γ is its unique simplest positive representative. Hence the basis vectors are selected by the defining simplicity hierarchy of surreal numbers, not by an external well ordering.*

Principle 4.10 (Multiplicative grading). *The index law*

$$\omega^\gamma \omega^\delta = \omega^{\gamma+\delta}$$

turns the additive exponent class into the multiplicative monomial class. The basis therefore interacts exactly with the field multiplication.

Principle 4.11 (Leading-coordinate order). *For nonzero x , the sign and Archimedean scale of x are determined by the coefficient and index of its largest supported monomial. Thus the basis is adapted to both the total order and the natural valuation.*

Principle 4.12 (Truncation stability). *Every subset of a reverse-well-ordered support is reverse well ordered. Consequently every coordinate truncation of a surreal normal form is again a surreal number.*

Principle 4.13 (Strong uniqueness). *The expansion remains unique even though arbitrary set-sized ordinal lengths are allowed. This is substantially stronger than a nonunique asymptotic expansion.*

No arbitrary Hamel basis supplied by choice is known to enjoy this package. Indeed, a Hamel basis must somehow encode infinite-support normal forms into individual basis vectors, destroying the one-monomial-per-scale geometry.

[write] Source 52 makes the qualification precise: the simplicity relation is extra structure beyond the ordered real-vector-space reduct, and a shear fixing $\mathbb{R}$ moves monomials while preserving order, leading values and leading coefficients ([section 9.4](#)).

4.4 Changing the scalar field

The special role of $\mathbb{R}$ can also be characterized by coefficient normalization.

Proposition 4.14 (Hahn bases over smaller coefficient fields). *Let $k \subseteq \mathbb{R}$ be a subfield and let C be a Hamel basis of $\mathbb{R}$ over k . Then*

$$\{c\omega^\gamma : c \in C, \gamma \in \mathbf{No}\} \quad (4.5)$$

is a Hahn basis of $\mathbf{No}$ over k : every surreal has a unique strongly summable expansion in these vectors, with finitely many C -coordinates at each exponent.

Proof. Write each real normal-form coefficient r_γ uniquely as a finite k -linear combination of elements of C . Replacing $r_\gamma\omega^\gamma$ by those finitely many terms preserves the union of exponent supports and local finiteness. Conversely, collecting the finitely many terms at each exponent recovers the unique real coefficient and then the unique Conway normal form. $\square$

Over $\mathbb{R}$, the coefficient basis is canonically $\{1\}$. Over $\mathbb{Q}$, for example, an additional highly noncanonical Hamel basis of $\mathbb{R}$ is unavoidable. This is another sense in which $\mathbb{R}$ is the natural scalar field for the monomial coordinates.

4.5 A first warning about topology

It is tempting to enumerate the support in decreasing order and call the normal form the limit of its finite initial sums. In ordinary Hahn fields this may work for selected supports and selected valuation topologies, but it fails globally for $\mathbf{No}$. The full reason is proved in [section 11](#); already the geometric series (3.10) is not the limit of its ordinary partial sums in the full natural valuation topology, because the tail valuations $1, 2, 3, \dots$ never exceed the surreal threshold ω . Strong summation, not sequential convergence, is the primitive operation.

5 Hamel bases, class size, and local dimensions

5.1 No set-sized family can span

The first obstruction to ordinary linear algebra is stronger than infinite-dimensionality.

Theorem 5.1 (No set spans $\mathbf{No}$). *If $S \subseteq \mathbf{No}$ is a set, then*

$$\text{span}_{\mathbb{R}}(S) \neq \mathbf{No}.$$

More precisely, there is an exponent $\gamma \in \mathbf{No}$ such that $\omega^\gamma \notin \text{span}_{\mathbb{R}}(S)$.

Proof. The union

$$U = \bigcup_{x \in S} \text{supp}(x)$$

is a set, by Replacement and Union. The support of every finite real linear combination of members of S is contained in U , since addition and scalar multiplication create no new exponents. The class $\mathbf{No}$ is proper, so choose $\gamma \notin U$. Then ω^γ has support $\{\gamma\}$ and cannot lie in the span. $\square$

Second proof (source 53; source 52 gives the same argument in [proposition 5.4](#)). If B is a set, the collection of finite sequences from B , finite sequences from $\mathbb{R}$, and their evaluated linear combinations is a set by Replacement and Union. Hence $\text{span}_{\mathbb{R}}(B)$ is a set. It cannot equal the proper class $\mathbf{No}$. $\square$

This argument is stronger than saying merely that the dimension is “infinite.” There is no set cardinal that can be the dimension of the entire class.

Corollary 5.2. *Every Hamel basis of $\mathbf{No}$ over $\mathbb{R}$, if one is available in the chosen class theory, is a proper class. In particular, $\mathbf{No}$ has no cardinal-valued Hamel dimension in the ordinary set-theoretic sense.*

The proof is support-theoretic and does not use birthdays, saturation, or field multiplication. It applies to many proper-class Hahn spaces.

5.2 Existence with global choice

The ordinary vector-space basis theorem applies only to sets. Nevertheless, NBG+GC supplies a class analogue by a transparent stage construction.

For an ordinal α , let $\mathbf{No}_{<\alpha}$ be the set of surreal numbers born before day α , and put

$$V_\alpha = \text{span}_{\mathbb{R}}(\mathbf{No}_{<\alpha}). \quad (5.1)$$

Each V_α is a set-sized real vector space, $V_\alpha \subseteq V_\beta$ for $\alpha < \beta$, and

$$\mathbf{No} = \bigcup_{\alpha \in \mathbf{On}} V_\alpha. \quad (5.2)$$

At a limit ordinal λ , every finite linear combination of elements born below λ already occurs at some earlier stage, so $V_\lambda = \bigcup_{\alpha < \lambda} V_\alpha$.

Theorem 5.3 (Existence of a Hamel class basis). *In NBG with global choice, $\mathbf{No}$ has a Hamel class basis over $\mathbb{R}$.*

Proof. Construct bases B_α by transfinite class recursion. Start with the empty basis of V_0 . At a successor stage, use the ordinary basis-extension theorem inside the set $V_{\alpha+1}$ to extend B_α to a basis $B_{\alpha+1}$ of $V_{\alpha+1}$. Global choice permits these extensions to be selected uniformly through all ordinals. At a limit λ , set $B_\lambda = \bigcup_{\alpha < \lambda} B_\alpha$; this is a basis of V_λ because every vector and every finite dependence occurs at an earlier stage. Finally let $B = \bigcup_{\alpha \in \mathbf{On}} B_\alpha$. Every surreal lies in some V_α and has a finite B_α -expansion, while every finite dependence among members of B occurs in one stage. Thus B is a Hamel class basis. $\square$

Proposition 5.4 (A Hamel basis may contain all monomials). *In NBG with global choice, the linearly independent class $\mathfrak{M}$ extends to a Hamel class basis B of $\mathbf{No}$. Every such extension has a proper class of additional, infinite-support vectors.*

Proof. Global choice gives a class enumeration $(x_\alpha)_{\alpha \in \mathbf{On}}$ of $\mathbf{No} \setminus \mathfrak{M}$. Recursively build a set $C_\alpha \subseteq \{x_\beta : \beta < \alpha\}$: at stage α , add x_α exactly when it is not in the finite real span of $\mathfrak{M} \cup C_\alpha$. Each stage adds at most one vector, so the previously added part is a set and elementary transfinite recursion applies. Put $C = \bigcup_{\alpha \in \mathbf{On}} C_\alpha$ and $B = \mathfrak{M} \cup C$.

A finite dependence in B would have a last-added member of C , which was added only because it was outside the span of all earlier members and $\mathfrak{M}$; hence B is independent. Every enumerated vector was either added or already in the span at its stage, so B spans $\mathbf{No}$. Finally, [corollary 5.8](#) shows that $B \setminus \mathbf{No}_{\text{fin}}$, and therefore also the supplement beyond the monomials, is a proper class. $\square$

[write] Source 52 states, as its proposition “What choice does and does not provide”:

Every set-sized real vector space has a Hamel basis under the axiom of choice, and every linearly independent subset extends to one. No set of surreal numbers spans $\mathbf{No}$ over $\mathbb{R}$. In class theory with global choice, there is a class Hamel basis of $\mathbf{No}$ containing all Conway monomials.

The first clause is the ordinary basis theorem and the second is [theorem 5.1](#). Source 52 works in Gödel–Bernays class theory with global choice, the same foundation as above, and proves the last clause differently.

Second proof (source 52), by a set-like global well-order. The set-sized assertion is the usual maximal-independent-set argument. If S is a set, its real linear span is the image of the set of finite lists from $\mathbb{R} \times S$ under finite summation, and is therefore a set. It cannot equal **No**.

For the last assertion, fix a set-like global well-order of the surreal class: every initial segment is a set. Traverse this order and select a surreal x exactly when

$$x \notin \text{span}_{\mathbb{R}}(\mathfrak{M} \cup C_{<x}),$$

where $C_{<x}$ is the set of previously selected elements. Membership in this span means existence of a *finite* combination and is an elementary formula with the class parameter $\mathfrak{M}$. On each set initial segment, ordinary transfinite recursion produces a unique selection; these constructions agree on overlaps and hence define a class C . Each skipped element is spanned, and each selected element enlarges the span. Every finite relation in $\mathfrak{M} \cup C$ is contradicted by its last selected member, or by independence of $\mathfrak{M}$ if it has none. Thus $\mathfrak{M} \cup C$ is a class Hamel basis. $\square$

This construction depends on a global well-order and is not a canonical coordinate formula. It also does not provide an algorithm for computing the finite Hamel expansion of a given surreal. The set-like construction is explained to avoid an invalid application of set-sized Zorn’s lemma to a proper class.

[write] Source 53 proves the same statement by staged Hahn fields. It writes:

The set-sized theorem “every vector space has a basis” does not automatically produce a basis for a proper class. A clean positive result is available once the class-recursion principle is stated explicitly.

Source 53’s statement.

Assume a class theory with global choice and enough elementary transfinite recursion to make coherent choices through all ordinals (for example, MK with global choice, or GBC augmented by a suitable ETR principle). Then **No** has a class Hamel basis. Moreover, one can construct such a basis containing every monomial ω^a .

[write] *Correction of the hypothesis.* “MK with global choice, or GBC augmented by a suitable ETR principle” is more than the construction needs. In GBC (NBG with global choice) the recursion below is set-valued with a class parameter: every stage B_α is a set. NBG proves such recursions, and global choice selects the extensions, so neither MK nor an ETR principle is needed. This is the foundation of [theorem 5.3](#) and of the two proofs above; the repository uses the same principle ([surcomplex/trigonometry](#), its set-valued class recursion). Source 53’s construction is kept as a third route, and its question on the weakest principle is re-scoped in [research question 21.1](#).

Third proof (source 53), by staged Hahn fields. For each ordinal α , let Γ_α be the set-sized additive subgroup of **No** generated by the exponents born before α , and let

$$W_\alpha = \mathbb{R}((\omega^{\Gamma_\alpha})) \subseteq \mathbf{No}.$$

Each W_α is a set-sized real vector space, the chain is increasing, and $\bigcup_{\alpha \in \mathbf{On}} W_\alpha = \mathbf{No}$: the support of any one surreal is a set, so the birthdays of its exponents have an ordinal bound.

Use class recursion to choose nested bases B_α of W_α . At successor stages, first adjoin all newly available monomials; their distinct new exponents make them independent modulo the previous Hahn field. Extend this independent set to a basis by the ordinary set-sized basis theorem. At a limit stage, the union of the earlier bases is independent in W_α and may again be extended to a basis. The class union $B = \bigcup_\alpha B_\alpha$ is independent and spans every W_α , hence all of $\mathbf{No}$, and contains all monomials. $\square$

Remark 5.5. The proposition is an existence statement, not a naturality statement. The extension steps make vast arbitrary choices. Conway normal form does not single out the added basis vectors, and two such Hamel bases need not bear any useful relation to the order, valuation, multiplication, simplicity hierarchy, or exponent symmetries.

Remark 5.6 (Existence is not naturality). The construction makes class-many arbitrary extension choices. It proves that a Hamel basis exists under a stated foundation; it does not identify a basis respected by normal-form truncations, multiplication, order, the omega map, exponentiation, or the canonical derivations. The Conway monomials are canonical precisely because they solve a different, strong-summation basis problem.

5.3 The quotient by finite support is already class-large

The failure of the monomials to Hamel-span $\mathbf{No}$ is not repaired by adding only a set of infinite-support vectors. We give an explicit witness.

For each ordinal α , define the disjoint reverse-well-ordered block

$$E_\alpha = \{-(\omega\alpha + n) : n < \omega\} \subseteq \mathbf{No}, \quad (5.3)$$

where $\omega\alpha$ denotes ordinal multiplication, viewed as a surreal ordinal, and put

$$u_\alpha = \sum_{n < \omega}^H \omega^{-(\omega\alpha + n)}. \quad (5.4)$$

The intervals $[\omega\alpha, \omega\alpha + \omega)$ are disjoint, hence so are the supports E_α .

Theorem 5.7 (A proper-class independent quotient family). *The cosets*

$$\{u_\alpha + \mathbf{No}_{\text{fin}} : \alpha \in \mathbf{On}\}$$

are real-linearly independent in $\mathbf{No}/\mathbf{No}_{\text{fin}}$.

Proof. Take distinct ordinals $\alpha_1, \dots, \alpha_m$ and real numbers $c_1, \dots, c_m$. Because the supports E_{α_j} are pairwise disjoint, no cancellation occurs between different blocks. If some $c_j \neq 0$, the support of $\sum_j c_j u_{\alpha_j}$ contains the entire infinite block E_{α_j} , and therefore the sum does not belong to $\mathbf{No}_{\text{fin}}$. Hence a linear combination represents zero in the quotient only when every coefficient is zero. $\square$

Corollary 5.8. *Every Hamel class basis B of $\mathbf{No}$ contains a proper class of vectors outside $\mathbf{No}_{\text{fin}}$.*

Proof. The images in $\mathbf{No}/\mathbf{No}_{\text{fin}}$ of the members of $B \setminus \mathbf{No}_{\text{fin}}$ span the quotient. If $B \setminus \mathbf{No}_{\text{fin}}$ were a set, its real span would be a set and could not contain the proper-class family of distinct independent cosets from [theorem 5.7](#). $\square$

This result sharpens the slogan that a Hamel basis is nonmonomial. It must contain class-many genuinely infinite normal forms.

No set-sized repair of the monomial family (source 52)

Write $\mathbf{No}_{\text{fin}} = \text{span}_{\mathbb{R}}\{t^\gamma : \gamma \in \mathbf{No}\}$. This is a class, so ordinary cosets of it are generally proper classes rather than elements of a set quotient.

Theorem 5.9 (No set-sized repair of the monomial family). *For every set $S \subseteq \mathbf{No}$,*

$$\text{span}_{\mathbb{R}}(\mathfrak{M} \cup S) \neq \mathbf{No}.$$

There is also a class-indexed family $(u_\alpha)_{\alpha \in \mathbf{On}}$ such that every nontrivial finite linear combination lies outside $\mathbf{No}_{\text{fin}}$.

Proof. The union T of the valuation supports of the members of S is a set of surreals, hence has an upper bound. Choose a positive ordinal η above all its elements. Then

$$u = \sum_{n \geq 0}^H t^{\eta+n}$$

has infinite support disjoint from T . Any vector in $\text{span}_{\mathbb{R}} S$ is supported in T , so no finite-monomial correction of such a vector equals u .

For the family assertion, let $\eta_\alpha = \omega^{\alpha+1}$ and put

$$u_\alpha = \sum_{n \geq 0}^H t^{\eta_\alpha+n}.$$

For $\alpha \neq \beta$, $\eta_\alpha - \eta_\beta$ is not an integer, so these supports lie in disjoint cosets modulo $\mathbb{Z}$. A nontrivial finite combination retains an infinite support. $\square$

In particular, any class Hamel basis containing $\mathfrak{M}$ has a proper class of additional elements. This is a statement about finite relations and set-sized repairs, not an illicit cardinal calculation with a proper-class quotient.

5.4 Every Hamel basis must contain enormous supports

Theorem 5.10 (Unavoidable support complexity). *Let B be any class Hamel basis of $\mathbf{No}$ over $\mathbb{R}$. For every infinite cardinal κ , some $b \in B$ has*

$$|\text{supp}(b)| \geq \kappa.$$

Thus no fixed set cardinal bounds the support sizes of the elements of a Hamel basis.

Proof. For an infinite initial ordinal κ , the series

$$x_\kappa = \sum_{\alpha < \kappa} \omega^{-\alpha}$$

is a surreal with support of cardinality κ . Write $x_\kappa = \sum_{j=1}^n r_j b_j$ with $b_j \in B$. Coefficientwise addition gives

$$\text{supp}(x_\kappa) \subseteq \bigcup_{j=1}^n \text{supp}(b_j).$$

A finite union of sets all of cardinality less than the infinite cardinal κ has cardinality less than κ . Hence at least one b_j has support cardinality at least κ . $\square$

Corollary 5.11. *There is no class Hamel basis all of whose elements have countable support, bounded ordinal support length, or support cardinality below any prescribed infinite cardinal.*

Remark 5.12 ([write] Three clauses of one phenomenon). The three sources prove three strengths of the same obstruction. [Theorem 5.10](#) (source 53) implies [corollary 5.8](#) (source 51): if the members of a Hamel basis outside $\mathbf{No}_{\text{fin}}$ formed a set, their support cardinalities would be bounded by a set cardinal. [Corollary 5.8](#) implies the final sentence of [theorem 5.9](#) (source 52), because $\mathfrak{M} \subseteq \mathbf{No}_{\text{fin}}$. The two explicit **On**-indexed families differ: source 51 uses the disjoint blocks $E_\alpha = \{-(\omega\alpha + n)\}$, source 52 the cosets $\eta_\alpha + \mathbb{Z}$ with $\eta_\alpha = \omega^{\alpha+1}$, in t -exponents.

5.5 Set-sized support windows

Class size should not hide the familiar cardinal arithmetic visible in every set-sized coordinate window. Let $A \subseteq \mathbf{No}$ be a reverse-well-ordered set and define

$$H(A) = \{x \in \mathbf{No} : \text{supp}(x) \subseteq A\}. \quad (5.5)$$

Every subset of A is reverse well ordered, so coefficient extraction gives an ordinary vector-space isomorphism

$$H(A) \cong \mathbb{R}^A. \quad (5.6)$$

The monomials indexed by A form a Hahn basis of $H(A)$, but only a Hamel basis of its finite-support subspace.

[write] Source 53 calls $H(A)$ a support slice and also names its finite part:

Global class dimension has no set cardinal value, but natural set-sized pieces do. They already show how far a Hamel basis is from the monomial coordinate family.

Definition 5.13 (Support slice). Let $A \subseteq \mathbf{No}$ be a set that is reverse well ordered. Define

$$H(A) = \{x \in \mathbf{No} : \text{supp}(x) \subseteq A\}, \quad H(A)^{\text{fin}} = H(A) \cap \mathbf{No}_{\text{fin}}.$$

Because every subset of a reverse-well-ordered set is again reverse well ordered, there is no further support restriction inside A .

Theorem 5.14 (Full product theorem for a support slice). *The coefficient map restricts to a canonical vector-space isomorphism*

$$H(A) \cong \mathbb{R}^A, \quad H(A)^{\text{fin}} \cong \mathbb{R}^{(A)},$$

where $\mathbb{R}^{(A)}$ denotes the finitely supported functions.

Proof. For any function $f : A \rightarrow \mathbb{R}$, the nonzero support of f is a subset of A and is therefore reverse well ordered. Hence it defines a surreal normal form. The inverse map is coefficient extraction. The finite-support assertion is immediate. $\square$

Theorem 5.15 (Hamel dimension of a support window). *If A is an infinite reverse-well-ordered set and $\kappa = |A|$, then*

$$\dim_{\mathbb{R}} H(A) = |\mathbb{R}|^\kappa, \quad (5.7)$$

$$\dim_{\mathbb{R}} H(A)^{\text{fin}} = \kappa, \quad (5.8)$$

$$\dim_{\mathbb{R}} (H(A)/H(A)^{\text{fin}}) = |\mathbb{R}|^\kappa. \quad (5.9)$$

[write] (5.7) is source 51's statement; source 53 (its "Exact local Hamel dimension") states all three.

Proof of (5.7) (source 51). By (5.6), this is the Erdős–Kaplansky theorem $\dim_D D^I = |D^I|$ for an infinite index set I and division ring D [13, 5]. Taking $D = \mathbb{R}$ and $I = A$ gives the formula. $\square$

Proof of all three (source 53). By theorem 5.14, the first equality is the Erdős–Kaplansky theorem

$$\dim_K K^I = |K|^{|I|}$$

for an infinite index set I [15]. The coordinate vectors form a Hamel basis of the direct sum, giving (5.8). Put $\mu = |\mathbb{R}|^\kappa$. Since $\mu \geq 2^\kappa > \kappa$, the dimension identity

$$\dim H(A) = \dim H(A)^{\text{fin}} + \dim(H(A)/H(A)^{\text{fin}})$$

for cardinal dimensions forces the quotient dimension to be μ . $\square$

Corollary 5.16 (A countable window has continuum dimension). *Let $A = \{\gamma_0 > \gamma_1 > \dots\}$ be countably infinite. Then*

$$\dim_{\mathbb{R}} H(A) = 2^{\aleph_0}.$$

An explicit continuum-sized independent family is

$$x_s = \sum_{n < \omega}^H s^n \omega^{\gamma_n}, \quad s \in \mathbb{R}. \quad (5.10)$$

Proof. The space has cardinality $|\mathbb{R}|^{\aleph_0} = 2^{\aleph_0}$, so its dimension is at most the continuum. To prove the matching lower bound, take distinct $s_1, \dots, s_m \in \mathbb{R}$ and suppose $\sum_{j=1}^m c_j x_{s_j} = 0$. Comparing the first m coordinates gives

$$\sum_{j=1}^m c_j t_j^n = 0, \quad n = 0, \dots, m-1.$$

The Vandermonde matrix (s_j^n) is invertible, hence all $c_j = 0$. Thus $\{x_s : s \in \mathbb{R}\}$ is linearly independent. $\square$

Remark 5.17 (Source 53’s countable tail). [write] Source 53 states, as a corollary of theorem 5.15: for

$$A = \{0, -1, -2, \dots\},$$

the space of series $\sum_{n \geq 0} r_n \omega^{-n}$ has real Hamel dimension $2^{\aleph_0}$, whereas its monomial span has countable dimension. The quotient by finite Laurent polynomials also has dimension $2^{\aleph_0}$.

Its theorem “Geometric-series Vandermonde family” is the family (5.10) on this window, $g_s = 1/(1 - s\omega^{-1}) = \sum_{n \geq 0} s^n \omega^{-n}$ for $s \in \mathbb{R}$, proved by the same Vandermonde determinant $\prod_{i < j} (s_j - s_i) \neq 0$.

This is the exact analogue of the distinction between the sequence space $\mathbb{R}^{\mathbb{N}}$ and its finitely supported subspace. The surreal order and field multiplication add structure, but they do not reduce the algebraic dimension.

Remark 5.18 (Source 53 on the Vandermonde family). The same family is used in standard proofs about product-space dimensions. Here it has a direct surreal interpretation: a one-parameter family of inverses of linear infinitesimal perturbations already produces continuum many independent vectors.

Example 5.19 (The gap between coordinate and Hamel dimensions). On a countable support window, the canonical monomial Hahn basis has cardinality $\aleph_0$, while every Hamel basis has cardinality continuum. The difference is exactly the information carried by arbitrary infinite coordinate strings.

5.6 Product topology versus algebraic dimension

Via $H(A) \cong \mathbb{R}^A$, one may put the product topology on a support slice. Then $H(A)^{\text{fin}}$ is dense: every basic neighborhood specifies only finitely many coordinates, which a finite-support vector can match. If A is uncountable, this density generally requires nets rather than sequences. This coefficientwise topology is radically different from the natural valuation topology studied in [section 11](#), where the same finite-support subspace is closed.

5.7 The full Hahn field of a set exponent group (source 52)

[write] Source 52 measures the defect of the monomial span in the whole Hahn field H_Γ , not in a fixed window. Its Vandermonde family $q_r = (1 - rt^\delta)^{-1}$ is the family [\(5.10\)](#) up to a change of window.

Theorem 5.20 (The missing part has full dimension). *Let Γ be a nonzero set-sized ordered abelian group. Then*

$$\dim_{\mathbb{R}}(H_\Gamma/\text{Fin}_\Gamma) = \dim_{\mathbb{R}} H_\Gamma = |H_\Gamma|. \quad (5.11)$$

Proof. Fix $\delta > 0$. For $r \in \mathbb{R}_{>0}$ put

$$q_r = \sum_{n \geq 0}^H r^n t^{n\delta} = \frac{1}{1 - rt^\delta}.$$

We first show that the cosets $q_r + \text{Fin}_\Gamma$ are linearly independent. If $r_1, \dots, r_m$ are distinct and

$$\sum_{j=1}^m a_j q_{r_j} \in \text{Fin}_\Gamma,$$

then $\sum_j a_j r_j^n = 0$ for all sufficiently large n . Taking m consecutive values $n = N, \dots, N + m - 1$ gives a matrix with determinant

$$\left(\prod_{j=1}^m r_j^N \right) \prod_{i < j} (r_j - r_i) \neq 0.$$

Thus all a_j vanish, proving $\dim_{\mathbb{R}}(H_\Gamma/\text{Fin}_\Gamma) \geq \mathfrak{c}$.

If $|\Gamma| > \mathfrak{c}$, choose representatives A for $\Gamma/\mathbb{Z}\delta$. Since each coset is countable, $|A| = |\Gamma|$. The series

$$u_a = \sum_{n \geq 0}^H t^{a+n\delta} \quad (a \in A)$$

have pairwise disjoint infinite supports. Every nontrivial finite linear combination has infinite support, so their cosets modulo Fin_Γ are independent. Together with the previous paragraph this proves

$$d := \dim_{\mathbb{R}}(H_\Gamma/\text{Fin}_\Gamma) \geq \max(\mathfrak{c}, |\Gamma|).$$

The dimension formula for a subspace and quotient gives

$$\dim_{\mathbb{R}} H_\Gamma = \dim_{\mathbb{R}} \text{Fin}_\Gamma + d = |\Gamma| + d = d.$$

Finally, an infinite-dimensional vector space over $\mathbb{R}$ with basis cardinal d has cardinality $\max(\mathfrak{c}, d)$. Since $d \geq \mathfrak{c}$, this equals d , proving the theorem. $\square$

The Vandermonde argument is the familiar cardinal-linear-algebra mechanism appearing in the Erdős–Kaplansky context; compare [\[15\]](#). The application modulo the finite-support span and the disjoint-coset argument above are given explicitly rather than inferred from cardinality alone.

Corollary 5.21 (Countable and real exponent groups). *If $\Gamma \neq 0$ is countable, or if $0 \neq \Gamma \subseteq \mathbb{R}$, then*

$$\dim_{\mathbb{R}} H_{\Gamma} = \dim_{\mathbb{R}} (H_{\Gamma} / \text{Fin}_{\Gamma}) = \mathfrak{c}.$$

Proof. For countable Γ , there are at most $\mathfrak{c}$ coefficient functions. For $\Gamma \subseteq \mathbb{R}$, every well-ordered subset is countable: assign a rational number in the interval between each nonfinal element and its successor. These intervals are disjoint, so there are only countably many nonfinal elements, and at most one final element. There are at most $\mathfrak{c}$ countable supports and $\mathfrak{c}$ coefficient assignments on each. Constants give the reverse cardinal inequality. Apply Theorem 5.20. $\square$

For $\Gamma = \mathbb{Z}$ the monomials form a countable independent family, but one must add continuum many Hamel directions. For $\Gamma = \mathbb{R}$, there are already continuum many monomials and one must still add continuum many directions. Equal infinite cardinality never proves that an independent family spans.

5.8 What a “dimension of No” can mean

There is no set cardinal κ with $\dim_{\mathbb{R}} \mathbf{No} = \kappa$, because no set spans. Three replacements are meaningful.

1. The *class Hamel dimension* is represented by any Hamel basis class, but comparing proper-class sizes requires the chosen class theory and is too coarse to capture support.
2. The *graded dimension* is one at each exponent (section 6).
3. The *window dimension profile* $A \mapsto |\mathbb{R}|^{|A|}$ records the ordinary Hamel dimensions of set-sized Hahn blocks.

The second and third are canonical and structurally informative; the first is mostly an existence invariant.

Part II

Canonical linear structure

6 Canonical projections, flags, and the associated graded

The monomial coordinates provide much more than a list of generators. They split $\mathbf{No}$ functorially by arbitrary exponent classes and resolve its natural valuation into one-dimensional layers.

6.1 Support projections

For a subclass $A \subseteq \mathbf{No}$, define

$$P_A(x) = \sum_{\gamma \in \text{supp}(x) \cap A}^H [\omega^{\gamma}] x \omega^{\gamma}. \quad (6.1)$$

This is well defined because $\text{supp}(x) \cap A$ is a subset of a reverse-well-ordered set.

Theorem 6.1 (Canonical support splitting). *For every subclass $A \subseteq \mathbf{No}$, P_A is a strongly summation-preserving real-linear idempotent and*

$$\mathbf{No} = \mathbf{No}[A] \oplus \mathbf{No}[\mathbf{No} \setminus A], \quad \mathbf{No}[A] = \{x : \text{supp}(x) \subseteq A\}. \quad (6.2)$$

The projections satisfy

$$P_A P_B = P_{A \cap B}, \quad (6.3)$$

$$P_{\mathbf{No} \setminus A} = \text{id} - P_A, \quad (6.4)$$

$$P_{A \cup B} = P_A + P_B - P_{A \cap B}. \quad (6.5)$$

Thus exponent subclasses act as a Boolean algebra of canonical linear projections.

Proof. Coefficientwise linearity and idempotence are immediate. Every normal form splits uniquely into the terms with exponent in A and those outside A , proving (6.2). If (x_i) is Hahn summable, the union of the supports of $(P_A x_i)$ is a subset of the original reverse-well-ordered union, and local finiteness at every exponent is preserved. Hence P_A is strong. The Boolean identities are checked on each coordinate. $\square$

Important special cases include:

1. the single-coordinate projection $P_{\{\gamma\}}x = [\omega^\gamma]x \omega^\gamma$;
2. initial and final truncations at an exponent δ ;
3. the decomposition into positive, zero, and negative exponents;
4. projections onto ordinal exponents, real exponents, or any definable scale family.

No Hamel-basis choice is needed.

[write] Source 53 proves the same calculus coordinatewise, listing also $P_\emptyset = 0$, $P_{\mathbf{No}} = \text{id}$ and $P_{S \cup T} = P_S + P_T$ for disjoint S, T , and derives:

Corollary 6.2 (Cut decomposition). *For every exponent a ,*

$$\mathbf{No} = \mathbf{No}_{>a} \oplus \mathbb{R}\omega^a \oplus \mathbf{No}_{<a}, \quad (6.6)$$

where the three summands consist of series supported above a , at a , and below a . The projections are $P_{(a, \infty)}$, $P_{\{a\}}$, and $P_{(-\infty, a)}$.

Setting $a = 0$ recovers purely infinite, real, and infinitesimal parts. More generally, a set-indexed partition $(S_i)_{i \in I}$ of the exponent class gives, for each x , a strongly summable decomposition

$$x = \sum_{i \in I}^H P_{S_i} x;$$

only set-many summands can be nonzero because $\text{supp}(x)$ is a set.

Coordinate-resolved subspaces of a Hahn workspace (source 52)

[write] Source 52 states the same splitting on H_Γ (in t -exponents) and classifies the subspaces it resolves.

For $A \subseteq \Gamma$ let

$$H[A] = \{x \in H_\Gamma : \text{supp}_t x \subseteq A\}, \quad P_A(x) = \sum_{\gamma \in A}^H x_\gamma t^\gamma.$$

Every P_A is a strong projection, $P_A P_B = P_{A \cap B}$, and

$$H_\Gamma = H[A] \oplus H[\Gamma \setminus A].$$

The subsets need not be subgroups; these are vector-space decompositions, not generally field decompositions.

Proposition 6.3 (Classification with all coordinate projections). *A real subspace $V \subseteq H_\Gamma$ is closed under all admissible Hahn sums and invariant under every one-coordinate projection if and only if $V = H[A]$ for a unique $A \subseteq \Gamma$.*

Proof. Let $A = \{\gamma : t^\gamma \in V\}$. If $x \in V$ and $x_\gamma \neq 0$, its coordinate projection puts t^γ in V , so $\text{supp}_t x \subseteq A$. Conversely every series supported in A is an admissible sum of monomials already in V . The reverse implication follows immediately from the support definitions. $\square$

This theorem provides an exact sense in which support geometry replaces arbitrary basis selection. Strong closure without coordinate invariance is a different, more difficult subspace problem.

Warning 6.4. Support projections are real-linear, not generally field homomorphisms. For example, constant-term extraction is multiplicative on certain support subrings but not on all of $\mathbf{No}$, because exponents from different factors can add to zero.

6.2 The dominance flag

For $\gamma \in \mathbf{No}$, define

$$F_{\leq \gamma} = \{x \in \mathbf{No} : \text{supp}(x) \subseteq (-\infty, \gamma]\}, \quad (6.7)$$

$$F_{< \gamma} = \{x \in \mathbf{No} : \text{supp}(x) \subseteq (-\infty, \gamma)\}. \quad (6.8)$$

Equivalently, a nonzero x belongs to $F_{\leq \gamma}$ exactly when $\text{lead}(x) \leq \gamma$. These subspaces form an increasing proper-class flag.

Proposition 6.5 (One-dimensional graded quotients). *For every $\gamma \in \mathbf{No}$, coefficient extraction induces a canonical isomorphism*

$$F_{\leq \gamma}/F_{< \gamma} \cong \mathbb{R}, \quad x + F_{< \gamma} \mapsto [\omega^\gamma]x. \quad (6.9)$$

The class of ω^γ is the distinguished basis vector of this quotient.

Proof. The kernel of the coefficient map on $F_{\leq \gamma}$ is precisely $F_{< \gamma}$, and every real coefficient occurs as $r\omega^\gamma$. $\square$

[write] Source 53 puts the same point this way:

A Hamel basis is noncanonical because it tries to encode an entire infinite tail by finitely many chosen vectors. The valuation filtration avoids that problem by retaining one leading scale at a time.

Remark 6.6 (Source 53's form, with the canonical splitting). [write] Source 53 states [proposition 6.5](#) as a split exact sequence.

For every $a \in \mathbf{No}$, coefficient extraction induces a short exact sequence

$$0 \longrightarrow F_{< a} \longrightarrow F_{\leq a} \xrightarrow{c_a} \mathbb{R} \longrightarrow 0. \quad (6.10)$$

It is canonically split by $r \mapsto r\omega^a$. Hence

$$F_{\leq a} = F_{< a} \oplus \mathbb{R}\omega^a, \quad F_{\leq a}/F_{< a} \cong \mathbb{R}.$$

Proof. The kernel of c_a on $F_{\leq a}$ consists precisely of the series with no a -term, hence with leading exponent below a . Surjectivity and the splitting follow from $c_a(r\omega^a) = r$. $\square$

Thus $\mathbf{No}$ has one real direction at every Archimedean scale, plus the support-completion needed to combine set-many scales.

6.3 Associated graded algebra

Let

$$\mathrm{gr}(\mathbf{No}) = \bigoplus_{\gamma \in \mathbf{No}} F_{\leq \gamma} / F_{< \gamma}. \quad (6.11)$$

The direct sum is algebraic: an element has only finitely many nonzero homogeneous coordinates. Multiplication of leading terms makes this a graded $\mathbb{R}$ -algebra.

Theorem 6.7 (The associated graded is the group algebra). *There is a canonical graded $\mathbb{R}$ -algebra isomorphism*

$$\mathrm{gr}(\mathbf{No}) \cong \mathbb{R}[\mathbf{No}_{\mathrm{add}}] \cong \mathbf{No}_{\mathrm{fin}}, \quad (6.12)$$

where $\mathbb{R}[\mathbf{No}_{\mathrm{add}}]$ is the finite-support group algebra of the additive surreal class. Under this isomorphism the homogeneous symbol of ω^γ is the group-algebra element e_γ , with

$$e_\gamma e_\delta = e_{\gamma+\delta}.$$

Proof. By [proposition 6.5](#), the grade at γ is one-dimensional with basis the symbol of ω^γ . If $x = r\omega^\gamma + \text{lower terms}$ and $y = s\omega^\delta + \text{lower terms}$, then

$$xy = rs\omega^{\gamma+\delta} + \text{terms of exponent below } \gamma + \delta.$$

Hence multiplication of homogeneous symbols is exactly the group-algebra law. The finite-support normal forms identify that group algebra with $\mathbf{No}_{\mathrm{fin}}$. $\square$

Remark 6.8 (What completion adds). The associated graded remembers every individual dominance layer and their multiplicative law, but it forgets the admissible infinite assembly of those layers. Recovering $\mathbf{No}$ from $\mathrm{gr}(\mathbf{No})$ therefore requires the reverse-well-ordered support rule. This is the precise sense in which $\mathbf{No}$ is a Hahn completion of its graded skeleton, rather than an ordinary linear span.

[write] Source 53 proves [theorem 6.7](#) from the multiplicativity of the filtration, writing T^a for the group-algebra monomial e_a :

The filtration is multiplicative:

$$F_{\leq a} F_{\leq b} \subseteq F_{\leq a+b}.$$

Therefore it has an associated graded class algebra

$$\mathrm{gr} \mathbf{No} = \bigoplus_{a \in \mathbf{No}} F_{\leq a} / F_{< a}.$$

Corollary 6.9 (Principal symbol). *For $x \neq 0$, the initial form*

$$\mathrm{in}(x) = \mathrm{lc}(x) T^{\mathrm{lead}(x)}$$

satisfies

$$\mathrm{in}(xy) = \mathrm{in}(x) \mathrm{in}(y).$$

This is the strongest canonical ordinary-basis statement available: one monomial basis vector survives in each scale quotient, and multiplication of leading terms is exact.

6.4 Canonical coefficient exact sequences

For any exponent subclass A , restriction of coefficients gives a split exact sequence of real vector classes

$$0 \longrightarrow \mathbf{No}[\mathbf{No} \setminus A] \longrightarrow \mathbf{No} \xrightarrow{P_A} \mathbf{No}[A] \longrightarrow 0. \quad (6.13)$$

For a singleton $A = \{\gamma\}$, composition with $r\omega^\gamma \mapsto r$ gives

$$0 \longrightarrow \ker[\omega^\gamma] \longrightarrow \mathbf{No} \xrightarrow{[\omega^\gamma]} \mathbb{R} \longrightarrow 0. \quad (6.14)$$

These canonical splittings will supply the simplest nonzero members of the strong dual in [section 10](#). [write] The next subsection is source 53's. The classification is classical (Hahn's embedding theory and Conrad's theory of convex subgroups of ordered groups: convex subgroups of a Hahn group correspond to initial segments of its value chain); source 53 gives a direct proof and claims no priority ([section 1.5](#)).

6.5 Convex subspaces are exactly exponent cuts

A real subspace $V \subseteq \mathbf{No}$ is *convex* if $x, z \in V$ and $x \leq y \leq z$ imply $y \in V$. For an initial (downward-closed) subclass $C \subseteq \mathbf{No}$, put

$$V_C = \{0\} \cup \{x \neq 0 : \text{lead}(x) \in C\}.$$

Since C is downward closed, this is equivalently the class of all normal forms supported in C .

Theorem 6.10 (Classification of convex real subspaces). *The assignments*

$$C \mapsto V_C, \quad V \mapsto C_V := \{a \in \mathbf{No} : \omega^a \in V\}$$

are mutually inverse order isomorphisms between

- (i) *downward-closed subclasses of the exponent order $\mathbf{No}$, and*
- (ii) *convex $\mathbb{R}$ -subspaces of the ordered vector space $\mathbf{No}$.*

Proof. First let C be downward closed. If $0 \leq x \leq y$ and $y \in V_C$, comparison of leading terms gives $\text{lead}(x) \leq \text{lead}(y) \in C$, so $x \in V_C$. Thus V_C is convex; closure under addition and real scalars follows from support containment.

Conversely let V be convex. If $a \in C_V$ and $b < a$, then $0 < \omega^b < \omega^a$, so convexity gives $\omega^b \in V$; hence C_V is downward closed. Now take $x \neq 0$ with leading exponent a . Its leading-term factorization has the form

$$x = \omega^a r(1 + \varepsilon), \quad r \in \mathbb{R}^\times, \quad |\varepsilon| < 1/n \text{ for all } n \geq 1.$$

Thus $|x|$ and ω^a bound positive real multiples of one another. A convex real subspace contains x if and only if it contains ω^a . Therefore $x \in V$ if and only if $a \in C_V$, proving $V = V_{C_V}$ and the inverse identity. $\square$

Corollary 6.11. *The principal filtration pieces $F_{<a}$ and $F_{\leq a}$ are convex. Nonprincipal cuts in the exponent class give additional convex subspaces. Hence the lattice of convex real subspaces is canonically the lattice of cuts of the exponent order.*

This result explains why the monomials are natural even without a basis theorem: they parameterize the Archimedean scale layers and completely control convex linear geometry.

7 Diagonal operators, commutants, and positivity

[write] Source 51 introduces the diagonal maps alongside strong maps (section 10); source 53 develops them into a spectral calculus. Both are printed here; “strong” refers to definition 10.1.

7.1 Diagonal maps and support projections

For an arbitrary class function $a : \mathbf{No} \rightarrow \mathbb{R}$, define

$$D_a \left(\sum_{\gamma}^H r_{\gamma} \omega^{\gamma} \right) = \sum_{\gamma}^H a(\gamma) r_{\gamma} \omega^{\gamma}. \quad (7.1)$$

The output support is contained in the input support, so D_a is strong.

Proposition 7.1 (Diagonal operator calculus). *For class functions $a, b : \mathbf{No} \rightarrow \mathbb{R}$,*

$$D_a D_b = D_{ab}, \quad (7.2)$$

$$D_a + D_b = D_{a+b}. \quad (7.3)$$

Moreover:

1. D_a is injective exactly when $a(\gamma) \neq 0$ for every γ ;
2. D_a is a strong automorphism exactly when every $a(\gamma) \neq 0$, with inverse $D_{1/a}$;
3. D_a is idempotent exactly when $a(\gamma) \in \{0, 1\}$ for every γ .

In particular, the support projection P_A is D_{1_A} .

Proof. All assertions follow by checking each monomial coordinate. Over the field $\mathbb{R}$, the scalar equation $a^2 = a$ has only the solutions 0 and 1. $\square$

[write] Source 53 proves the same calculus (its “Diagonal algebra”, with $D_1 = \text{id}$) and writes D_{λ} for D_a ; its corollary that a diagonal operator is idempotent exactly when it is a support projection is clause 3 of proposition 7.1.

The coefficient projections behave like a complete family of one-dimensional spectral projections. This analogy can be made exact without introducing an inner product or a norm.

Definition 7.2 (Diagonal operator). For a class function $\lambda : \mathbf{No} \rightarrow \mathbb{R}$, define

$$D_{\lambda} x = \sum_{a \in \text{supp}(x)} \lambda(a) c_a(x) \omega^a. \quad (7.4)$$

The support can only shrink, so this is a well-defined surreal number.

Theorem 7.3 (Commutant theorem). *Let $T : \mathbf{No} \rightarrow \mathbf{No}$ be an $\mathbb{R}$ -linear class map. The following are equivalent.*

- (i) T commutes with every one-coordinate projection $P_{\{a\}}$.
- (ii) There is a unique class function $\lambda : \mathbf{No} \rightarrow \mathbb{R}$ such that $T = D_{\lambda}$.

Thus the commutant of the complete monomial projection family is exactly the coefficientwise diagonal algebra.

Proof. Every diagonal operator plainly commutes with every coordinate projection. Conversely, suppose $TP_{\{a\}} = P_{\{a\}}T$. Since $P_{\{a\}}\omega^a = \omega^a$, we have

$$P_{\{a\}}T(\omega^a) = T(\omega^a),$$

so $T(\omega^a)$ lies in the one-dimensional range $\mathbb{R}\omega^a$. Write $T(\omega^a) = \lambda(a)\omega^a$. For arbitrary x ,

$$P_{\{a\}}T(x) = T(P_{\{a\}}x) = T(c_a(x)\omega^a) = c_a(x)\lambda(a)\omega^a.$$

Hence the a -coefficient of $T(x)$ is $\lambda(a)c_a(x)$ for every a . Coefficients separate points by [corollary 3.7](#), so $T = D_\lambda$. Uniqueness follows by evaluation on ω^a . $\square$

[write] Source 52 proves a different theorem with the same conclusion: on a real-exponent workspace, every real-linear map commuting with the single Euler operator $\mathfrak{D}$ is a coefficient multiplier ([theorem 14.1](#)).

7.2 Which diagonal maps preserve order?

A linear map T on an ordered vector space is *positive* if $x \geq 0$ implies $T(x) \geq 0$. Because surreal order is determined by the first nonzero coefficient, positivity of D_λ has a sharp and slightly surprising classification.

Theorem 7.4 (Positive diagonal operators). *Let $\lambda : \mathbf{No} \rightarrow \mathbb{R}$, and put*

$$U_\lambda = \{a \in \mathbf{No} : \lambda(a) \neq 0\}.$$

Then D_λ is positive if and only if

- (i) $\lambda(a) \geq 0$ for every a , and*
- (ii) U_λ is a final (upward-closed) subclass of the exponent order.*

Furthermore, D_λ is an order automorphism if and only if $\lambda(a) > 0$ for every a .

Proof. Assume first that D_λ is positive. Applying it to the positive monomial ω^a gives $\lambda(a)\omega^a \geq 0$, hence $\lambda(a) \geq 0$. If $b < a$, $\lambda(b) > 0$, and $\lambda(a) = 0$, then

$$x = \omega^a - \omega^b > 0$$

but

$$D_\lambda x = -\lambda(b)\omega^b < 0,$$

a contradiction. Thus nonvanishing propagates upward, so U_λ is final.

Conversely, let $x > 0$ have leading exponent a and positive leading coefficient. If $a \in U_\lambda$, the leading coefficient of $D_\lambda x$ is multiplied by the positive number $\lambda(a)$, so $D_\lambda x > 0$. If $a \notin U_\lambda$, finality implies that no exponent below a lies in U_λ ; hence every term of x is killed and $D_\lambda x = 0$. This proves positivity.

If every $\lambda(a) > 0$, then the same argument shows that D_λ preserves and reflects the sign of every nonzero x , and [proposition 7.1](#) gives the positive inverse $D_{1/\lambda}$. Conversely an order automorphism cannot kill or reverse any positive monomial, so all $\lambda(a) > 0$. $\square$

Corollary 7.5 (Positive support projections). *The support projection P_S is positive if and only if S is a final subclass of $\mathbf{No}$.*

For example, projection onto exponents > 0 is positive, whereas projection onto negative exponents is not: the positive surreal $1 - \omega^{-1}$ is sent to the negative infinitesimal $-\omega^{-1}$.

7.3 Covariance under monomial symmetries

Multiplication by a monomial defines an $\mathbb{R}$ -linear order automorphism

$$S_b(x) = \omega^b x,$$

which translates every exponent by b . Therefore

$$S_b D_\lambda S_{-b} = D_{\lambda_b}, \quad \lambda_b(a) = \lambda(a - b). \quad (7.5)$$

The diagonal algebra and the shift group form a natural crossed-product-like operator system.

More generally, every ordered additive automorphism $e : (\mathbf{No}, +, <) \rightarrow (\mathbf{No}, +, <)$ induces the coefficient-reindexing field automorphism

$$\sigma_e \left(\sum_a r_a \omega^a \right) = \sum_a r_a \omega^{e(a)}. \quad (7.6)$$

It fixes $\mathbb{R}$, sends ω^a to $\omega^{e(a)}$, and satisfies

$$\sigma_e P_S \sigma_e^{-1} = P_{e(S)}, \quad (7.7)$$

$$\sigma_e D_\lambda \sigma_e^{-1} = D_{\lambda \circ e^{-1}}. \quad (7.8)$$

The PROVEIT source contains a Lean implementation of these exponent automorphisms for the actual sign-sequence surreal field. This covariance is a strong sense in which the monomial Hahn basis is natural: every exponent symmetry merely permutes its vectors.

[write] “Every exponent symmetry merely permutes its vectors” holds for the field automorphisms σ_e induced by exponent automorphisms. It does not hold for all order-preserving strong linear automorphisms: source 52’s reweightings $T_{\gamma,a}$ rescale one monomial (theorem 9.6) and its shear sends t^γ to $t^\gamma + t^\delta$ (section 9.4). Monomials are canonical relative to the omega map.

Corollary 7.6 (Projection-and-shift rigidity). *If an $\mathbb{R}$ -linear map $T : \mathbf{No} \rightarrow \mathbf{No}$ commutes with every coordinate projection $P_{\{a\}}$ and every monomial shift S_b , then $T = r \text{ id}$ for a unique $r \in \mathbb{R}$.*

Proof. By theorem 7.3, $T = D_\lambda$. Commutation with S_b and (7.5) give $\lambda(a) = \lambda(a - b)$ for all a, b . Taking $b = a$ shows $\lambda(a) = \lambda(0)$, so λ is constant. $\square$

8 Finite real linear algebra transfers coefficientwise

A principal advantage of the coordinate model is that every finite real linear system over surreals decomposes into the same real system at each exponent.

Let $A = (A_{ij}) \in M_{m \times n}(\mathbb{R})$ and define

$$A_{\mathbf{No}} : \mathbf{No}^n \rightarrow \mathbf{No}^m, \quad (A_{\mathbf{No}} x)_i = \sum_{j=1}^n A_{ij} x_j.$$

Theorem 8.1 (Coefficientwise linear-system criterion). *For $x = (x_1, \dots, x_n) \in \mathbf{No}^n$,*

$$A_{\mathbf{No}} x = 0 \iff A(c_a(x_1), \dots, c_a(x_n))^T = 0 \text{ for every } a \in \mathbf{No}.$$

Equivalently, the coefficient vector at every exponent belongs to the ordinary real nullspace $\ker_{\mathbb{R}} A$.

Proof. For every row i and exponent a , linearity of coefficient extraction gives

$$c_a \left(\sum_{j=1}^n A_{ij} x_j \right) = \sum_{j=1}^n A_{ij} c_a(x_j).$$

The result follows because coefficients separate points. $\square$

Theorem 8.2 (Rank transfer). *If A has real rank r , then, as class real vector spaces,*

$$\ker A_{\mathbf{No}} \cong \mathbf{No}^{n-r}, \quad \text{im } A_{\mathbf{No}} \cong \mathbf{No}^r.$$

After choosing real change-of-basis matrices, $A_{\mathbf{No}}$ is equivalent to the block map

$$(x_1, \dots, x_n) \mapsto (x_1, \dots, x_r, 0, \dots, 0).$$

Proof. Choose $P \in \text{GL}_m(\mathbb{R})$ and $Q \in \text{GL}_n(\mathbb{R})$ with

$$PAQ = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

The same finite matrices act invertibly on $\mathbf{No}^m$ and $\mathbf{No}^n$, since their inverses also have real entries. Conjugating $A_{\mathbf{No}}$ gives the displayed block map, from which the kernel and image descriptions follow. $\square$

Corollary 8.3 (Scale-by-scale solution space). *If $u_1, \dots, u_{n-r}$ is a real basis of $\ker A$, every surreal solution is a strongly summable scale decomposition*

$$x = \sum_{a \in S}^H \omega^a \left(\sum_{k=1}^{n-r} \rho_{a,k} u_k \right),$$

where S is reverse well ordered and $\rho_{a,k} \in \mathbb{R}$. Conversely every such admissible family is a solution.

This theorem is the finite-dimensional version of “tensoring real linear algebra with the surreal coefficient space.” It requires no choice of a global Hamel basis. The PROVEIT files on coefficient extraction and module kernel bases formalize closely related statements for the actual purely infinite omnific ideal.

[write] The declarations meant are `Surreal.Foundations.SignSequence.omnificPurelyInfiniteCoeff`, `omnificPurelyInfinite_eq_zero_iff` and `omnificPurelyInfinite_linear_coeff_iff` (`Surreal/Foundations/OmnificPurelyInfiniteCoefficients.lean`), on the purely infinite omnific ideal only, and the generic kernel-basis API of `Surreal.ModuleKernelBasis` (`Surreal/Algebra/ModuleKernelBasis.lean`). [Theorems 8.1](#) and [8.2](#) on all of $\mathbf{No}$ are not formalized.

9 Valuation bases and the limits of naturality (source 52)

[write] This section is source 52’s, in its t -convention. Its graded statement [\(9.1\)](#) is [proposition 6.5](#) read in t -exponents. The finite-dimensional reduced basis and the absence of a valuation basis are classical in valued-vector-space theory (Brown, Kuhlmann [\[16\]](#)); source 52 gives direct Hahn proofs and claims no priority ([section C.1](#)).

9.1 Leading values and the associated graded space

For $x \neq 0$ define

$$v(x) = \min \text{supp}_t x, \quad \text{lc}(x) = x_{v(x)}, \quad v(0) = \infty.$$

Then $v(rx) = v(x)$ for $r \neq 0$, and $v(x + y) \geq \min(v(x), v(y))$. Let

$$H_{\geq \gamma} = \{x : v(x) \geq \gamma\}, \quad H_{> \gamma} = \{x : v(x) > \gamma\}.$$

The coefficient map at γ induces a canonical isomorphism

$$H_{\geq \gamma} / H_{> \gamma} \simeq \mathbb{R}. \tag{9.1}$$

Thus the associated graded real vector space is

$$\text{gr}_v H_\Gamma \simeq \bigoplus_{\gamma \in \Gamma} \mathbb{R}.$$

It has a canonical one-dimensional line at each scale. Passing to the associated graded space forgets all tails; it does not identify H_Γ itself with that direct sum.

Definition 9.1 (Valuation independence). A family (b_i) is valuation-independent over $\mathbb{R}$ if every nonzero finite combination satisfies

$$v\left(\sum_i r_i b_i\right) = \min_{r_i \neq 0} v(b_i).$$

A valuation basis is a Hamel basis with this property. This is the trivially valued ground-field convention of valued-vector-space theory; see [16].

In a Hahn space with real coefficients a family is valuation-independent exactly when its nonzero members have pairwise distinct values. Distinct values prevent cancellation at the unique least one. Conversely, two equal values can be canceled by a suitable nonzero real linear combination of their leading coefficients.

Theorem 9.2 (Canonical reduced Hahn row echelon form). *Let $V \subseteq H_\Gamma$ be a d -dimensional real subspace. There are uniquely determined values*

$$\gamma_1 < \cdots < \gamma_d$$

and a unique ordered basis $b_1, \dots, b_d$ of V such that

$$v(b_i) = \gamma_i, \quad (b_i)_{\gamma_j} = \delta_{ij}.$$

The set of values attained by $V \setminus \{0\}$ is exactly $\{\gamma_1, \dots, \gamma_d\}$.

Proof. Take any finite spanning family of V . The minimum of its leading values is the minimum value attained in V : no finite sum can have a smaller value. Call this minimum γ_1 . The nonzero coefficient functional at γ_1 has a kernel V_1 of dimension $d - 1$, and every nonzero vector in V_1 has value greater than γ_1 . Repeat on V_1 , then on the next kernel. The dimension drops by one each time and gives d intrinsic pivots.

The map

$$V \longrightarrow \mathbb{R}^d, \quad x \longmapsto (x_{\gamma_1}, \dots, x_{\gamma_d})$$

is an isomorphism: its successive kernels are exactly those just constructed. Let b_i be the inverse image of the i th coordinate vector. It has value γ_i and the required coefficients. This also proves uniqueness. Any nonzero combination of the b_i has value the first pivot with nonzero coefficient. $\square$

This is a useful positive answer to the user’s question at finite dimension: the normal-form coordinates determine a canonical basis of each finite-dimensional subspace, even when its vectors have infinite support. No global well-order or arbitrary tie-breaking is required.

[write] [Lemma 9.3](#) is the case $V = \mathbb{R}$ of the three-duals report’s Proposition 5.1 (spherical completeness of the full Hahn space V_Γ); source 52’s proof is printed as a second route.

9.2 Spherical completeness and the obstruction

Lemma 9.3 (Set-indexed spherical completeness). *Every nonempty set-sized nest of closed balls*

$$B(a_i, \gamma_i) = \{x : v(x - a_i) \geq \gamma_i\}$$

in H_Γ has nonempty intersection. The same assertion holds in the full surreal Hahn class for set-sized nests.

Second proof (source 52), direct. Membership in $B(a_i, \gamma_i)$ prescribes the coefficients at exponents $\delta < \gamma_i$. Nestedness makes these prescriptions compatible: two centers agree below the smaller radius. Let S be the union of the nonzero prescribed supports and give each member its prescribed coefficient, with zero elsewhere.

We verify the one nontrivial point, well-ordering. Given a nonempty $A \subseteq S$, choose $s \in A$ and a prescription with $s < \gamma_i$. Every member of S at most s belongs to the corresponding initial support of a_i , by compatibility. Hence the nonempty set $A \cap (-\infty, s]$ has a least member, which is also least in A . Thus S is well ordered. The constructed series lies in every ball. For **No**, the union remains a set because the nest and all individual supports are sets. $\square$

Theorem 9.4 (No global valuation-independent Hamel basis). *If $\Gamma \neq 0$, H_Γ has no valuation basis over $\mathbb{R}$. Neither does the full surreal class have a class valuation basis over $\mathbb{R}$.*

Proof. Suppose B were such a basis and fix $\delta > 0$. Because values of finite combinations are minima of basis values, for each $n \geq 1$ there is a basis vector b_n with $v(b_n) = n\delta$. Put $s_n = b_1 + \cdots + b_n$. The balls

$$B_n = B(s_n, (n+1)\delta)$$

are nested: $v(s_{n+1} - s_n) = (n+1)\delta$. By [Lemma 9.3](#), choose $x \in \bigcap_n B_n$.

A Hamel expansion of x uses only finitely many members of B . But $v(x - s_n) \geq (n+1)\delta$, together with valuation independence, forces the coefficient of every b_i , $1 \leq i \leq n$, in x to equal 1. This holds for every n , a contradiction. The same proof uses only a countable nest for **No**. $\square$

It is important that we did *not* assume the family (b_n) itself is Hahn-summable. Increasing leading values alone do not ensure summability in arbitrary rank. The nested-ball proof avoids that error.

Proposition 9.5 (No best finite-monomial approximation). *For $x \notin \text{Fin}_\Gamma$, the set $\{v(x - e) : e \in \text{Fin}_\Gamma\}$ has no largest member.*

Proof. Given e , the difference $z = x - e$ is nonzero. Replace e by $e + \text{lc}(z)t^{v(z)}$. This remains in Fin_Γ and strictly improves the value of the error. $\square$

The inclusion $\text{Fin}_\Gamma \subseteq H_\Gamma$ is therefore immediate in the valued-vector-space sense: every leading residue is already represented by a monomial. It is not “nice” in the best-approximation sense used in [\[16\]](#). The quotient in [Theorem 5.20](#) is an *algebraic* quotient; no best-approximation quotient valuation has silently been assumed.

9.3 A precise symmetry obstruction

For $\gamma \in \Gamma$ and $a > 0$, define a coordinate reweighting

$$T_{\gamma,a}(x) = x + (a - 1)x_\gamma t^\gamma.$$

It is a strong, order-preserving, real-linear automorphism. If $\gamma \neq 0$, it fixes the entire real subfield. It is generally not a field automorphism.

Theorem 9.6 (No symmetry-invariant projective Hamel basis). *For $\Gamma \neq 0$, there is no Hamel basis B of H_Γ whose set of real lines $\{\mathbb{R}b : b \in B\}$ is invariant under all $T_{\gamma,a}$ with $\gamma \neq 0$ and $a > 0$. The assertion also holds for the full surreal class.*

Proof. Suppose a basis vector b has at least two nonzero coordinates. Choose a nonzero exponent γ in its support. As a ranges over three distinct positive reals, the vectors

$$b + (a - 1)b_\gamma t^\gamma$$

lie in the two-dimensional span of b and t^γ but determine three distinct lines. Invariance would make those three lines basis lines, contradicting linear independence. Hence every basis vector would have one-point support. Such vectors span only Fin_Γ , not H_Γ . $\square$

This is stronger than observing that there are many Hamel bases, but narrower than claiming that no conceivable enriched structure can define one. It rules out naturality with respect to a specified elementary family of strong ordered-vector-space symmetries, even when bases are regarded only up to rescaling.

9.4 Why the simplicity relation matters

If $\gamma < \delta$ and $\gamma \neq 0$, the shear

$$S(x) = x + x_\gamma t^\delta$$

is another strong order-preserving real-linear automorphism fixing $\mathbb{R}$. Its inverse is $x \mapsto x - x_\gamma t^\delta$. It preserves every leading value and leading coefficient, but sends t^γ to $t^\gamma + t^\delta$. Thus even the distinguished monomial representative of a value is not recoverable from the ordered-vector-space structure alone.

The monomials are canonical in the *surreal structure with simplicity*, or in a Hahn presentation with a specified cross-section. There is no contradiction between that canonicity and the shear example. They refer to different categories of structure-preserving maps.

Part III

Strong maps, duals, and topology

10 Strong linear maps and the intrinsic real dual

A Hamel-linear map is determined by a Hamel basis, hence inherits all of its choice dependence. A map adapted to Conway normal form should instead commute with the distinguished strong sums.

10.1 Strong linearity

Definition 10.1 (Strong real-linear map). An $\mathbb{R}$ -linear class map $T : \mathbf{No} \rightarrow \mathbf{No}$ is *strong* if for every Hahn-summable set-indexed family $(x_i)_{i \in I}$, the family $(Tx_i)_{i \in I}$ is Hahn summable and

$$T\left(\sum_{i \in I}^H x_i\right) = \sum_{i \in I}^H Tx_i. \quad (10.1)$$

The definition includes both preservation of admissibility and commutation with the sum. Merely writing a coefficientwise formula is not enough unless its output supports satisfy the Hahn conditions.

Definition 10.2 (Hahn-compatible column family). A class family $(y_\gamma)_{\gamma \in \mathbf{No}}$ in $\mathbf{No}$ is *Hahn compatible* if for every reverse-well-ordered set $S \subseteq \mathbf{No}$, the family $(y_\gamma)_{\gamma \in S}$ is Hahn summable.

Write

$$y_\gamma = \sum_{\delta}^H a_{\delta\gamma} \omega^\delta, \quad a_{\delta\gamma} \in \mathbb{R}. \quad (10.2)$$

Then compatibility says exactly that, for every reverse-well-ordered set S ,

1. $\bigcup_{\gamma \in S} \{\delta : a_{\delta\gamma} \neq 0\}$ is reverse well ordered, and
2. for each row exponent δ , only finitely many $\gamma \in S$ have $a_{\delta\gamma} \neq 0$.

This is the surreal-Hahn analogue of a column-finiteness condition.

Theorem 10.3 (Hahn-matrix classification of strong maps). *The assignments*

$$T \mapsto (T\omega^\gamma)_{\gamma \in \mathbf{No}} \quad (10.3)$$

and

$$(y_\gamma) \mapsto \left[\sum_{\gamma \in S}^H r_\gamma \omega^\gamma \mapsto \sum_{\gamma \in S}^H r_\gamma y_\gamma \right] \quad (10.4)$$

are inverse bijections between strong $\mathbb{R}$ -linear endomorphisms of $\mathbf{No}$ and Hahn-compatible column families.

Proof. Let T be strong. For a reverse-well-ordered set S , the monomial family $(\omega^\gamma)_{\gamma \in S}$ is Hahn summable. Therefore $(T\omega^\gamma)_{\gamma \in S}$ is Hahn summable, so the columns are compatible. Applying strong linearity to a normal form gives (10.4), establishing uniqueness.

Conversely, suppose (y_γ) is compatible and define T by (10.4). The output sum exists by compatibility and is plainly real linear. Let $(x_i)_{i \in I}$ be Hahn summable and write $x_i = \sum_{\gamma}^H r_{i\gamma} \omega^\gamma$. Put $S = \bigcup_i \text{supp}(x_i)$. Compatibility says that $(y_\gamma)_{\gamma \in S}$ has reverse-well-ordered union of supports and is locally finite in every output exponent. For a fixed γ , only finitely many i have $r_{i\gamma} \neq 0$. Hence the flattened family $(r_{i\gamma} y_\gamma)_{(i,\gamma)}$ is Hahn summable: at each output exponent only finitely many γ 's contribute, and for each such γ only finitely many i 's contribute. Associativity of finite coefficient sums now yields

$$T\left(\sum_i^H x_i\right) = \sum_{i,\gamma}^H r_{i\gamma} y_\gamma = \sum_i^H Tx_i.$$

Thus T is strong. □

Remark 10.4. The theorem is a universal property of the Conway–Hahn basis. A Hamel basis determines all algebraic linear maps; the monomial Hahn basis determines all strong linear maps, with the compatibility condition recording exactly which images of basis vectors extend to every normal form.

[write] The three-duals report proves the K -linear analogue for $V((t^\Gamma))$: its Theorem 3.2 (“Global-support classification”) classifies strong K -linear maps between Hahn vector spaces. [Theorem 10.3](#) is the real-linear, proper-class version for the scalar class $\mathbf{No}$ itself. It answers source 53’s question on strongly linear matrices ([remark 21.42](#)).

10.2 The strong real-valued dual

Embed $\mathbb{R}$ in $\mathbf{No}$ as the exponent-zero line. A family of real numbers is Hahn summable in this subspace exactly when only finitely many members are nonzero, because all nonzero terms occupy the same exponent 0.

Definition 10.5 (Summation-finite exponent class). A subclass $A \subseteq \mathbf{No}$ is *summation finite* if

$$A \cap S \text{ is finite for every reverse-well-ordered set } S \subseteq \mathbf{No}. \quad (10.5)$$

A weight $a : \mathbf{No} \rightarrow \mathbb{R}$ is summation finite if its active class $\text{supp}(a) = \{\gamma : a(\gamma) \neq 0\}$ is summation finite.

Proposition 10.6 (Sequential characterization). *A subclass $A \subseteq \mathbf{No}$ is summation finite if and only if it contains no strictly decreasing sequence $a_0 > a_1 > a_2 > \dots$ indexed by ω .*

Proof. A strictly decreasing sequence has reverse-well-ordered range: every nonempty subset of its range has a largest member, corresponding to the least index that occurs. Hence such a sequence violates summation finiteness. Conversely, if $A \cap S$ is infinite for a reverse-well-ordered set S , enumerate that intersection in decreasing order. Its order type is an infinite ordinal and its first ω members form a strictly decreasing sequence in A . $\square$

Theorem 10.7 (Classification of the strong real dual). *An $\mathbb{R}$ -linear map $\Lambda : \mathbf{No} \rightarrow \mathbb{R}$ is strong if and only if there is a summation-finite weight $a : \mathbf{No} \rightarrow \mathbb{R}$ such that*

$$\Lambda(x) = \sum_{\gamma \in \text{supp}(x) \cap \text{supp}(a)} a(\gamma) [\omega^\gamma] x. \quad (10.6)$$

The sum in (10.6) is always finite. The weight is unique and equals

$$a(\gamma) = \Lambda(\omega^\gamma). \quad (10.7)$$

Proof. Suppose Λ is strong and define a by (10.7). For every reverse-well-ordered set S , the family $(\omega^\gamma)_{\gamma \in S}$ is Hahn summable. Its image family $(a(\gamma))_{\gamma \in S}$ must be summable in $\mathbb{R}$, hence only finitely many of those reals are nonzero. Thus $\text{supp}(a)$ is summation finite. Applying Λ to the strong normal form of x yields (10.6).

Conversely, let a be summation finite and define Λ by (10.6). It is real linear. If $(x_i)_{i \in I}$ is Hahn summable and $S = \bigcup_i \text{supp}(x_i)$, then $F = S \cap \text{supp}(a)$ is finite. For each $\gamma \in F$, only finitely many i have a nonzero γ -coefficient. Therefore only finitely many $\Lambda(x_i)$ can be nonzero, and finite rearrangement of coefficients gives $\Lambda(\sum_i^H x_i) = \sum_i \Lambda(x_i)$. Hence Λ is strong. Uniqueness follows by evaluation on monomials. $\square$

Example 10.8 (Coordinate functionals). For every γ , the coefficient map

$$\pi_\gamma(x) = [\omega^\gamma]x$$

is strong. It corresponds to the weight supported on the singleton $\{\gamma\}$.

The next example shows that a strong functional need not use only finitely many coordinates globally.

Lemma 10.9 (Well-ordered active classes are summation finite). *If a subclass $A \subseteq \mathbf{No}$ is well ordered by the inherited order, then it is summation finite.*

Proof. For a reverse-well-ordered set S , the intersection $A \cap S$ is both well ordered and reverse well ordered. An infinite linear order cannot have both properties: an infinite ordinal, or any infinite set of ordinal type, has a subset with no greatest element. Hence $A \cap S$ is finite. $\square$

Example 10.10 (The ordinal-coefficient trace). The class $\mathbf{On}$ is well ordered, so

$$\mathrm{Tr}_{\mathbf{On}}(x) = \sum_{\gamma \in \mathrm{supp}(x) \cap \mathbf{On}} [\omega^\gamma]x \quad (10.8)$$

is a well-defined strong $\mathbb{R}$ -linear functional. Its active exponent class is the proper class $\mathbf{On}$. Every individual normal form contains only finitely many ordinal exponents, even though the functional is active at all ordinals.

The strong dual of a set-sized slice (source 52)

[write] Source 52 proves the same classification for a set-sized workspace H_Γ . Its weights have reverse-well-ordered support in t -exponents, that is well-ordered support in ω -exponents; by [lemma 10.9](#) and [proposition 10.6](#) (and dependent choice) this is exactly summation-finiteness. So for $\Gamma \subseteq \mathbf{No}$ the theorem is [theorem 10.7](#) restricted to H_Γ ; source 52 adds that strong functionals are valuation-continuous.

10.3 The strong dual has an explicit support description

When the target is $\mathbb{R}$ alone, a Hahn-summable family of target vectors means that only finitely many are nonzero. This convention differs from ordinary convergence of a numerical real series.

Theorem 10.11 (Strong real dual). *Let Γ be set-sized. Strong real-linear maps $f : H_\Gamma \rightarrow \mathbb{R}$ correspond exactly to weight functions $a : \Gamma \rightarrow \mathbb{R}$ whose nonzero support is reverse well ordered. The formula is*

$$f(x) = \sum_{\gamma \in \mathrm{supp}_t x \cap \mathrm{supp}_t a} a_\gamma x_\gamma, \quad (10.9)$$

where the displayed sum is always finite.

Proof. If f is strong, let $a_\gamma = f(t^\gamma)$. An infinite increasing sequence in $\mathrm{supp}_t a$ would give a summable monomial family with infinitely many nonzero images. Therefore $\mathrm{supp}_t a$ has no such sequence. Under choice, this is equivalent to reverse well-ordering: a nonempty subset with no largest member recursively yields an increasing sequence. Strongness applied to the monomial expansion of x now gives [\(10.9\)](#).

Conversely, a well-ordered set and a reverse-well-ordered set have finite intersection; an infinite intersection would contain an increasing sequence. For a summable family (x_i) , its joint support meets $\mathrm{supp}_t a$ in a finite set, and each of those exponents occurs in only finitely many x_i . Thus only finitely many $f(x_i)$ can be nonzero, and all summations in the required identity reduce to a finite calculation. $\square$

Every strong functional is valuation-continuous: the support of a , when nonempty, has a largest member, so the functional vanishes on a sufficiently high tail. Coordinate extraction is the simplest example.

Remark 10.12 (Three duals already appear). There are at least three very different notions of real dual for **No**:

1. the algebraic dual determined by a choice of Hamel basis;
2. the intrinsic strong dual classified by [theorem 10.7](#);
3. the order-bounded dual, which is zero by [theorem 11.2](#).

This mirrors, at the scalar-class level, the repository's separation of represented, strong, and valuation-continuous duals for set-sized Hahn vector spaces.

11 Order duality and the failure of Schauder convergence

The strong dual is nontrivial because it is adapted to formal support sums. Demanding compatibility with the total order has the opposite effect: every real-valued order-bounded linear functional vanishes.

11.1 The order-bounded real dual is zero

Definition 11.1 (Order-bounded linear map). An $\mathbb{R}$ -linear map $L : \mathbf{No} \rightarrow \mathbb{R}$ is *order bounded* if for every $y \geq 0$ there is a real number M_y such that

$$|x| \leq y \implies |L(x)| \leq M_y.$$

Equivalently, it maps every surreal order interval to a bounded subset of $\mathbb{R}$.

Theorem 11.2 (Collapse of the order-bounded real dual). *Every order-bounded $\mathbb{R}$ -linear map $L : \mathbf{No} \rightarrow \mathbb{R}$ is zero.*

Proof. Fix $x \in \mathbf{No}$. If $x = 0$ there is nothing to prove. Put $y = \omega|x| > 0$. For every positive integer n , one has $n|x| < \omega|x| = y$, so $nx \in [-y, y]$. Order boundedness gives a real M_y with

$$n|L(x)| = |L(nx)| \leq M_y \quad (n \in \mathbb{N}_{>0}).$$

No positive real number can satisfy these inequalities for every n . Hence $L(x) = 0$. Since x was arbitrary, $L = 0$. $\square$

Corollary 11.3 (No positive real linear state). *Every positive $\mathbb{R}$ -linear map $L : \mathbf{No} \rightarrow \mathbb{R}$ is zero. In particular, there is no positive real-linear retraction of **No** onto its real subfield.*

Proof. If L is positive and $-y \leq x \leq y$, then $-L(y) \leq L(x) \leq L(y)$. Thus L is order bounded and [theorem 11.2](#) applies. $\square$

Example 11.4 (Constant term is strong but not positive). The coefficient functional

$$\text{ct}(x) = [\omega^0]x$$

is strong and restricts to the identity on $\mathbb{R}$, but it is not positive: $\omega - 1 > 0$ while $\text{ct}(\omega - 1) = -1$. This example shows exactly why [corollary 11.3](#) does not contradict the existence of canonical coefficient retractions.

Corollary 11.5 (No nonzero order-compatible real seminorm). *Suppose $p : \mathbf{No} \rightarrow \mathbb{R}_{\geq 0}$ is absolutely homogeneous over $\mathbb{R}$ and monotone with respect to surreal magnitude:*

$$|x| \leq |y| \implies p(x) \leq p(y).$$

Then $p = 0$.

Proof. For $x \neq 0$ and every $n \in \mathbb{N}_{>0}$, $|nx| < |\omega x|$. Hence

$$np(x) = p(nx) \leq p(\omega x),$$

where the right side is one fixed real number. Therefore $p(x) = 0$. $\square$

A choice of Hamel basis can of course be declared orthonormal, producing a noncanonical real-valued norm. The corollary says that no such norm can respect the ambient surreal order in the displayed sense.

[write] Source 52 proves the vanishing of positive functionals in every non-Archimedean ordered field containing $\mathbb{R}$; [corollary 11.3](#) is the case $K = \mathbf{No}$, and [theorem 11.2](#) strengthens it, for $\mathbf{No}$, from positive to order-bounded maps.

Proposition 11.6 (No global positive real observable). *Let $K \supsetneq \mathbb{R}$ be a non-Archimedean ordered field containing $\mathbb{R}$. Every positive real-linear map $L : K \rightarrow \mathbb{R}$ is zero. In particular this holds for H_Γ , $\Gamma \neq 0$, and for $\mathbf{No}$.*

Proof. Choose $u > n$ for every positive integer n . For $x > 0$, positivity gives $L(ux) \geq nL(x)$ for every n . Both values are real, and $L(x) \geq 0$, so $L(x) = 0$. Differences of positive elements give $L = 0$. $\square$

On the finite-element valuation ring $\mathcal{O} = H_{\geq 0}$, the coefficient map $x \mapsto x_0$ is instead the unique positive real-linear map fixing 1. An infinitesimal ε satisfies $-1/n < \varepsilon < 1/n$ for every n , so positivity forces its image to zero; the real part is fixed. The same coefficient map extends linearly to all H_Γ , but not positively: $t^{-\delta} - 1 > 0$ has constant coefficient -1 .

This explains why a real scalar field alone does not provide a Hilbert-space interpretation of all surreal scales. Arbitrary Hamel-based inner products on set-sized slices can be introduced, but they are extra choices and cannot turn the surreal order into ordinary real norm geometry.

11.2 The fine valuation topology

The natural valuation [\(3.12\)](#) gives fine balls

$$B_\gamma(0) = \{x \in \mathbf{No} : v(x) > \gamma\}, \quad \gamma \in \mathbf{No}. \quad (11.1)$$

Translates of these balls define the full natural valuation topology. It is “fine” because every surreal threshold is allowed, not merely thresholds from a fixed set-sized subgroup.

The following elementary fact about $\mathbf{No}$ is decisive.

Lemma 11.7 (Every set of surreals is bounded). *Every set $A \subseteq \mathbf{No}$ has a surreal upper bound and a surreal lower bound.*

Proof. The Conway cut $\{A \mid \emptyset\}$ is greater than every member of A . Applying this to $-A$ gives a lower bound. $\square$

Theorem 11.8 (Set-indexed convergence is eventually constant). *Let $(x_i)_{i \in I}$ be a net in $\mathbf{No}$ whose directed index I is a set. If $x_i \rightarrow x$ in the full natural valuation topology, then $x_i = x$ eventually.*

Proof. Consider the set

$$A = \{v(x_i - x) : i \in I, x_i \neq x\} \subseteq \mathbf{No}.$$

By lemma 11.7, choose $\beta \in \mathbf{No}$ strictly above every member of A . Convergence to x in the neighborhood $x + B_\beta(0)$ implies that eventually $v(x_i - x) > \beta$. No nonzero difference in the net has such a value by the choice of β , so eventually $x_i - x = 0$. $\square$

Remark 11.9 (Non-discrete but net-rigid). The topology is not discrete: no single threshold γ excludes every nonzero infinitesimal, because $\omega^{-\delta} \in B_\gamma(0)$ for sufficiently large δ . Nevertheless, every set-indexed net has only a set of valuation values, hence a common upper bound. This is a proper-class phenomenon: the neighborhood system itself has proper-class cofinality.

Corollary 11.10 (Normal forms are not Schauder expansions). *If a surreal has infinite normal-form support, no set-indexed net of its finite support truncations converges to it in the full natural valuation topology. Consequently the Conway monomials are not a Schauder basis for that topology.*

Proof. A convergent net of finite truncations would be eventually equal to its limit by theorem 11.8, forcing the limit to have finite support. $\square$

The same argument applies to the fine order topology.

Corollary 11.11 (Fine order convergence is also eventually constant). *Every set-indexed net convergent in the order topology generated by all surreal intervals $(x - \varepsilon, x + \varepsilon)$, with $\varepsilon > 0$, is eventually constant.*

Proof. For the set of nonzero distances $D = \{|x_i - x| : x_i \neq x\}$, the cut $\{0 \mid D\}$ gives a positive surreal ε smaller than every member of D . Eventual membership in the ε -interval around x therefore forces equality. $\square$

[write] Source 52 proves a stronger form of corollary 11.11: every set is uniformly discrete, and Cauchy nets are also eventually constant. It works with interval neighbourhoods only.

We use “order topology on $\mathbf{No}$ ” only through its interval neighborhoods. We do not treat a proper-class collection of open subclasses as an ordinary set topology.

Theorem 11.12 (Every set is uniformly discrete in the full surreal line). *For every set $A \subseteq \mathbf{No}$, there is $\epsilon > 0$ in $\mathbf{No}$ such that*

$$a \neq b \text{ in } A \implies |a - b| > \epsilon.$$

Every set-indexed convergent net in the full surreal order topology is eventually constant. Every set-indexed Cauchy net is also eventually constant.

Proof. The positive pairwise distances form a set D . The cut $\{0 \mid D\}$ is positive and smaller than every member of D ; if D is empty choose any positive ϵ . This proves uniform discreteness. For convergence to x , apply the result to the range of the net together with x , then use the ϵ -neighborhood of x . For a Cauchy net, apply it to the range and use the Cauchy condition at this fixed ϵ : all entries in a sufficiently late tail must coincide. $\square$

In particular, every set-sized Hahn subfield has the *discrete* topology induced from the full surreal order topology, although its internal valuation topology is non-discrete. The sequence ω^{-n} converges to zero in $H_{\mathbb{Z}}$ internally, but not in the full surreal line, where $\omega^{-\omega}$ is a smaller positive scale than every term.

Corollary 11.13 (The full finite-monomial class is closed). $\mathbf{No}_{\text{fin}}$ is closed in the interval sense in $\mathbf{No}$: for every $x \notin \mathbf{No}_{\text{fin}}$ there is an interval about x missing $\mathbf{No}_{\text{fin}}$.

Proof. Choose a valuation exponent δ above the entire set $\text{supp}_t x$. For any finite-support e , at least one exponent of the infinite support of x remains uncanceled. Hence $v(x - e) < \delta$ and $|x - e| > t^\delta$. The interval of radius t^δ about x works. $\square$

Warning 11.14 (Not a topological vector space over the usual reals). For nonzero Γ , the scalar copy of $\mathbb{R}$ is discrete in the internal valuation topology, and also in the full surreal order topology. The inclusion of the usual Euclidean $\mathbb{R}$ is not continuous: a sufficiently small valuation neighborhood meets it only at zero. Thus scalar multiplication is not continuous when $\mathbb{R}$ has its usual topology. These topologies give vector spaces over the *discrete* real field, not ordinary real locally convex spaces.

11.3 Closedness of the finite span and the trichotomy (source 53)

[write] Source 53 shows that no approximation by finite-support vectors at all, not only by truncations, reaches an infinite-support surreal. Its neighbourhoods are $U_\gamma = B_{-\gamma}(0)$.

The most important conceptual trap is to treat a Conway normal form as an ordinary limit of its finite truncations. It is a legitimate Hahn sum, but generally not a limit in the full natural valuation topology.

For $\gamma \in \mathbf{No}$, let

$$U_\gamma = \{0\} \cup \{x \neq 0 : \text{lead}(x) < \gamma\}.$$

These classes form the zero-neighborhood basis corresponding to the valuation $v(x) = -\text{lead}(x)$: a difference is small when all of its exponents lie far below the chosen threshold.

Theorem 11.15 (Closedness of the finite monomial span). *The finite-support subspace $\mathbf{No}_{\text{fin}}$ is closed in the full natural valuation topology on $\mathbf{No}$. Equivalently,*

$$\overline{\text{span}_{\mathbb{R}}\{\omega^a : a \in \mathbf{No}\}}^v = \mathbf{No}_{\text{fin}}.$$

In particular, no infinite-support surreal is the natural-valuation limit of a net of finite-support surreals.

Proof. Let $x \notin \mathbf{No}_{\text{fin}}$ and put $S = \text{supp}(x)$, an infinite set. Every set of surreals has a strict lower bound: for example, Conway's cut $\{\mid S\}$ is less than every member of S . Choose such a γ , so $\gamma < a$ for all $a \in S$.

For any $y \in \mathbf{No}_{\text{fin}}$, the finite set $\text{supp}(y)$ cannot contain S . Choose $a \in S \setminus \text{supp}(y)$. The coefficient of ω^a in $x - y$ is then $c_a(x) \neq 0$, so

$$\text{lead}(x - y) \geq a > \gamma.$$

Hence $x - y \notin U_\gamma$. The neighborhood $x + U_\gamma$ is disjoint from $\mathbf{No}_{\text{fin}}$, proving that the complement of $\mathbf{No}_{\text{fin}}$ is open. $\square$

Example 11.16 (The geometric series does not converge valuation-topologically). Let

$$x = \sum_{n \geq 0} \omega^{-n}, \quad s_N = \sum_{n=0}^N \omega^{-n}.$$

Then

$$\text{lead}(x - s_N) = -(N + 1), \quad v(x - s_N) = N + 1.$$

Although $N + 1$ tends to infinity among the natural numbers, it never eventually exceeds the surreal threshold ω . Thus (s_N) does not converge to x in the full natural valuation topology. Its Hahn sum is nevertheless perfectly well defined and equals $(1 - \omega^{-1})^{-1}$.

Corollary 11.17 (No natural-valuation Schauder basis). *The monomial family is not a Schauder basis, transfinite Schauder basis, or net basis for $\mathbf{No}$ when “convergence” is interpreted using the full natural valuation topology and approximants are finite linear combinations.*

Theorem 11.18 (Algebraic–topological–summation trichotomy). *Let $\mathfrak{M} = \{\omega^a : a \in \mathbf{No}\}$. Then*

$$\begin{aligned} &\text{algebraic span of } \mathfrak{M} = \mathbf{No}_{\text{fin}}, \\ &\text{natural-valuation closure of that span} = \mathbf{No}_{\text{fin}}, \\ &\text{smallest class subspace containing } \mathfrak{M} \text{ and closed under strongly summable set-families} = \mathbf{No}. \end{aligned}$$

Proof. The first line is [proposition 4.5](#); the second is [theorem 11.15](#). For the third, every surreal is the strongly summable family of its normal-form monomials by [theorem 4.2](#). $\square$

Remark 11.19 (Why no contradiction arises). On a fixed support slice $H(A) \cong \mathbb{R}^A$, the product topology makes finite-support vectors dense. The natural valuation topology compares a tail against *every* surreal scale, including thresholds below the entire set A . It is therefore much finer in the direction relevant to the closure theorem. Coarsened valuations used in transseries or rank-one Hahn fields may restore ordinary convergence for selected series; the theorem concerns the full natural valuation with value group $\mathbf{No}$.

11.4 How nontrivial convergence can be recovered

The anti-Schauder theorem concerns the *full* value group $\mathbf{No}$. Useful analytic work generally chooses a set-sized Hahn workspace $k((t^\Gamma))$, a set-sized cofinal scale family, or a coarser topology. Within such a workspace, a support can be cofinal in the chosen value group, and partial sums may converge. This is why the repository’s analytical reports carefully distinguish set-sized workspaces, strong summation, and fine surreal topology. A future theory of “Schauder bases for $\mathbf{No}$ ” must specify which scale system has been retained and which proper-class neighborhoods have been discarded.

12 Completion of the monomial span on a slice (source 52; the three-duals report)

[write] The statements of this section are not new. They are the diagonal case of the report *Three Duals at Surreal Scales* ([surcomplex/three-duals-of-hahn-vector-spaces](#); “three-duals” below), and source 52 calls that report “an explicit precedent, not a novelty”. Source 52’s direct proofs are printed as second routes.

The dictionary. Let $V = \mathbb{R}^{(\Gamma)}$ with basis $(e_\gamma)_{\gamma \in \Gamma}$, and let

$$\iota : H_\Gamma \longrightarrow V((t^\Gamma)), \quad \iota\left(\sum_\gamma^H x_\gamma t^\gamma\right) = \sum_\gamma^H t^\gamma (x_\gamma e_\gamma).$$

ι is an $\mathbb{R}$ -linear embedding preserving supports, valuations and strong sums, and the coefficients of $\iota(x)$ below a cut span a space of dimension equal to the number of support points of x below that cut. Hence, by three-duals Theorem 4.2 (finite rank below every cut),

$$\text{Fin}_\Gamma = \iota^{-1}(E_\Gamma(V)), \quad \text{Lf}_\Gamma = \iota^{-1}(C_\Gamma(V)).$$

The witnesses in the proofs of three-duals Theorems 4.5 and 4.6 have the form $\sum_n^H t^{\gamma_n} e_n$ with independent e_n ; choosing $e_n = e_{\gamma_n}$ puts them in the image of ι . So [theorems 12.1](#) and [12.2](#) are

three-duals Theorems 4.2, 4.5 and 4.6 (with Lemma 4.4) pulled back along ι , and [proposition 12.3](#) is three-duals Theorem 10.1 (cofinal preservation and noncofinal collapse) pulled back. Do not read $\text{Fin}_\Gamma, \text{Lf}_\Gamma$ as three-duals' $E_\Gamma(\mathbb{R}), C_\Gamma(\mathbb{R})$, which are all of H_Γ .

The internal valuation topology on H_Γ has neighborhoods $H_{>\gamma}$ of zero. We call a support *left-finite* if it has finite intersection with $(-\infty, \gamma]$ for every $\gamma \in \Gamma$.

Theorem 12.1 (Exact completion of the monomial span). *For $\Gamma \neq 0$,*

$$\text{Lf}_\Gamma = \overline{\text{Fin}_\Gamma}^v = \{x \in H_\Gamma : (\forall \gamma \in \Gamma) \mid \text{supp}_t x \cap (-\infty, \gamma] \mid < \infty\}. \quad (12.1)$$

This space is complete and is a realization of the valuation completion of Fin_Γ . The net of all finite monomial subsums of x converges to x exactly when $x \in \text{Lf}_\Gamma$.

Second proof (source 52), direct. To have $v(x - e) > \gamma$, the finite-support series e must agree with every coefficient of x at exponents at most γ . Such an e exists exactly when there are finitely many of these coefficients. This proves the closure formula and the finite-subsums criterion.

For completeness, let (x_i) be a Cauchy net in Lf_Γ . Below any fixed cutoff, its coefficients eventually agree exactly with those of one x_i and hence have finite support there. The stabilized coefficients define a function whose support is left-finite. Such a support is well ordered: for a nonempty subset, choose one member s and take the least of the finitely many members at most s . The resulting Hahn series is in Lf_Γ and is the net's limit. Density of Fin_Γ gives the asserted completion. $\square$

No growth bound on the real coefficients appears in this theorem. The topology regards nonzero real scalars as having value zero; coefficients cannot compensate for a support that fails to escape to higher valuations.

Theorem 12.2 (Cofinality and cyclicity classification). *For a nonzero ordered abelian group Γ :*

$$\begin{aligned} \text{Lf}_\Gamma \neq \text{Fin}_\Gamma &\iff \text{cf}(\Gamma) = \aleph_0, \\ \text{Lf}_\Gamma = H_\Gamma &\iff \Gamma \simeq_{\text{ord}} \mathbb{Z}. \end{aligned}$$

Second proof (source 52), direct. An infinite left-finite support can be listed in increasing order as an ω -sequence: each member has only finitely many predecessors. This sequence must be cofinal in Γ , or a bound would contain infinitely many support elements below it. Conversely, an increasing cofinal sequence is a left-finite infinite support. This proves the first equivalence.

In $\mathbb{Z}$, every well-ordered subset is bounded below, and has finite intersection with every bounded interval, so every support is left-finite.

Suppose Γ is not cyclic. If it has a least positive element ϵ but is not $\mathbb{Z}\epsilon$, there is a positive η greater than every $n\epsilon$. Indeed, any positive element bounded by some $n\epsilon$ must be an integer multiple of ϵ , by minimality of ϵ . Thus $\{n\epsilon : n \geq 0\}$ is an infinite bounded well-ordered support.

If there is no least positive element, choose a strictly decreasing positive sequence ϵ_n . Then $\{-\epsilon_n : n \geq 1\}$ is an infinite increasing, bounded support. Either construction yields a Hahn series outside Lf_Γ . $\square$

Exponent group	Finite span, completion, and full Hahn space
$\mathbb{Z}$	$\text{Fin}_\Gamma \subsetneq \text{Lf}_\Gamma = H_\Gamma$.
$\mathbb{Q}$ or $\mathbb{R}$	$\text{Fin}_\Gamma \subsetneq \text{Lf}_\Gamma \subsetneq H_\Gamma$.
$\mathbb{Z}^2$ lexicographically ordered	The same two strict inclusions; the support $\{(0, n) : n \geq 0\}$ is bounded by $(1, 0)$.
$\text{cf}(\Gamma) > \aleph_0$	$\text{Fin}_\Gamma = \text{Lf}_\Gamma \subsetneq H_\Gamma$.

Here $\text{Lf}_{\mathbb{Q}}$ is the familiar left-finite, Levi–Civita-type completion. The table concerns the internally chosen exponent group, not the topology induced from all of **No**.

12.1 Changing the available precision scales

Proposition 12.3 (Ambient enlargement). *Let $\Gamma \subseteq \Delta$ be ordered abelian groups. If Γ is cofinal in Δ , then $\text{Lf}_{\Delta} \cap H_{\Gamma} = \text{Lf}_{\Gamma}$. If it is not cofinal, then*

$$\text{Lf}_{\Delta} \cap H_{\Gamma} = \text{Fin}_{\Gamma}.$$

Second proof (source 52), direct. In the cofinal case, every Δ -cutoff is bounded above by a Γ -cutoff, so the two left-finiteness tests agree. Otherwise choose $\delta \in \Delta$ above all of Γ . The left-finiteness test at δ requires the entire support of a vector in H_{Γ} to be finite. $\square$

This is the monomial-span version of the noncofinal-enlargement phenomenon emphasized in the repository’s completion and tail-span reports [23, 24].

[write] The next theorem is the real-valued counterpart of three-duals Theorem 6.4 (separation of the completion defect), which separates the same defect by K -valued functionals. The two theorems have different codomains; source 52’s proof is independent of that report and says so.

Theorem 12.4 (Exactly what monomial-invisible continuous functionals detect). *Let $\mathcal{N}$ be the space of valuation-continuous real-linear maps $f : H_{\Gamma} \rightarrow \mathbb{R}$ vanishing on Fin_{Γ} , with $\mathbb{R}$ given its usual topology. Then*

$$\bigcap_{f \in \mathcal{N}} \ker f = \text{Lf}_{\Gamma}.$$

In particular, $\mathcal{N} = 0$ if and only if $\Gamma \simeq_{\text{ord}} \mathbb{Z}$, for $\Gamma \neq 0$.

Proof. Continuity forces each such kernel to contain the closure Lf_{Γ} . Conversely, if $x \notin \text{Lf}_{\Gamma}$, there is a γ for which

$$x \notin \text{Fin}_{\Gamma} + H_{>\gamma}.$$

On the real vector-space quotient by $\text{Fin}_{\Gamma} + H_{>\gamma}$ choose a linear functional taking the nonzero class of x to 1, extending from its one-dimensional span by a Hamel basis. Its pullback annihilates Fin_{Γ} and the open subspace $H_{>\gamma}$, so is continuous. It detects x .

A continuous real-linear functional must in fact vanish on some open tail: choose a tail whose image lies in $(-1, 1)$, and use closure of that tail under all real scalars. Thus the codomain topology used here introduces no hidden bounded-but-nonzero possibility. The final equivalence follows from Theorem 12.2. $\square$

A nonzero functional in $\mathcal{N}$ is not strong, since strongness and zero monomial values would force it to vanish everywhere. This elementary real-linear separation argument parallels, but does not require, the more elaborate Hahn-valued duality of [23].

Part IV

The Euler quotient (source 52)

[write] This Part is source 52’s Sections 8–12, whole, in its t -convention (section 2.2), for a nonzero additive subgroup $\Gamma \subseteq \mathbb{R}$. The Euler operator $\mathfrak{D}$ itself is not new to the collection: it is the Euler

derivation of the tail-span report, equation (7.2) there (for $\Gamma = \mathbb{Q}$), the derivation ∂_λ of the omnific–Diophantine report, (16.1) and Lemma 16.2 there, with $\lambda = -\text{id}$ (no divisibility of Γ is needed there), and the differential-equations report’s ∂_τ on real exponents. In ω -exponents it is D_d of [theorem 18.5](#) with $d(\beta) = -\beta$, so $D_{\text{ct}} = -\mathfrak{D}$ on H_Γ . What this Part adds is the module-theoretic package: torsion, finite resonance, the $\mathbb{R}(X)$ -structure of $Q_\Gamma = H_\Gamma/\text{Fin}_\Gamma$ and its dimension, the automatic strongness of the Euler commutant, nonsplitting, the lift ideal J_Γ , and the two layers. No other report of the collection proves these; source 52 does not certify their priority ([section C.1](#)).

13 The Euler quotient: canonical extra scalars

Throughout this section and the next two, Γ is a nonzero additive subgroup of $\mathbb{R}$. This restriction is essential: the exponent itself will be used as a *real* coefficient.

13.1 A strong derivation diagonal in Hahn coordinates

Definition 13.1 (Euler derivation). Define

$$\mathfrak{D}(x) = \sum_{\gamma \in \Gamma}^{\text{H}} \gamma x_\gamma t^\gamma.$$

It is real-linear and strong, since it does not enlarge supports. It is a derivation: in a product coefficient at exponent $\eta = \gamma + \delta$, the identity $\eta = \gamma + \delta$ gives the Leibniz rule term by term. Hahn convolution has only finitely many contributions at each exponent, so this computation is legitimate.

No global surreal derivative is being redefined. On an appropriate rational-exponent slice, this is the familiar Euler operator $t d/dt$. The repository already uses such operators [24]. Here the object of study is their action on real vector-space quotients.

Proposition 13.2 (All algebraic spectral torsion is finite-monomial). *Regard H_Γ as an $\mathbb{R}[X]$ -module by $Xx = \mathfrak{D}x$. Its torsion submodule is exactly Fin_Γ :*

$$\{x : (\exists p \in \mathbb{R}[X] \setminus \{0\}) p(\mathfrak{D})x = 0\} = \text{Fin}_\Gamma.$$

Its nonzero eigenspaces are precisely $\mathbb{R}t^\gamma$, with eigenvalue $\gamma \in \Gamma$.

Proof. The equation $p(\mathfrak{D})x = 0$ is the coefficientwise condition $p(\gamma)x_\gamma = 0$. A nonzero real polynomial has finitely many roots, so x has finite support. Conversely, a vector with support $\{\gamma_1, \dots, \gamma_m\}$ is killed by $\prod_j (\mathfrak{D} - \gamma_j)$. The same argument for $p(X) = X - \gamma$ gives the eigenspaces. $\square$

Thus the direct sum of all eigenspaces is a proper subspace, even though $\mathfrak{D}$ is diagonal in the Hahn representation. Infinite-coordinate diagonalization is not algebraic diagonalization by a Hamel eigenbasis.

13.2 Exact polynomial solution operators

For $p \in \mathbb{R}[X] \setminus \{0\}$ put

$$Z_p = \{\gamma \in \Gamma : p(\gamma) = 0\}, \quad P_{Z_p}x = \sum_{\gamma \in Z_p} x_\gamma t^\gamma,$$

and define

$$G_p(x) = \sum_{\substack{\gamma \in \Gamma \\ p(\gamma) \neq 0}}^{\text{H}} \frac{x_\gamma}{p(\gamma)} t^\gamma. \tag{13.1}$$

These maps preserve supports, so they are strong and preserve both Fin_Γ and Lf_Γ .

Theorem 13.3 (Finite resonance is the only obstruction). *For every nonzero p ,*

$$p(\mathfrak{D})G_p = G_p p(\mathfrak{D}) = I - P_{Z_p}. \quad (13.2)$$

Furthermore,

$$\ker p(\mathfrak{D}) = H[Z_p], \quad \text{im } p(\mathfrak{D}) = \{y : y_\gamma = 0 \text{ for all } \gamma \in Z_p\}.$$

Consequently $p(\mathfrak{D})$ has finite-dimensional kernel and cokernel of the same dimension $|Z_p|$; in the algebraic Fredholm sense its index is zero.

Proof. At a nonresonant exponent, multiplication by $p(\gamma)$ and division by it cancel. At a resonant exponent both compositions vanish. This proves (13.2). The kernel and image assertions follow from the same coefficient equations; the cokernel is represented by the missing finitely many coordinate values. $\square$

Repeated roots of p contribute only one real coordinate each. Jordan chains do not arise: $\mathfrak{D}$ is already coefficientwise scalar on each monomial line.

Theorem 13.4 (Canonical rational-function vector space). *Let $Q_\Gamma = H_\Gamma/\text{Fin}_\Gamma$. There is a unique $\mathbb{R}(X)$ -vector-space structure on Q_Γ extending its real scalar structure and making X act by the induced Euler operator. For $p, q \in \mathbb{R}[X]$, $q \neq 0$, it is given by*

$$\frac{p(X)}{q(X)} \cdot [x] = [p(\mathfrak{D})G_q(x)]. \quad (13.3)$$

Proof. Every $p(\mathfrak{D})$ preserves Fin_Γ and descends to the quotient. By Theorem 13.3, the induced maps of $p(\mathfrak{D})$ and G_p are mutually inverse because P_{Z_p} has image in Fin_Γ . Hence every nonzero polynomial acts invertibly. The commuting $\mathbb{R}[X]$ -action therefore extends uniquely to its field of fractions, with the displayed formula.

Directly, different representatives of a rational function give identical coefficient multipliers outside the finite set of their poles, so their difference on any vector has finite support. Likewise, changing x by an element of Fin_Γ changes the result by an element of Fin_Γ . Thus neither fraction representatives nor vector representatives introduce ambiguity. $\square$

The quotient is not a quotient *field*: Fin_Γ contains 1 and is not an ideal of H_Γ . The new scalars are *operators*, not multiplication by surreal representatives of rational functions. This distinction prevents a common but serious misreading of the construction.

13.3 Exact dimension over the larger scalar field

Lemma 13.5 (Exponential-polynomial independence). *Let $0 < r_1 < \dots < r_m$ be distinct real numbers and let $p_1, \dots, p_m \in \mathbb{R}[X]$. If $\delta > 0$ and*

$$\sum_{j=1}^m p_j(n\delta) r_j^n = 0$$

for all sufficiently large integers n , then all the polynomials are zero.

Proof. Choose the largest r_j whose polynomial is nonzero, and let that polynomial have degree d and leading coefficient $a \neq 0$. Divide the identity by $r_j^n n^d$. Its own contribution tends to $a\delta^d \neq 0$. Every smaller-base contribution tends to zero, because an exponential decay dominates any fixed power of n . This is a contradiction. $\square$

Theorem 13.6 (Continuum many independent rational-calculus directions). *For $0 \neq \Gamma \subseteq \mathbb{R}$,*

$$\dim_{\mathbb{R}(X)} Q_\Gamma = \mathfrak{c}.$$

For any fixed $\delta > 0$ in Γ , the explicit family

$$\left\{ \left[\frac{1}{1 - rt^\delta} \right] : r > 0 \right\}$$

is $\mathbb{R}(X)$ -linearly independent.

Proof. A finite rational-function relation may be multiplied by a common nonzero polynomial denominator, which is invertible on Q_Γ . We obtain

$$\sum_j p_j(\mathfrak{D}) q_{r_j} \in \text{Fin}_\Gamma.$$

At the exponents $n\delta$, this says $\sum_j p_j(n\delta) r_j^n = 0$ for all sufficiently large n . Lemma 13.5 forces all $p_j = 0$, and hence all original rational coefficients vanish. This gives the lower bound $\mathfrak{c}$. The cardinality bound $|H_\Gamma| = \mathfrak{c}$ from Corollary 5.21 gives the upper bound. $\square$

These witnesses are simple rational expressions in t^δ . Their independence is linear over a field of *Euler operators*; it is not algebraic independence over the ambient Hahn field. Continuum cardinality does not imply that this displayed family is a basis, as the following explicit counterexample shows.

Proposition 13.7 (An explicit direction missed by all geometric classes). *Fix $\delta > 0$ in Γ . The vector*

$$f = \sum_{n \geq 0}^H n! t^{n\delta}$$

has a class in Q_Γ outside the $\mathbb{R}(X)$ -span of all $[q_r]$, $r > 0$. More generally, the enlarged family

$$\left\{ \left[\sum_{n \geq 0}^H (n!)^\rho r^n t^{n\delta} \right] : \rho \geq 0, r > 0 \right\}$$

is $\mathbb{R}(X)$ -linearly independent. Thus the previously displayed geometric family is definitely not a basis, despite already having the correct cardinality.

Proof. All displayed supports are $\{n\delta : n \geq 0\}$, hence are Hahn-admissible. The positive real powers $(n!)^\rho$ are ordinary real numbers. After clearing rational-function denominators, a finite relation modulo Fin_Γ gives an eventual identity

$$\sum_{j=1}^m p_j(n\delta) (n!)^{\rho_j} r_j^n = 0,$$

where the pairs (ρ_j, r_j) are distinct. Suppose some polynomial is nonzero. Among the corresponding pairs choose the largest lexicographically, first in ρ and then in r . Let d be the degree of its polynomial. Divide by its $(n!)^\rho r^n$. Its contribution tends to its nonzero leading coefficient times δ^d . Terms with the same ρ but smaller r tend to zero exponentially. Terms with smaller ρ tend to zero because, for every $\epsilon > 0$, $A > 0$, and fixed integer k ,

$$\frac{n^k A^n}{(n!)^\epsilon} \rightarrow 0.$$

For instance, $n! \geq (n/2)^{\lfloor n/2 \rfloor}$ makes the latter bound immediate. This contradicts the relation. Taking $(\rho, r) = (1, 1)$ together with finitely many pairs $(0, r_j)$ proves the first assertion. $\square$

This is a linear independence statement for operator scalars, not an asymptotic restriction on allowed Hahn coefficients. Factorial growth creates no difficulty for Hahn summability: only the support matters.

14 Automatic strongness and nonsplitting

14.1 The whole algebraic commutant is coordinatewise

The next theorem begins with an arbitrary everywhere-defined real-linear map. Neither continuity nor compatibility with Hahn sums is assumed.

Theorem 14.1 (Automatic strongness of the Euler commutant). *For $0 \neq \Gamma \subseteq \mathbb{R}$,*

$$\{T \in \text{End}_{\mathbb{R}}(H_{\Gamma}) : T\mathfrak{D} = \mathfrak{D}T\} \simeq \mathbb{R}^{\Gamma} \quad (14.1)$$

as real algebras. The isomorphism sends (a_{γ}) to the coefficient multiplier

$$M_a(x) = \sum_{\gamma}^H a_{\gamma} x_{\gamma} t^{\gamma}.$$

Every such endomorphism is strong and satisfies $v(M_a x) \geq v(x)$.

Proof. Fix $\gamma \in \Gamma$. Because T commutes with $\mathfrak{D}$, it sends the one-dimensional eigenspace $\mathbb{R}t^{\gamma}$ to itself; write $T(t^{\gamma}) = a_{\gamma} t^{\gamma}$.

For arbitrary x , define

$$z = \sum_{\eta \neq \gamma}^H \frac{x_{\eta}}{\eta - \gamma} t^{\eta}.$$

This is a Hahn series because its support is contained in $\text{supp}_t x$. It gives the exact decomposition

$$x = x_{\gamma} t^{\gamma} + (\mathfrak{D} - \gamma)z.$$

Applying T and taking the coefficient at γ yields

$$(Tx)_{\gamma} = a_{\gamma} x_{\gamma},$$

because every vector in the image of $\mathfrak{D} - \gamma$ has zero γ -coefficient. This works at every exponent and proves $T = M_a$.

Conversely, every coefficient multiplier commutes with $\mathfrak{D}$. Supports cannot increase, so it preserves summability, commutes with all allowed sums, and cannot decrease valuation. Composition corresponds to pointwise multiplication of the weights. $\square$

This explains a useful rigidity phenomenon: although arbitrary Hamel-linear maps on H_{Γ} can be highly non-coordinatewise, a single natural operator $\mathfrak{D}$ forces every commuting map back into Hahn coordinate geometry.

Corollary 14.2 (Euler-compatible projections are exactly support projections). *A real-linear idempotent $P : H_{\Gamma} \rightarrow H_{\Gamma}$ commutes with $\mathfrak{D}$ if and only if $P = P_A$ for a subset $A \subseteq \Gamma$.*

Proof. By Theorem 14.1, $P = M_a$. Idempotence means $a_{\gamma}^2 = a_{\gamma}$ in $\mathbb{R}$, so each weight is 0 or 1. $\square$

The spectral coordinate projections are therefore determined by the algebraic operator, even without a strongness hypothesis. What fails is an algebraic projection onto the *sum of all* its monomial eigenspaces.

14.2 No compatible Hamel complement

Theorem 14.3 (Nonsplitting of the finite-coordinate extension). *The exact sequence*

$$0 \longrightarrow \text{Fin}_\Gamma \longrightarrow H_\Gamma \longrightarrow Q_\Gamma \longrightarrow 0$$

splits as real vector spaces but not as $\mathbb{R}[X]$ -modules with $X = \mathfrak{D}$. Equivalently, there is no real-linear projection onto Fin_Γ commuting with $\mathfrak{D}$, and no $\mathfrak{D}$ -invariant algebraic complement of Fin_Γ .

Proof. A real-linear complement exists by extending a Hamel basis of Fin_Γ . Suppose a commuting projection onto Fin_Γ existed. Since it fixes every monomial, its weights in Theorem 14.1 would all equal 1. It would be the identity on all of H_Γ , whose image is not Fin_Γ .

If a $\mathfrak{D}$ -invariant complement existed, both it and Fin_Γ would be invariant, making the associated projection commute with $\mathfrak{D}$. Conversely, a commuting projection gives such a complement. This proves all formulations. $\square$

Proposition 14.4 (No nonzero divisible submodule upstairs). *The $\mathbb{R}[X]$ -module H_Γ has no nonzero divisible submodule. Here divisible means that every nonzero polynomial acts surjectively on the submodule.*

Proof. Let W be divisible and $w \in W$. For each $\gamma \in \Gamma$, write $w = (\mathfrak{D} - \gamma)z$ with $z \in W$. Then $w_\gamma = 0$. All coordinates vanish, so $w = 0$. $\square$

The quotient Q_Γ is divisible and torsion-free by Theorem 13.4, whereas no nonzero divisible submodule can represent it inside H_Γ . This is a second proof and a conceptual strengthening of the nonsplitting result.

14.3 Many quotient operators cannot be lifted

Define the ideal

$$J_\Gamma = \{a \in \mathbb{R}^\Gamma : \text{supp}_t a \text{ is reverse well ordered}\}.$$

It is an ideal for pointwise multiplication: subsets and finite unions of reverse-well-ordered sets are reverse well ordered.

Proposition 14.5 (The exact algebra of Euler-compatible lifts). *The coefficient multipliers induce a faithful real-algebra embedding*

$$\mathbb{R}^\Gamma / J_\Gamma \hookrightarrow \text{End}_{\mathbb{R}(X)}(Q_\Gamma). \quad (14.2)$$

This embedding is not surjective. Consequently some $\mathbb{R}(X)$ -linear endomorphisms of Q_Γ have no real-linear lift to H_Γ commuting with $\mathfrak{D}$.

Proof. A coefficient multiplier preserves Fin_Γ and commutes with rational Euler actions on the quotient. It induces zero exactly when it maps every Hahn series to a finite-support series. If its weight support is reverse well ordered, its intersection with each Hahn support is finite, proving this condition. If the weight support is not reverse well ordered, it contains an infinite increasing sequence (γ_n) . The series $\sum_n t^{\gamma_n}$ is a Hahn series whose image has infinite support. This proves that the kernel is exactly J_Γ .

The source of (14.2) is commutative. The target is not: by Theorem 13.6, Q_Γ has at least two independent vectors over $\mathbb{R}(X)$; extend them to a basis and define two noncommuting elementary matrix endomorphisms. Thus the image cannot be the whole target. Every commuting lift is a coefficient multiplier by Theorem 14.1, proving the final assertion. $\square$

Warning 14.6 (The kernel is larger than finite-support weights). For $\Gamma = \mathbb{Z}$, the indicator of the negative integers belongs to J_Γ although it has infinite support. Every Laurent series has only finitely many negative terms, so the corresponding projection always lands in Fin_Γ . Replacing J_Γ by the ideal of finitely supported weight functions would give a false lifting theorem.

This ideal is the same support condition that appeared in the strong real dual, but here it controls which diagonal endomorphisms disappear after quotienting by all finite-monomial vectors.

15 Two distinct infinite-coordinate layers

Continue to assume $0 \neq \Gamma \subseteq \mathbb{R}$. The completion Lf_Γ is preserved by every coefficient multiplier, including $\mathfrak{D}$ and all G_p .

Theorem 15.1 (Rational scalar structure on both completion layers). *There is a canonical exact sequence of $\mathbb{R}(X)$ -vector spaces*

$$0 \longrightarrow \text{Lf}_\Gamma / \text{Fin}_\Gamma \longrightarrow H_\Gamma / \text{Fin}_\Gamma \longrightarrow H_\Gamma / \text{Lf}_\Gamma \longrightarrow 0. \quad (15.1)$$

The first term has dimension $\mathfrak{c}$ over $\mathbb{R}(X)$. The last term is zero for cyclic Γ and has dimension $\mathfrak{c}$ for noncyclic Γ .

Proof. The operators G_p preserve Lf_Γ , and their errors in (13.2) have finite-support image. Hence all three quotients carry the rational scalar actions and all maps in (15.1) respect them.

For any $\delta > 0$ in Γ , the support $\{n\delta : n \geq 0\}$ is cofinal in Γ because $\Gamma \subseteq \mathbb{R}$. Thus every q_r in Theorem 13.6 belongs to Lf_Γ . Their classes remain independent in $\text{Lf}_\Gamma / \text{Fin}_\Gamma$, giving dimension $\mathfrak{c}$.

If Γ is cyclic, $\text{Lf}_\Gamma = H_\Gamma$. If it is noncyclic, it is dense in $\mathbb{R}$: absence of a least positive element gives positive group elements smaller than every prescribed real ϵ , whose integer translates meet every real interval. For each $n \geq 1$, choose

$$\gamma_n \in \Gamma \cap \left(-\frac{1}{n}, -\frac{1}{n+1} \right). \quad (15.2)$$

Then γ_n is strictly increasing to zero and $|\gamma_n|$ is comparable to $1/n$. For $r > 0$ let

$$y_r = \sum_{n \geq 1}^H r^n t^{\gamma_n}.$$

These are valid Hahn series, all supported below zero.

Suppose a finite rational relation between their classes modulo Lf_Γ holds. Clear denominators to obtain

$$z = \sum_j p_j(\mathfrak{D}) y_{r_j} \in \text{Lf}_\Gamma.$$

Since its support is bounded above by zero, left-finiteness forces z to have finite support. Therefore

$$\sum_j p_j(\gamma_n) r_j^n = 0$$

for all sufficiently large n . Let r_j be the largest base with $p_j \neq 0$. Write $p_j(X) = X^{d_j} u_j(X)$ with $u_j(0) \neq 0$. By (15.2), $|p_j(\gamma_n)|$ is bounded above and below by positive constants times n^{-d_j} . For every smaller base, the ratio

$$\frac{p_i(\gamma_n) r_i^n}{p_j(\gamma_n) r_j^n}$$

tends to zero, since it is bounded by a fixed power of n times $(r_i/r_j)^n$. Dividing the relation by the largest-base term gives $1 + o(1) = 0$, a contradiction. Thus the classes $[y_r]$ are $\mathbb{R}(X)$ -independent. The cardinality upper bound is again $\mathfrak{c}$. $\square$

For noncyclic real exponent groups, the two layers have genuinely different support behavior. The first consists of infinite expansions whose finite subsums converge in the chosen valuation topology. The second records infinite supports accumulating below a fixed precision cutoff. Both survive as full continuum-dimensional spaces even after the new rational-function scalars are admitted.

Corollary 15.2 (Splitting after quotient, nonsplitting before quotient). *The exact sequence (15.1) admits a noncanonical $\mathbb{R}(X)$ -linear splitting. By contrast, if Γ is noncyclic, the sequence*

$$0 \longrightarrow \text{Lf}_\Gamma \longrightarrow H_\Gamma \longrightarrow H_\Gamma/\text{Lf}_\Gamma \longrightarrow 0$$

has no $\mathbb{R}[X]$ -linear splitting.

Proof. Every exact sequence of vector spaces over a field splits by choosing bases, giving the first assertion. The last quotient is nonzero and divisible over $\mathbb{R}[X]$. A module splitting would embed it as a nonzero divisible submodule of H_Γ , contradicting Proposition 14.4. $\square$

The contrast is instructive. A complement becomes available after the finite-coordinate torsion is removed, but not as a compatible subspace of the original Hahn space. The obstruction is structural, not merely a failure to find a clever basis.

16 Concrete examples and computational consequences

16.1 A Laurent slice: infinite coordinates without a completion gap

Take $\Gamma = \mathbb{Z}$ and $t = \omega^{-1}$. Then

$$H_{\mathbb{Z}} = \mathbb{R}((t)), \quad \text{Fin}_{\mathbb{Z}} = \mathbb{R}[t, t^{-1}], \quad \text{Lf}_{\mathbb{Z}} = H_{\mathbb{Z}}.$$

Every Laurent series is the internal valuation limit of its finite truncations, yet

$$\dim_{\mathbb{R}}(H_{\mathbb{Z}}/\text{Fin}_{\mathbb{Z}}) = \dim_{\mathbb{R}(X)}(H_{\mathbb{Z}}/\text{Fin}_{\mathbb{Z}}) = \mathfrak{c}.$$

Thus completion may add maximally many Hamel directions even when it supplies all admissible series. The Euler operator is $\mathfrak{D} = t d/dt$, and the quotient has no real eigenvalues: $\mathfrak{D} - \lambda$ is invertible there for every $\lambda \in \mathbb{R}$.

Consider $x = (1 - 2t)^{-1} = \sum_{n \geq 0} 2^n t^n$. For $p(X) = X - 2$, the equation $(\mathfrak{D} - 2)y = x$ has no solution in $H_{\mathbb{Z}}$, because the t^2 coefficient of the left side is always zero while that of x equals 4. The canonical nonresonant solution is

$$y = \sum_{\substack{n \geq 0 \\ n \neq 2}}^H \frac{2^n}{n - 2} t^n, \quad (\mathfrak{D} - 2)y = x - 4t^2.$$

Modulo $\text{Fin}_{\mathbb{Z}}$, this is the unique solution class to $(\mathfrak{D} - 2)[y] = [x]$. Finite resonance, not an analytic convergence issue, is the entire obstruction.

16.2 A rational-exponent slice: the completion gap is visible

For $\Gamma = \mathbb{Q}$, let

$$u = \sum_{n \geq 1}^H t^{-1/n}.$$

This is a valid Hahn series with increasing support bounded above by zero. It is not in $\text{Lf}_{\mathbb{Q}}$, because infinitely many support exponents lie below zero. Its finite subsums never approximate it beyond the cutoff zero, although their Hahn sum is perfectly well defined.

Since the exponent $-1/n$ is nonzero,

$$\mathfrak{D} \left(- \sum_{n \geq 1}^H n t^{-1/n} \right) = u.$$

The unbounded coefficient factor n causes no support problem. This illustrates why support criteria, rather than ordinary coefficient magnitude estimates, govern the Euler calculus here.

16.3 A finite-dimensional canonical basis with infinite tails

Let

$$a = 1 + t + t^2 + t^3 + \cdots, \quad b = 1 + 2t + 3t^2 + 4t^3 + \cdots,$$

and $V = \text{span}_{\mathbb{R}}\{a, b\} \subseteq \mathbb{R}((t))$. The pivot exponents are 0 and 1. The unique reduced basis from Theorem 9.2 is

$$\begin{aligned} b_0 &= 2a - b = 1 - t^2 - 2t^3 - 3t^4 - \cdots, \\ b_1 &= b - a = t + 2t^2 + 3t^3 + \cdots. \end{aligned}$$

Their coefficients at exponents 0 and 1 are $(1, 0)$ and $(0, 1)$. Every nonzero member of V has leading exponent 0 or 1, despite the infinitely many later coordinates. Finite-dimensional leading-coordinate elimination is exact, not an approximation by dropping tails.

16.4 Recommended symbolic representation

For computation, the normal-form representation should be the primary interface. A series object needs an exponent domain, a support specification with an admissibility argument, and a coefficient rule. It should not merely be an unconstrained dictionary indexed by every surreal.

The natural primitive operations are finite-support arithmetic, coordinate extraction, support projection, certified Hahn summation, leading-term elimination, and diagonal coefficient operators. Polynomial Euler inversion additionally returns the finite resonance projection $P_{Z_p}x$ as an explicit obstruction certificate. This is more informative than asking a symbolic system for an unspecified Hamel basis.

For finitely many explicitly represented Hahn vectors, the row-echelon algorithm successively finds the first nonzero coefficient functional, normalizes a pivot, and eliminates it. In exact symbolic infinite-support input, termination of each search requires an effective leading-term or zero test. The abstract finite-dimensional theorem by itself does not provide such an algorithm for arbitrary computable coefficient streams.

17 Verification and formalization boundaries

The accompanying program `code/52-euler-quotient-verify.py` uses exact rational arithmetic. It checks finite Hahn-row reduction, the Vandermonde tail determinant, the Euler resonance identities,

the finite matrix commutant calculation for distinct exponents, and the identities underlying the elementary reweighting and shear examples. It also checks finite prefixes of the two independent-family constructions. All 39 exact finite checks passed in the supplied run; the recorded result is in `data/52-euler-quotient-verification.json`.

[write] Source 52 named the program `code/verify.py` and the record `data/verification.json`; they are shipped under the names above, byte-identical. The program always writes `../data/verification.json` relative to its own directory, so a rerun must use a copy laid out under the delivered names (the report’s `README.md` gives the recipe).

These tests are useful for catching coefficient, sign, pivot, and resonance mistakes. They do not prove well-ordering of infinite supports, global choice, infinite dimension, spherical completeness, or the universal quantifier over all real-linear operators. Those statements are supported by the written proofs, not by finite experiments.

A formalization should begin at the set-sized Hahn-space level. The most reusable steps are the support-preserving diagonal-map construction, the exact kernel and image of $\mathfrak{D} - \gamma$, the finite-resonance identity, and the commutant proof. The rational-function action can then be obtained by localization of the $\mathbb{R}[X]$ -module. The more general class-basis discussion should remain a separate layer, rather than contaminating the set-sized proof architecture.

Conway normal form has already been formally studied in Mizar by Pąk and Kaliszuk [20]. That is background evidence for a formalization route, not verification of this manuscript. No new Lean or Mizar proof is claimed in this package.

Part V

Multiplicative structure, synthesis, and outlook

18 Monomial symmetries, derivations, and algebraic independence

The Hahn basis is not only linear. Because its indices form the additive surreal group, linear transformations of the monomials often extend to field automorphisms and derivations.

18.1 Monomial-preserving ordered-field automorphisms

Let $\phi : (\mathbf{No}, +, <) \rightarrow (\mathbf{No}, +, <)$ be an ordered additive class automorphism, and let

$$\chi : (\mathbf{No}, +) \longrightarrow (\mathbb{R}_{>0}, \cdot)$$

be a positive real character. Define

$$T_{\chi, \phi} \left(\sum_{\gamma \in S}^H r_{\gamma} \omega^{\gamma} \right) = \sum_{\gamma \in S}^H r_{\gamma} \chi(\gamma) \omega^{\phi(\gamma)}. \quad (18.1)$$

[write] *Repository results.* The forward direction of the next theorem is already in the collection. For $\chi = 1$ the map $T_{1, \phi}$ is the canonical Hahn lift of the exponent-rigidity report (`surreal/exponential-automorphism-rigidity`, Proposition 8.3, “Canonical field lift”), formalized as `Surreal.Foundations.SignSequence.exponentAutomorphism` (`Surreal/Foundations/NormalFormExponentAutomorphisms.lean`). With a character, $T_{\chi, \phi}$ is the monomial map $M_{\chi, \tau}$ of the omnific-preserving report (`surreal/omnific-preserving-automorphisms`, (5.1)), written there in t -exponents. The converse is the case of that report’s Theorem 5.1 (strong valued factorization

$\sigma = M_{\chi, \tau} \circ u$, after Kuhlmann–Serra) in which the factor u fixes every monomial and is therefore the identity; there it is stated for set-sized Hahn fields. Source 51 cites neither result (its repository table, [section 20](#), names the exponent-rigidity report only as complementary). Its proof is printed as a second route; only the class-level formulation on all of **No** is not already in the collection.

Theorem 18.1 (Monomial automorphisms). *The map $T_{\chi, \phi}$ is a strong $\mathbb{R}$ -linear ordered-field automorphism of **No**. Conversely, every strong $\mathbb{R}$ -linear ordered-field automorphism that permutes the monomial lines $\{\mathbb{R}\omega^\gamma : \gamma \in \mathbf{No}\}$ has this form for unique χ and ϕ .*

Proof (source 51); a second route to the results cited above. Because ϕ is an order isomorphism, it sends reverse-well-ordered sets to reverse-well-ordered sets. The columns in (18.1) have singleton supports with distinct output exponents, so [theorem 10.3](#) gives strong real linearity. The inverse is obtained from ϕ^{-1} and the character $\gamma \mapsto \chi(\phi^{-1}\gamma)^{-1}$.

For monomials,

$$\begin{aligned} T_{\chi, \phi}(\omega^\gamma \omega^\delta) &= \chi(\gamma + \delta) \omega^{\phi(\gamma + \delta)} \\ &= \chi(\gamma) \chi(\delta) \omega^{\phi(\gamma) + \phi(\delta)} \\ &= T_{\chi, \phi}(\omega^\gamma) T_{\chi, \phi}(\omega^\delta). \end{aligned}$$

Strong distributivity extends this identity to arbitrary normal forms. Positive real coefficients and the increasing map ϕ preserve the sign of the leading term, hence the order.

Conversely, suppose T strongly permutes monomial lines. Write uniquely

$$T(\omega^\gamma) = \chi(\gamma) \omega^{\phi(\gamma)}, \quad \chi(\gamma) > 0.$$

Bijectivity on lines makes ϕ a bijection. Multiplicativity of T gives $\phi(\gamma + \delta) = \phi(\gamma) + \phi(\delta)$ and $\chi(\gamma + \delta) = \chi(\gamma)\chi(\delta)$. Preservation of the order of positive monomials makes ϕ increasing. Strong linearity then forces T to act on every normal form by (18.1). $\square$

This elementary subgroup is one of the reasons the monomial basis is useful: its symmetries are controlled by familiar group data. The full automorphism group is richer; Kaplan, Krapp, and Serra give a systematic Hahn-field decomposition and distinguish strongly linear automorphisms from general ones [14].

Example 18.2 (Exponent dilations). For every positive surreal a , multiplication

$$\phi_a(\gamma) = a\gamma$$

is an ordered additive automorphism of **No**. Taking $\chi = 1$ gives

$$\sigma_a\left(\sum_{\gamma}^{\mathbf{H}} r_{\gamma} \omega^{\gamma}\right) = \sum_{\gamma}^{\mathbf{H}} r_{\gamma} \omega^{a\gamma}. \quad (18.2)$$

Then $\sigma_a(\omega) = \omega^a$, $\sigma_a^{-1} = \sigma_{a^{-1}}$, and $\sigma_a \sigma_b = \sigma_{ab}$. These are ordered-field automorphisms fixing $\mathbb{R}$. Preservation of the surreal exponential or of the omega map as an additional named operation is a separate and much more rigid requirement.

[write] For a rational $a = r \neq 1$ the exponent-rigidity report proves more (its Theorem 8.4, “Rational dilations do not lift”): no automorphism of **No** commuting with the exponential induces the value action $\gamma \mapsto r\gamma$. In particular σ_r does not commute with $\exp$. This answers part of [research question 21.14](#).

Theorem 18.3 (Dilation-invariant strong functionals). *Let σ_a be the exponent-dilation automorphism (18.2). A strong real-linear functional $\Lambda : \mathbf{No} \rightarrow \mathbb{R}$ satisfies*

$$\Lambda \circ \sigma_a = \Lambda \quad \text{for every } a \in \mathbf{No}_{>0}$$

if and only if Λ is a real multiple of constant-term extraction $\text{ct} = [\omega^0]$.

Proof. Let $w(\gamma) = \Lambda(\omega^\gamma)$. Invariance gives $w(a\gamma) = w(\gamma)$ for all $a > 0$. Any two positive exponents are related by multiplication by a positive surreal, so w is constant on $\mathbf{No}_{>0}$; similarly it is constant on $\mathbf{No}_{<0}$. If the positive constant were nonzero, the active set would contain the decreasing sequence $1 > 1/2 > 1/3 > \dots$, contradicting [proposition 10.6](#). If the negative constant were nonzero, it would contain $-1 > -2 > -3 > \dots$, again a contradiction. Thus only the weight at exponent 0 may survive, giving a multiple of ct . The converse is immediate because every σ_a fixes exponent 0 and its coefficient. $\square$

Example 18.4 (Coefficient characters). If $d : (\mathbf{No}, +) \rightarrow (\mathbb{R}, +)$ is an additive class homomorphism, then $\chi_d(\gamma) = \exp(d(\gamma))$ is a positive character. The resulting automorphism fixes every monomial line and rescales it by a positive real. A concrete choice is $d = \text{ct}$, constant-term extraction applied to the exponent γ :

$$\omega^\gamma \mapsto e^{\text{ct}(\gamma)} \omega^\gamma.$$

18.2 Diagonal strong derivations

The same additive characters produce derivations rather than automorphisms.

[write] The dilation-invariant dual ([theorem 18.3](#)) is not in the collection. The derivations of the next theorem are: for a set-sized Hahn field they are the Euler derivations ∂_λ of the omnific–Diophantine report ([surreal/omnific-diophantine-geometry](#), (7.1) and Lemma 7.1 for divisible H and $\mathbb{Q}$ -linear λ ; (16.1) and Lemma 16.2 for any Γ and additive λ), with $\lambda = d$. Source 51 proves the forward direction on the whole class and adds the converse, which is new but short.

Theorem 18.5 (Diagonal derivations). *Let $d : (\mathbf{No}, +) \rightarrow (\mathbb{R}, +)$ be additive. Then*

$$D_d\left(\sum_{\gamma}^H r_{\gamma} \omega^{\gamma}\right) = \sum_{\gamma}^H r_{\gamma} d(\gamma) \omega^{\gamma} \quad (18.3)$$

is a strong $\mathbb{R}$ -linear derivation of $\mathbf{No}$:

$$D_d(xy) = xD_d(y) + yD_d(x).$$

Conversely, every strong $\mathbb{R}$ -linear derivation satisfying $D(\omega^\gamma) \in \mathbb{R}\omega^\gamma$ for every γ is uniquely of this form.

Proof (source 51); its forward direction is a second route. The support of $D_d x$ is contained in $\text{supp}(x)$, so the map is strong. For monomials,

$$\begin{aligned} D_d(\omega^{\gamma+\delta}) &= d(\gamma + \delta) \omega^{\gamma+\delta} \\ &= (d(\gamma) + d(\delta)) \omega^{\gamma+\delta} \\ &= \omega^\gamma D_d(\omega^\delta) + \omega^\delta D_d(\omega^\gamma). \end{aligned}$$

Hahn distributivity and local finiteness extend the Leibniz identity to all normal forms.

Conversely, write $D(\omega^\gamma) = d(\gamma) \omega^\gamma$. The Leibniz identity on a product of two monomials forces $d(\gamma + \delta) = d(\gamma) + d(\delta)$. Strong linearity determines D on every normal form, giving (18.3). $\square$

Example 18.6 (A surreal Euler-type derivation). With $d = \text{ct}$,

$$D_{\text{ct}}(\omega^r) = r\omega^r \quad (r \in \mathbb{R}),$$

so in particular $D_{\text{ct}}(\omega) = \omega$, while D_{ct} kills real constants. It is diagonal in the Conway–Hahn basis. It is not asserted here to commute with the surreal exponential; differential compatibility is an additional structure studied by Berarducci–Mantova and later work [4, 2].

[write] Restricted to a real-exponent workspace, $D_{\text{ct}} = -\mathfrak{D}$, where $\mathfrak{D}$ is source 52’s Euler operator of Part IV; the sign comes from $t^\gamma = \omega^{-\gamma}$.

18.3 Algebraic independence of monomials

Linear coordinates also give a clean field-theoretic criterion.

Theorem 18.7 (Rational exponents versus algebraic monomials). *For $\gamma_1, \dots, \gamma_n \in \mathbf{No}$, the monomials $\omega^{\gamma_1}, \dots, \omega^{\gamma_n}$ are algebraically independent over $\mathbb{R}$ if and only if the exponents $\gamma_1, \dots, \gamma_n$ are linearly independent over $\mathbb{Q}$.*

Proof. Suppose first that the exponents are $\mathbb{Q}$ -dependent. Clearing denominators gives a nonzero integer relation $\sum_i m_i \gamma_i = 0$. Moving negative terms to the other side gives an equality between two distinct monomials in the ω^{γ_i} , hence a nonzero polynomial relation over $\mathbb{R}$.

Conversely, assume the exponents are $\mathbb{Q}$ -independent and let

$$P(X_1, \dots, X_n) = \sum_{\mathbf{k} \in F} c_{\mathbf{k}} X_1^{k_1} \cdots X_n^{k_n}$$

be a nonzero polynomial with finite exponent set $F \subseteq \mathbb{N}^n$. Distinct multiindices $\mathbf{k} \neq \mathbf{l}$ give distinct surreal exponents

$$\sum_i k_i \gamma_i \neq \sum_i l_i \gamma_i,$$

by rational independence. Therefore

$$P(\omega^{\gamma_1}, \dots, \omega^{\gamma_n}) = \sum_{\mathbf{k} \in F} c_{\mathbf{k}} \omega^{\sum_i k_i \gamma_i}$$

is a nonzero finite normal form. No cancellation between distinct exponents is possible. $\square$

Corollary 18.8. *Every nontrivial monomial ω^γ , with $\gamma \neq 0$, is transcendental over $\mathbb{R}$.*

Corollary 18.9 (An explicit proper-class independent family). *The family*

$$\{\omega^{\omega^\alpha} : \alpha \in \mathbf{On}\} \tag{18.4}$$

is algebraically independent over $\mathbb{R}$.

Proof. The exponent family $\{\omega^\alpha : \alpha \in \mathbf{On}\}$ is real-linearly, hence rationally, independent: every finite relation would be a finite normal form with distinct exponents α . Apply theorem 18.7. $\square$

Thus the field extension $\mathbf{No}/\mathbb{R}$ has proper-class transcendence complexity in the evident class-theoretic sense. An actual transcendence basis again requires class choice and is not supplied canonically by normal form; the explicit family (18.4) gives a natural lower bound without such a choice.

[write] The independent-surreal-copies report (`surreal/independent-surreal-copies`) proves algebraic independence of other explicit families, including a class-level statement for support subfields; it selects no proper-class Hamel basis. Its results and corollary 18.9 concern different families, and neither is used for the other.

19 What the natural basis should mean

The preceding results support a precise hierarchy of answers.

- A. The **canonical basis-like object** is the monomial Hahn basis $\{\omega^a : a \in \mathbf{No}\}$ with coefficient functionals c_a and strong summation.
- B. The **canonical ordinary basis** exists only after passing to the associated graded algebra: the degree- a line is generated by $[\omega^a]$.
- C. A **global Hamel basis** is possible only under a stated class-choice framework. It is noncanonical and necessarily contains vectors with supports of unbounded set cardinality.
- D. The monomials are **not a natural-valuation topological basis**; their finite span is already closed.

This hierarchy reconciles two common but apparently conflicting statements: “Conway normal form gives coordinates” and “the monomials do not form a vector space basis.” Both are correct once the permitted support and summation law are specified.

19.1 Why a Hamel basis is the wrong invariant for most surreal analysis

A Hamel basis deliberately forgets order, valuation, support, multiplication, and the simplicity hierarchy. In a finite-dimensional real vector space that loss may be harmless because all norms and bases are equivalent up to manageable linear transformations. In $\mathbf{No}$ it is catastrophic. A Hamel basis must compress a normal form of arbitrarily large support into a finite list of basis vectors; by [theorem 5.10](#), the complexity is merely relocated into the basis vectors themselves.

By contrast, the monomial Hahn basis simultaneously exposes:

- the total order, through the greatest exponent with nonzero coefficient;
- the valuation, through the leading exponent;
- multiplication, through addition of exponents and convolution of coefficients;
- convex subspaces, through downward-closed exponent cuts;
- canonical projections and diagonal operators;
- strong summability, through reverse-well-ordered supports and finite coefficient fibres; and
- exponent symmetries, through permutation of monomials.

No ordinary Hamel basis is known or expected to package all these features.

20 Relation to the ProveIt collection and the literature

20.1 Repository map

The repository snapshot `f7e7d86` contains a broad surreal-number program [\[21\]](#). The present article is designed to complement, rather than repeat, the following components.

Repository item	Principal object	Relation to this article
<code>docs/surreal/vector-and-tensor-fields</code>	Finite-dimensional K^n , tensor fields, and support-controlled calculus over set-sized surreal subfields	Uses surreal numbers as <i>scalars</i> ; does not ask for a basis of No over $\mathbb{R}$.
<code>docs/surcomplex/three-duals-of-hahn-vector-spaces</code>	$V((t^\Gamma))$, algebraic scalar extension, valuation completion, strong and continuous duals	Supplies the closest set-sized analogy to the strong map and dual distinctions in section 10 .
<code>docs/surcomplex/infinite-dimensional-hahn-spectral-theory</code>	Hahn-summable vector families and Hahn-orthonormal synthesis	Explicitly distinguishes Hahn bases from Hamel and Hilbert bases; the present report applies that distinction to No itself.
<code>docs/surcomplex/hahn-hilbert-geometry</code>	Set-sized Hahn–Hilbert workspaces at surreal scales	Illustrates why analytical work selects a set-sized workspace rather than the full fine class topology.
<code>docs/surreal/exponential-automorphism-rigidity</code>	Ordered, valuation, and exponential automorphisms of No	Complements the monomial-preserving strong automorphisms in section 18 ; extra exponential structure is substantially more rigid.

[write] Exact correspondences, established when this report was written: the three-duals report’s Theorem 3.2 is the K -linear analogue of [theorem 10.3](#), and source 52’s completion results are its diagonal case ([section 12](#)); the exponent-rigidity report’s Proposition 8.3 and Theorem 8.4 contain the monomial automorphisms with $\chi = 1$ and the exponential obstruction for rational dilations ([section 18](#)).

The repository states that its reports are AI-assisted, unrefereed drafts and separately records which results have Lean formalizations. The same claim firewall is adopted here: no theorem newly proved in this report is presented as machine checked merely because related foundations exist in the repository.

20.2 Classical sources

Conway introduced **No** and its normal forms; Gonshor developed the sign- sequence and omega-map theory in detail [6, 11]. Hahn’s generalized series and Neumann’s support lemma provide the ambient algebraic technology [12, 19]. Van den Dries and Ehrlich related the surreal field and exponentiation to model-theoretic and Hahn-field structure [25]; Ehrlich’s surveys explain the role of **No** as a universal ordered continuum [10]. Berarducci–Mantova and Aschenbrenner–van den Dries–van der Hoeven developed the differential and transseries aspects [4, 2]. Mantova–Matusinski give a modern survey of Conway normal form and surreal analysis [17].

The automorphism viewpoint used in [section 18](#) is now developed systematically by Kaplan, Krapp, and Serra, who work in NBG with global choice, view **No** as a Hahn field, decompose its automorphism group, and isolate strongly linear automorphisms [14]. Their work also notes that choice-dependent bases enter descriptions of additive homomorphisms; this reinforces the separation between intrinsic Hahn coordinates and arbitrary Hamel coordinates.

The set-sized dimension formula in [theorem 5.15](#) is the Erdős–Kaplansky theorem. We cite Jacobson’s classical account and Bergman’s later discussion and extension [13, 5].

20.3 What is genuinely specific to the vector-space question

The literature generally invokes Conway normal form to study order, fields, valuations, derivations, or automorphisms. The present organization instead starts from the linear question and follows its consequences:

1. it separates Hamel, Hahn, graded, and Schauder meanings of basis;
2. it proves the no-set-spanning theorem directly from support;
3. it quantifies the local Hamel-dimension explosion inside support windows;
4. it exposes the quotient $\mathbf{No}/\mathbf{No}_{\text{fin}}$ by an explicit proper-class independent family;
5. it derives an intrinsic strong real dual and contrasts it with the zero order dual;
6. it explains why the full proper-class valuation topology cannot realize normal forms as set-indexed limits.

These are primarily structural deductions from classical coordinates. Their value is to prevent category errors when importing finite-dimensional or Banach-space intuitions into surreal analysis.

[write] Source 53 maps its results to the Lean development as follows. Its paths were checked to exist at the pin and at the placement commit; the fully qualified declarations are listed in [section 20.5](#).

20.4 Connection with the ProveIt repository

The repository was inspected on the `main` branch at commit `f7e7d8607d3b1c49c93b844cae20a77ab3689fe7` (1 October 2026). Its `Algebra/SurrealNumbers` project contains both research reports and a Lean 4 development of a universe-relative surreal field. The following files are particularly relevant.

Repository path	Role in the present report
<code>Surreal/Foundations/SmallNormalFormFieldEquiv.lean</code>	Packages canonical extraction and evaluation as an ordered ring equivalence between the actual sign-sequence surreal field and a small formal Hahn field. This is the formal counterpart of theorem 3.6 .
<code>Surreal/Foundations/SignSequenceStrongSummation.lean</code>	Defines strong summability by joint support well-ordering and finite coefficient fibres, constructs the actual strong sum, and proves exact coefficient formulas. It explicitly distinguishes coefficientwise Hahn summation from topological convergence.
<code>Surreal/Foundations/OmnificPurelyInfiniteModule.lean</code>	Transports a real module structure to the actual purely infinite omnific ideal and proves that scalar multiplication is literal multiplication by the embedded real.
<code>Surreal/Foundations/OmnificPurelyInfiniteCoefficients.lean</code>	Defines fixed Conway-coefficient extraction as an $\mathbb{R}$ -linear functional, proves that all coefficients vanishing detects zero, and proves a coefficientwise criterion for finite real linear equations. This is a formalized special case of corollary 3.7 and theorem 8.1 .
<code>Surreal/Algebra/ModuleKernelBasis.lean</code>	Develops generic kernel bases for systems with vector-space-valued unknowns, matching the mechanism behind theorem 8.2 .

Repository path	Role in the present report
<code>Surreal/Foundations/NormalFormExponentAutomorphisms.lean</code>	Transports ordered additive exponent automorphisms to actual surreal field automorphisms, fixes real constants, and maps ω^a to $\omega^{e(a)}$ as in (7.6).
<code>README.md</code> and <code>docs/FORMALIZATION.md</code>	State the scope of the project, its machine-checked foundations, and the mapping from research statements to declarations.

The repository therefore supplies unusually strong foundations for formalizing the present article. It already has the coefficient API that would make most new proofs short. The remaining work is mainly to organize existing primitives into a dedicated linear-algebra layer.

20.4.1 Proposed Lean implementation plan

A practical formalization could proceed in the following order.

- L1.** `Surreal/Linear/MonomialSpan.lean`: prove monomial independence; define $\mathbf{No}_{\text{fin}}$; identify it with the finite-support group algebra; prove the geometric-series witness lies outside it.
- L2.** `Surreal/Linear/SupportProjection.lean`: define P_S for set-sized or predicate-defined exponent classes; prove the Boolean projection identities and cut decompositions.
- L3.** `Surreal/Linear/Diagonal.lean`: define D_λ ; prove the commutant, idempotent, invertibility, positivity, and covariance theorems.
- L4.** `Surreal/Linear/LeadingFiltration.lean`: define $F_{<a}$ and $F_{\leq a}$; prove the split exact sequence and the one-dimensional graded layer. A fully packaged proper-class associated graded algebra may require a universe-local formulation, but each degree statement is set-sized and direct.
- L5.** `Surreal/Linear/ConvexSubspaces.lean`: classify convex submodules by exponent cuts, using leading-term comparability.
- L6.** `Surreal/Linear/NaturalValuationClosure.lean`: formalize the lower bound of a set of support exponents and prove that $\mathbf{No}_{\text{fin}}$ is closed. The statement can first be made in terms of the explicit neighborhood predicates U_γ , avoiding heavy general topology.
- L7.** `Surreal/Linear/SupportSlices.lean`: for a fixed reverse-well-ordered set A , construct the equivalence $\mathbf{H}(A) \simeq \mathbb{R}^A$. The exact Hamel-dimension formula depends on the available cardinal-dimension API; the explicit Vandermonde family is a simpler first milestone.

Verification boundary. The repository paths above were inspected as source and are part of the repository’s stated Lean build. This report did not clone and rebuild the entire multi-gigabyte repository, and the new theorems in this manuscript have not yet been translated into Lean. The mathematical proofs here are self-contained once Conway normal form and standard Hahn arithmetic are granted.

20.5 Formal status

[write] **Nothing in this report is formalized.** Placement in `Algebra/SurrealNumbers`, beside its Lean development, confers no formal status, and the collection’s `FORMALIZATION.md` maps no

`rvs`: label to a declaration. The development formalizes the following special cases and ingredients (declarations checked to exist):

- `Surreal.Foundations.SmallNormalForm.cutEvaluationOrderRingIso` (`Surreal/Foundations/SmallNormalFormFieldEquiv.lean`): normal-form extraction and evaluation as an ordered ring isomorphism between the universe-relative sign-sequence field and a small Hahn field, the universe-relative form of [theorem 3.6](#);
- `Surreal.Foundations.SignSequence.StronglySummable`, `strongSum` and `coeff_normalForm_strongSum` (`Surreal/Foundations/SignSequenceStrongSummation.lean`): strong summation, [definition 3.8](#);
- `Surreal.Foundations.SignSequence.omnificPurelyInfiniteCoeff`, `omnificPurelyInfinite_eq_zero_iff` and `omnificPurelyInfinite_linear_coeff_iff` (`Surreal/Foundations/OmnificPurelyInfiniteCoefficients.lean`), with the real module structure of `Surreal/Foundations/OmnificPurelyInfiniteModule.lean`: coefficient separation and the coefficientwise criterion for finite real systems, on the purely infinite omnific ideal only ([corollary 3.7](#) and [theorem 8.1](#));
- `Surreal.ModuleKernelBasis` (`Surreal/Algebra/ModuleKernelBasis.lean`): generic kernel bases for systems with vector-space-valued unknowns, the mechanism of [theorem 8.2](#);
- `Surreal.Foundations.SignSequence.exponentAutomorphism` (`Surreal/Foundations/NormalFormExponentAutomorphisms.lean`): the automorphisms $T_{1,\phi} = \sigma_e$ of [theorem 18.1](#) and (7.6).

Source 53’s plan L1–L7 proposes new files under `Surreal/Linear/`; none exists, and the plan is a proposal, not a promise. Source 52’s program checks finite instances ([section 17](#)); it proves no infinite statement.

21 Further research questions and proposed program

The vector-class viewpoint opens a research program that is distinct from, but compatible with, the existing surreal analysis, automorphism, and formalization programs. The questions below are ordered roughly from foundational to analytic and computational.

21.1 Foundations and definability

Research question 21.1 (Choice strength for class bases). What is the weakest conservative class theory over ZFC that proves every class vector space whose carrier is bijective with **On** has a Hamel class basis? In particular, how much of global choice is needed for [theorem 5.3](#), and can the statement fail in natural models of NBG without global choice?

The staged proof uses global choice to select basis extensions through a proper class of stages. A reverse-mathematical analysis would separate the choice needed for this particular highly structured vector class from a general class-basis principle.

[write] Sources 53 and 52 ask the same question. Source 53 asked it for the hypothesis “MK, or GBC with ETR” of its staged construction; as corrected after [proposition 5.4](#), GBC suffices, and what remains open is the question above: how much of global choice is needed.

Source 53 (“Weakest basis principle”). What is the weakest natural class theory proving that every class vector space presented as an ordinal-indexed union of set-sized subspaces has a class Hamel

basis? Determine the exact strength needed for [proposition 5.4](#), separating global choice, elementary transfinite recursion, and stronger class choice schemes.

Source 52 (“Class foundations and definability of Hamel extensions”). Determine the exact foundational strength needed for a class Hamel basis of **No** extending all monomials, and distinguish existence from definability in enriched surreal structures. The argument here assumes a set-like global well-order. Which weaker class-choice principles suffice? Conversely, which naturality conditions are incompatible with definable selection even when existence is assumed? The explicit symmetry obstruction is a starting point, not a complete definability theorem.

[Added 2 October 2026, batch 80: source 52’s hypothesis, the set-like global well-order of **No** used in its second proof of [proposition 5.4](#), is equivalent to global choice over GB, without set choice. The surreal-well-orders report ([foundations-and-computation/surreal-well-orders](#), from batch-80 manuscripts 08, 09, 11 and 12) proves in its Theorem 13.1 ([swo:cf:choice](#), its source 11’s) that over GB without set choice global choice, a class well-order of **No**, a set-like class well-order of **No** and a class bijection **On** $\rightarrow$ **No** are equivalent; its Remark 13.4 ([swo:rem:answered-choice](#)) explains why no set choice is needed. So source 52’s “set-like global well-order” is the same hypothesis as the global choice assumed in [proposition 5.4](#), and the question above, how much of global choice a class Hamel basis of **No** needs and which weaker class-choice principles suffice, is unaffected: that report does not derive a well-order of **No** from a Hamel basis.]

Research question 21.2 (Definable Hamel bases). Does **No** admit a Hamel basis definable, without arbitrary parameters, in natural class universes such as $(V, \in)$, L , or HOD? Can one prove that no such basis is invariant under specified large subgroups of $\text{Aut}(\mathbf{No}, +, \cdot, <)$?

A useful intermediate target is a definable basis containing all monomials, as in [proposition 5.4](#), together with a definable rule for selecting the necessary infinite-support supplement. Symmetry may make this impossible in the full universe even when a basis exists abstractly.

[write] *Source 53 asks the same (“Definable Hamel bases”):* Can one define a Hamel basis of a set-sized surreal universe from its canonical well-order, and under what hypotheses can one define a global class Hamel basis? Study ordinal-definability, parameter-definability, and the behavior in $V = L$, HOD, or canonical inner models.

Research question 21.3 (Class matroids). Develop a class-matroid framework in which linear independence on **No**, basis extension, closure, and rank are formulated without forcing all ranks to be set cardinals. Which ordinary matroid theorems survive for set-finitary class matroids under global choice?

The linear matroid of **No** is finitary—dependence is witnessed by a finite subset—but its ground class and every basis are proper. This makes it a natural test case for class-sized combinatorial geometry.

21.2 Support strata and quotient geometry

Research question 21.4 (Structure of $\mathbf{No}/\mathbf{No}_{\text{fin}}$). Classify the real vector-class quotient $\mathbf{No}/\mathbf{No}_{\text{fin}}$ and its natural filtrations by support order type, cofinality, and leading exponent. Is there a canonical hierarchy of subquotients whose set-sized pieces have computable Hamel dimensions?

The block family in [theorem 5.7](#) gives only a lower bound. A finer analysis could distinguish countable support, bounded support order type, and support contained in specified exponent intervals. [write] [Theorem 5.9](#) and [theorem 5.10](#) give further lower bounds; the question is open.

Research question 21.5 (Support-rank dimension spectra). For an infinite cardinal κ , let $\mathbf{No}^{(<\kappa)}$ consist of surreals whose normal-form support has cardinality below κ , or order type below κ . Determine the Hamel dimensions of set-sized intersections of these classes and the strictness of the resulting hierarchy.

The countable-window theorem suggests cardinal-exponent formulas, but order type and ambient exponent geometry can interact when the allowed support window itself is not reverse well ordered.

Research question 21.6 (Canonical complements). Which naturally defined real subspaces of $\mathbf{No}$ admit canonical strong complements? Support-defined subspaces do by [theorem 6.1](#); what happens for birthday cutoffs, differential kernels, logarithmic strata, or fields generated by selected monomials?

The contrast between support subspaces and birthday subspaces may reveal which decompositions belong to the Hahn geometry and which depend on the genetic construction.

[write] For the Euler kernel and its finite-monomial part, source 52's [theorem 14.3](#) shows that no $\mathfrak{D}$ -invariant complement exists, although algebraic complements do.

21.3 Strong operator algebras and duality

Research question 21.7 (The algebra of Hahn-compatible matrices). Give intrinsic necessary and sufficient conditions for composition, invertibility, triangular factorization, and spectral decomposition of the class matrices in [theorem 10.3](#). Identify natural operator ideals: finite-row, finite-column, support-contracting, valuation-increasing, and nuclear analogues.

A promising route is to first solve the problem on every set-sized support window and then formulate uniform class conditions. The recent theory of automorphisms and derivations on algebras with formal infinite sums should be compared directly [\[3\]](#).

[write] Source 53's question "Strongly linear matrices" asked for the classification itself; it is answered by [theorem 10.3](#) ([remark 21.42](#)). The question above, on the algebra of these matrices, stays open.

Research question 21.8 (Invariant strong duals). For important automorphism groups G of $\mathbf{No}$, compute

$$(\mathbf{No}_{\text{str}}^{\vee})^G.$$

[Theorem 18.3](#) shows that invariance under all positive exponent dilations leaves exactly the constant-term line. What are the invariant strong functionals for valuation-preserving, omega-preserving, or exponential automorphism groups?

This is a tractable bridge between the linear theory developed here and the automorphism decomposition of Kaplan–Krapp–Serra.

Research question 21.9 (Duality beyond real scalars). Classify strong, order-bounded, and topology-continuous maps from $\mathbf{No}$ to other codomains: a set-sized Hahn field, $\mathbf{No}$ itself, the surcomplex field, or quotient spaces such as $\mathbf{No}/\mathbf{No}_{\text{fin}}$. Which codomains avoid the collapse of [theorem 11.2](#)?

Real-valued positive states vanish because the codomain is Archimedean. Surreal-valued positive maps may exist and could provide a better foundation for surreal measure, probability, and convex analysis.

21.4 Topology and generalized bases

Research question 21.10 (Coarser topologies with genuine Schauder expansions). Classify set-generated valuation or bornological structures on $\mathbf{No}$ for which selected classes of Conway normal forms are limits of their finite truncations. What is the largest natural subspace on which the monomials form a Schauder basis for a given set-sized scale system?

Any answer must evade [theorem 11.8](#) by replacing the full proper-class neighborhood basis with a set-sized cofinal family, restricting the domain, or allowing class-indexed convergence.

[write] *Source 53 asks the same (“Topologies making monomials a basis”):* Classify coarsenings of the natural valuation, coefficientwise topologies, and bornologies on $\mathbf{No}$ for which finite monomial sums are dense and every normal form is the limit of a canonical net of truncations. Determine when such a topology is Hausdorff, complete, spherically complete, compatible with multiplication, or compatible with surreal exponentiation.

Research question 21.11 (Class-indexed convergence). Can class-indexed nets or filters be developed safely enough to make every Conway normal form a topological sum without collapsing separation axioms or leaving conservative class theory? How does such convergence compare with strong Hahn summation?

Strong summation may already be the correct replacement. A topological reconstruction would be valuable only if it adds robust continuity, completion, or functional-analytic theorems.

Research question 21.12 (Bornological and locally convex analogues). Is there a useful class-bornology on $\mathbf{No}$ whose bounded linear dual lies strictly between the strong dual and the zero order-bounded real dual? Can one formulate barrelledness, nuclearity, or reflexivity using only set-sized bounded pieces?

The support windows $H(A)$ provide a canonical directed family of set-sized test spaces. An inductive-limit or bornological approach may be more faithful than the full fine topology.

21.5 Differential and exponential compatibility

Research question 21.13 (Derivation-adapted bases). How does the Berarducci–Mantova derivation act relative to the monomial Hahn basis, and is there a more refined basis—perhaps using log-atomic numbers or transseries trees—in which its matrix is triangular or locally finite?

The diagonal derivations of [theorem 18.5](#) are completely controlled by additive real characters, but the canonical surreal derivation changes scales in a subtler way. A basis adapted simultaneously to multiplication, exponentiation, and derivation would be highly significant.

[write] *Source 53 asks the same (“Matrix of the surreal derivation”):* Express the Berarducci–Mantova derivation in monomial coordinates [4]. Determine its support matrix, triangularity properties, and induced operator on the associated graded algebra. Identify the principal symbol of differentiation at each asymptotic scale.

Research question 21.14 (Exponential rigidity of monomial automorphisms). Determine exactly which pairs (χ, ϕ) in [theorem 18.1](#) commute with the surreal exponential or preserve the omega map. Existing rigidity results rule out broad subclasses; an explicit intersection with the monomial automorphism group should be computable.

A direct coefficient-and-leading-scale proof could provide an independent check on more general automorphism theorems.

[write] *Partly answered.* The exponent-rigidity report’s Theorem 8.4 shows that the rational dilations $\phi(\gamma) = r\gamma$, $r \in \mathbb{Q}_{>0} \setminus \{1\}$, have no exponential lift; so $(1, \phi_r)$ does not commute with the exponential. The question for general (χ, ϕ) stays open.

Research question 21.15 (Transcendence bases compatible with support). Can one construct a class transcendence basis of $\mathbf{No}/\mathbb{R}$ whose elements have controlled normal-form support, and then describe the algebraic closure of the generated field inside $\mathbf{No}$? How far is the explicit family $\{\omega^{\omega^\alpha}\}$ from a transcendence basis?

The answer should distinguish finite field generation from Hahn closure; normal-form completeness is far larger than rational-function generation.

21.6 Tensor constructions and geometry

Research question 21.16 (Completed tensor powers). Define and compare the algebraic tensor powers of the class vector space $\mathbf{No}$ with support-completed Hahn tensor products. Do the monomials $\omega^{\gamma_1} \otimes \cdots \otimes \omega^{\gamma_n}$ form a natural multi-Hahn basis, and which support conditions are forced by contraction and symmetrization?

This would connect the present scalar decomposition with the repository’s finite-dimensional tensor-field program and with infinite-dimensional Hahn Hilbert geometry.

Research question 21.17 (Exterior and symmetric class algebras). Describe the exterior, symmetric, and Clifford algebras of $\mathbf{No}$ over $\mathbb{R}$ under both Hamel and Hahn completions. Which constructions retain a canonical grading by surreal exponents?

The algebraic versions depend on a Hamel basis only through isomorphism, while support-completed versions may preserve much more of the normal-form geometry.

21.7 Formalization and computation

Research question 21.18 (Formalization strategy). Formalize the set-sized core of the article in Lean: reverse-well-ordered support spaces, strong sums, support projections, the matrix theorem, the strong-dual criterion, the Vandermonde lower bound, and the monomial algebraic-independence theorem. Then isolate the additional class-theoretic axioms needed for global statements.

Most proofs reduce to finite-support bookkeeping plus well-order properties and are good candidates for machine checking. The class-basis existence theorem should be kept logically separate from constructive normal-form results.

[write] *Source 52 asks the same (“A formal proof library for the basis distinction”):* Formalize the finite-support obstruction, the canonical finite-dimensional valuation basis, the full-codimension theorem, and the Euler resonance-commutant-nonsplitting chain. Each has a short conceptual proof but requires clean interfaces among supports, linear maps, quotients, and localization. The most promising initial target is the Laurent slice, where exponents are integers and the infinite support arguments are especially explicit.

Source 53 asks the same (“Machine-checked linear layer”): Implement the plan in [section 20.4](#), beginning with monomial independence, support projections, and diagonal maps. Use theorem names and APIs that make the results reusable by the repository’s omnific, surcomplex, Hahn-function, and transseries developments.

Research question 21.19 (Symbolic linear algebra for normal forms). Develop algorithms that operate on finite or certified transfinite truncations: support projection, leading-grade elimination, diagonal and reindexing operators, strong matrix application, and independence tests for monomial families. Produce proof certificates for all support conditions.

Such tools would turn the Hahn-basis viewpoint into a practical computer algebra interface. A certificate should state the support order type, local finiteness bounds, and the exact truncation region used.

Research question 21.20 (Repository integration). Add a dedicated ProveIt report and Lean ledger entry that distinguishes $\mathbf{No}^n$ over $\mathbf{No}$, set-sized Hahn vector spaces, and $\mathbf{No}$ over $\mathbb{R}$. Cross-link the terminology “Hamel basis,” “Hahn basis,” “strongly summable family,” and “associated graded” across all related reports.

This editorial step would prevent the same word “basis” from denoting three incompatible synthesis rules in neighboring parts of the project.

[write] *Partly answered.* This report is the dedicated report; a ledger entry in `FORMALIZATION.md` and the cross-linking of terminology across reports are not yet made.

21.8 Questions of source 52

[write] Source 52 lists its questions as subsections; they are numbered here. Its question on class foundations and definability is printed with [research question 21.1](#), and its formal proof library with source 51’s formalization question.

The following questions are proposed by this article. Unless explicitly stated, they are not being advertised as established open problems in the published literature.

Research question 21.21 (Priority and exact scope of the Euler package (source 52)). Locate the quotient $\mathbb{R}((t))/\mathbb{R}[t, t^{-1}]$ and its generalized-series analogues in the literature on divisible modules, diagonal operator commutants, and formal differential equations. Is the combination of rational scalar structure, automatic strongness, and nonsplitting already known in this form? The first deliverable should be a citation-level comparison, not another novelty assertion. The proofs here remain useful regardless of priority.

Research question 21.22 (An explicit spanning family for the Euler quotient (source 52)). The geometric classes $[(1 - rt^\delta)^{-1}]$, $r > 0$, are independent over $\mathbb{R}(X)$ and have the right cardinality, but Proposition 13.7 proves that they do not span. In the Laurent case, their span consists of coefficient streams that, outside a finite set, are finite sums of rational functions of n times positive real exponentials r^n . The enlarged factorial-exponential family is independent as well; characterize its span and construct explicit directions beyond it. Can a richer, naturally described family actually span the quotient? A “basis” claim must address spanning, not just cardinality.

Research question 21.23 (The extension class behind nonsplitting (source 52)). Study the sequence $0 \rightarrow \text{Fin}_\Gamma \rightarrow H_\Gamma \rightarrow Q_\Gamma \rightarrow 0$ in

$$\text{Ext}_{\mathbb{R}[X]}^1(Q_\Gamma, \text{Fin}_\Gamma).$$

Can one describe its extension class using the coordinate residues at the eigenvalues? Does the Hahn support restriction distinguish it from the unrestricted product module $\mathbb{R}^\Gamma$? The no-divisible-submodule theorem gives a concrete invariant that any classification must preserve.

Research question 21.24 (Higher-rank Euler systems (source 52)). A single injective real-valued exponent character is unavailable for non-Archimedean ordered groups. Replace $\mathfrak{D}$ by a commuting family

$$\mathfrak{D}_j(t^\gamma) = \ell_j(\gamma)t^\gamma$$

of additive real-valued characters separating exponents, when such a family is chosen. Determine when their joint commutant is automatically strong, and identify the correct multi-variable localization quotient. Polynomial zero sets in several variables can meet the exponent image infinitely, so simply quotienting by finite support will no longer remove every resonance.

Research question 21.25 (Finite fibers and matrix-valued coefficients (source 52)). For a noninjective character $\ell : \Gamma \rightarrow \mathbb{R}$, eigenspaces are whole fiber-support spaces. Classify all commuting real-linear maps and identify conditions forcing strongness within each fiber. With finite-dimensional vector coefficients, scalar weights become matrices, suggesting a block version of Theorem 14.1. Infinite-dimensional fibers are a genuine obstruction, not a cosmetic change of notation.

Research question 21.26 (Strong-invariant subspaces versus coordinate subspaces (source 52)). Proposition 6.3 requires invariance under every coordinate projection. What follows from strong closure and invariance under $\mathfrak{D}$ alone? More generally, which $\mathbb{R}[X]$ -submodules of H_Γ admit strong retractions, and which quotient subspaces lift to support-controlled subspaces upstairs? The ideal J_Γ and the nonliftability theorem provide test cases.

Research question 21.27 (Birthday-sensitive finite-dimensional elimination (source 52)). Given a finite-dimensional surreal subspace generated by vectors of bounded birthdays, bound the birthdays of its canonical reduced basis and the complexity of its pivot exponents. Ordinary row reduction gives algebraic control but not sharp surreal simplicity bounds. A useful project would begin with Laurent and rational-exponent slices before attempting arbitrary normal forms.

Research question 21.28 (Topologies that retain ordinary real scalar continuity (source 52)). The valuation and full order topologies do not make the usual $\mathbb{R}$ act continuously. Investigate set-sized topologies that combine ordinary coefficient convergence with support control. State explicitly which Hahn sums remain continuous and whether the Euler commutant retains automatic continuity. A successful topology should not erase the difference between unbounded support and bounded accumulating support.

21.9 Questions of source 53

[write] Source 53's questions keep its grouping. Six are printed above with source 51's (weakest basis principle, definable Hamel bases, topologies making monomials a basis, matrix of the surreal derivation, machine-checked linear layer) or answered (strongly linear matrices).

The linear viewpoint opens several programs that are more substantial than the bare question of basis existence. The items below are arranged from foundational and structural problems to analysis, computation, and formalization.

21.9.1 Class foundations, choice, and definability

[write] “Weakest basis principle”: see [research question 21.1](#).

Research question 21.29 (Extending the monomials). Over weak class theories, is “the monomial class extends to a Hamel basis of **No**” strictly stronger than “**No** has some class Hamel basis”? Can the gap be calibrated by standard subsystems of second-order set theory?

[write] “Definable Hamel bases”: printed with source 51’s question of the same name.

Research question 21.30 (Absoluteness). How do class Hamel bases of **No** behave between transitive models with the same ordinals? Which basis-existence, spanning, or definability statements are absolute, and which can change after forcing that adds real coefficients or ordinal sequences?

Research question 21.31 (No-go theorems for natural Hamel bases). Prove impossibility results for Hamel bases satisfying combinations of invariance under monomial shifts, ordered additive exponent automorphisms, truncations, conjugation in the surcomplexification, or simplicity-preserving maps. The rigidity in [corollary 7.6](#) suggests that strong naturality requirements may force a coordinate system to be the Hahn basis rather than a Hamel basis.

[write] Source 52’s [theorem 9.6](#) is one such no-go theorem: no projective Hamel basis is invariant under the coordinate reweightings.

21.9.2 Dimensions of canonical set-sized fragments

Research question 21.32 (Birthday-cutoff dimensions). For ordinals λ for which $\mathbf{No}_{<\lambda}$ is an $\mathbb{R}$ -vector space or a field (notably suitable additively indecomposable ordinals and epsilon numbers), compute

$$\dim_{\mathbb{R}} \mathbf{No}_{<\lambda}$$

in ZFC, with exact dependence on $|\lambda|$, $2^{<|\lambda|}$, and the cofinality of λ .

Research question 21.33 (Dimensions of Hahn subfields). For a set-sized ordered abelian subgroup $\Gamma \subseteq \mathbf{No}$ and support-length bound λ , determine the Hamel dimension of $\mathbb{R}((\omega^\Gamma))_{<\lambda}$ over $\mathbb{R}$ and over natural intermediate Hahn subfields. Identify cases where the answer is absolute and cases sensitive to cardinal arithmetic.

Research question 21.34 (Cut quotients). For exponent cuts $C \subseteq D$, compute the dimension and ordered-module structure of V_D/V_C . The principal one-step quotient is $\mathbb{R}$ by [remark 6.6](#); nonprincipal intervals may encode interesting cofinality and saturation data.

Research question 21.35 (Support ideals). Given a family $\mathcal{A}$ of admissible supports (bounded cardinality, bounded order type, prescribed cofinality, or membership in an ideal on the exponent class), determine when the corresponding series class is a vector space, algebra, field, truncation-closed subspace, or elementary substructure.

Research question 21.36 (Explicit large independent systems). Generalize [corollary 5.16](#). Construct explicit independent families of the optimal cardinality in $\mathbf{H}(A)$ for arbitrary infinite reverse-well-ordered A , preferably with algebraic, differential, or exponential meaning rather than pure set-theoretic almost-disjoint-family arguments.

21.9.3 Hahn bases, summation, and topology

Research question 21.37 (A category of Hahn-based vector spaces). Axiomatize real vector spaces equipped with an ordered coordinate class, admissible supports, and a strong summation operation. Determine the correct morphisms: coefficientwise continuous maps, maps preserving all strong sums, or maps preserving truncations. Establish analogues of bases, tensor products, exactness, completion, and duality.

[write] “Topologies making monomials a basis”: printed with source 51’s question on coarser topologies.

Research question 21.38 (Completion of the group algebra). For a chosen chain of convex subspaces or a quotient of the value group, compute the completion of $\mathbb{R}[\mathbf{No}]$. Which completions recover familiar fields such as Levi–Civita fields, set-sized Hahn fields, transseries fields, or natural surreal subfields, and which fail because supports cease to be admissible?

Research question 21.39 (Sequential versus net convergence). On support slices $H(A) \cong \mathbb{R}^A$, characterize the sequential closure of finite support in the product topology and compare it with net closure, strong summation, and valuation closure. The answer should depend sharply on $|A|$ and its cofinal structure.

Research question 21.40 (Universal property of strong summation). Can $\mathbf{No}$ be characterized as a free or initial strongly complete ordered $\mathbb{R}$ -algebra generated by the exponent group, subject to the Hahn support axioms? A satisfactory universal property would make the phrase “Hahn basis” categorical rather than merely descriptive.

21.9.4 Linear operators and invariant subspaces

Research question 21.41 (Continuous duals). The algebraic dual of a huge product-like space is intractable. Classify instead: (i) natural-valuation continuous linear functionals; (ii) product-topology continuous functionals on $H(A)$; (iii) strong-sum-preserving functionals; and (iv) truncation-continuous functionals. Determine when coefficient functionals span the relevant dual.

[write] *Partly answered.* Item (iii) is answered by [theorem 10.7](#) (on $\mathbf{No}$) and [theorem 10.11](#) (on a set-sized workspace). For (i), [theorem 12.4](#) describes the valuation-continuous functionals of a set-sized workspace that vanish on its finite-monomial part; the full natural valuation on $\mathbf{No}$ is open.

Remark 21.42 ([write] Answered: strongly linear matrices (source 53)). Source 53 asked:

Classify class matrices $K = (K_{ab})$ for which

$$(Tx)_a = \sum_b K_{ab} c_b(x)$$

defines a surreal for every x and preserves strong summability. The answer should involve row-finiteness, well-based unions of supports, and a class analogue of Hahn-series matrix multiplication.

[Theorem 10.3](#) (source 51) answers this: the strong maps are exactly those whose columns $(T\omega^b)_b$ form a Hahn-compatible family, that is, for every reverse-well-ordered set S of column indices the union of the column supports is reverse well ordered and each row meets only finitely many columns of S .

Research question 21.43 (Triangular operators and symbols). Develop a calculus of operators preserving the leading-exponent filtration. Define order, principal symbol, ellipticity, and parametrices. Determine when an operator with invertible graded symbol is itself invertible, and identify the appropriate completeness hypothesis for a transfinite Neumann-series argument.

Research question 21.44 (Shift commutants). Classify linear maps commuting with every monomial shift S_b . On ordinary group algebras these are convolution operators; on the full Hahn field, support admissibility makes the question subtler. Intersect this commutant with continuous, strongly linear, order-preserving, or truncation-preserving operators.

Research question 21.45 (Invariant subspaces). Classify subspaces invariant under: all support projections; all shifts; all ordered exponent automorphisms; multiplication by purely infinite elements; or the full diagonal algebra. Determine which invariance conditions force a subspace to be a support cut, a Hahn subfield, or one of a small number of canonical pieces.

[write] For invariance under all support projections, source 52’s [proposition 6.3](#) answers the question on a set-sized workspace for strongly closed subspaces; without strong closure it is open.

Research question 21.46 (Positive-operator theory). Extend [theorem 7.4](#) from diagonal maps to triangular and strongly linear maps. Is there a useful analogue of Perron–Frobenius theory, band-preserving operator theory, or Riesz-space spectral theory for the ordered class vector space **No**?

21.9.5 Derivations, exponentials, and transseries

[write] “Matrix of the surreal derivation”: printed with source 51’s question on derivation-adapted bases.

Research question 21.47 (Bases adapted to logarithmic structure). The monomials are adapted to multiplication and valuation, while log-atomic and κ - or λ -numbers are adapted to iterated logarithms. Develop mixed Hahn bases or nested coordinate systems that expose both structures and compare them with hyperserial normal forms.

Research question 21.48 (Linear differential equations). For a linear differential operator with surreal coefficients, give support-based criteria for existence, uniqueness, and the support complexity of solutions. Determine when coefficient recursion is well founded and when resonance creates new transseries scales.

Research question 21.49 (Exponentiation as a nonlinear coordinate transform). Study the map from the purely infinite additive subspace to the monomial group via $x \mapsto \exp(x)$. Its source is linear but its image is multiplicative. Can its differential, tangent filtration, and induced graded map be organized into a useful Lie-theoretic or logarithmic linearization framework?

21.9.6 Surcomplex and quaternionic extensions

Research question 21.50 (Surcomplex Hahn bases). For **No**[i], the same monomials with complex coefficients give a canonical Hahn basis over $\mathbb{C}$. Develop the analogues of support projections, diagonal operators, conjugation-equivariant maps, Hermitian forms on set-sized slices, and real versus complex Hamel dimensions.

Research question 21.51 (Real forms). Classify real subspaces and real forms of set-sized surcomplex Hahn spaces that are stable under truncation or exponent symmetries. Determine whether conjugation is characterized by naturality with respect to the monomial projection algebra.

Research question 21.52 (Surquaternionic modules). Investigate left and right vector-space structures over **No**, **No**[i], and surquaternions. Noncommutativity should force a distinction between left/right Hahn bases and between coefficientwise and convolutional operator algebras.

21.9.7 Formalization and computation

[write] “Machine-checked linear layer”: printed with source 51’s formalization question.

Research question 21.53 (Certified truncation algorithms). Design executable procedures for extracting bounded exponent windows, multiplying finite truncations with certified tail bounds, and deciding finite systems of coefficient constraints. Prove that the algorithms commute with the exact strong sum where the relevant support hypotheses hold.

Research question 21.54 (Automated support reasoning). Build tactics for goals of the form “the union of these supports is reverse well ordered” and “each coefficient fibre is finite.” Much of surreal formalization is presently support bookkeeping; automation could turn strong-summation proofs into routine algebra.

21.9.8 Highest-priority near-term projects

Among the preceding questions, the most accessible and broadly useful next steps are:

1. formalize support projections, diagonal operators, and the commutant theorem;
2. formalize the natural-valuation closedness of $\mathbf{No}_{\text{fin}}$;
3. compute dimensions of important set-sized support slices and birthday cutoffs;
4. classify strong-sum-preserving linear maps by admissible class matrices;
5. analyze the associated-graded symbol of the canonical surreal derivation;
6. find the weakest class theory needed for a monomial-extending Hamel basis;
7. study topologies or bornologies in which normal-form truncations converge; and
8. extend the entire theory to $\mathbf{No}[i]$ with conjugation and complex coefficients.

[write] Item 4 is answered by [theorem 10.3](#).

22 Conclusions

[write] Each source ends with its own conclusion; all three are printed.

22.1 Source 51

The surreal numbers are unquestionably a vector class over $\mathbb{R}$, but the ordinary phrase “choose a basis” suppresses the structure that makes them surreal. A Hamel basis sees only finite sums. It exists under global choice, may be chosen to contain every monomial, and is necessarily supplemented by a proper class of infinite-support vectors. Such a basis is useful for abstract algebraic existence arguments but not for normal-form computation, order, or asymptotics.

The natural answer is instead

$$\mathfrak{M} = \{\omega^\gamma : \gamma \in \mathbf{No}\}.$$

This class is canonical in three mutually reinforcing senses:

1. it is the Conway–Hahn basis giving every surreal a unique strongly summable real expansion;
2. it is the basis of the one-dimensional dominance grades, with $\text{gr}(\mathbf{No}) \cong \mathbb{R}[\mathbf{No}_{\text{add}}]$;
3. it is normalized by simplicity, since ω^γ is the simplest positive representative of its Archimedean class.

Its basis law is therefore not $\mathbf{No} = \bigoplus_\gamma \mathbb{R}\omega^\gamma$, but the sharper statement

$$\mathbf{No} = \prod_{\gamma \in \mathbf{No}}^H \mathbb{R}\omega^\gamma,$$

where the product sign denotes set-supported, reverse-well-ordered Hahn coordinates. (Equivalently, use (4.3); the displayed formula is conceptual rather than new notation.)

Several consequences follow immediately once the correct synthesis rule is fixed. Support subsets give canonical projections. Strong operators have Hahn matrices. Strong real functionals are precisely summation-finite weighted coefficient sums, while order-bounded real functionals vanish. Set-sized support windows reveal an enormous gap between Hahn and Hamel dimensions. The full fine topology is too refined for set-indexed partial sums, so normal forms are formal strong sums rather than Schauder series. Monomial symmetries and diagonal derivations reduce to group maps on the exponents, and rational exponent independence becomes algebraic monomial independence.

The resulting picture is analogous to several familiar distinctions, but is not identical to any one of them: Fourier series versus finite trigonometric polynomials, formal power series versus polynomials, completed graded objects versus their direct sums, and topological bases versus algebraic bases. The surreal case adds two features absent from those set-sized analogies: the exponent group is itself the proper class **No**, and every individual series still has only set support. That combination is exactly why the Conway–Hahn basis is both global and rigorously manageable.

22.2 Source 52

Surreal numbers are naturally a real vector *class with Hahn coordinates*, not a finite-coordinate enlargement of $\mathbb{R}$. Conway monomials are the right canonical coordinate objects, but not an algebraic basis. Their finite span omits a full-dimensional quotient in every nontrivial set-sized Hahn field, and no set of extra vectors repairs the full surreal class.

The positive finite-dimensional theorem and the negative global theorem fit together: every finite-dimensional subspace has a canonical reduced leading-coordinate basis, yet no valuation-independent Hamel basis exists for the whole Hahn space. Completion does not generally close the gap between finite spans and strong Hahn sums, and the full surreal topology is finer still.

The Euler operator gives the missing directions additional structure. On real exponent slices the quotient by finite monomial sums is canonically a continuum-dimensional vector space over $\mathbb{R}(X)$. All operators commuting with Euler are forced to be coefficientwise, but no Euler-invariant algebraic complement of the finite-monomial subspace exists. This combination supplies both a rigorous answer to the basis question and a concrete program for further research.

22.3 Source 53

The surreal numbers do form a vector space over $\mathbb{R}$, but ordinary basis language is too coarse to capture their structure. The canonical coordinate directions are the monomials ω^a . They are linearly independent, their coefficient maps separate points, and every surreal is uniquely assembled from them. Yet the assembly operation is a set-supported Hahn sum, not a finite Hamel combination and not, in the full natural valuation, a limit of finite combinations.

This distinction yields a coherent answer:

- $\{\omega^a\}$ is the natural Hahn basis of **No**;
- it is the natural homogeneous basis of $\text{gr } \mathbf{No}$;
- it is only a basis of $\mathbf{No}_{\text{fin}}$ in the Hamel sense;
- any Hamel basis of all **No** is a noncanonical proper class with vectors of unbounded support complexity; and

- the valuation, order, convex geometry, support projections, and natural operator algebra are all organized directly by the monomial coordinates.

The resulting linear theory is not merely terminological. It gives exact dimensions of support slices, explicit large independent families, a canonical projection and diagonal calculus, a classification of convex subspaces and positive diagonal maps, a coefficientwise reduction of finite linear systems, and a sharp separation between algebraic span, topological closure, and strong-summation closure. These results make the monomial Hahn basis the correct starting point for a systematic “linear analysis of **No**” and provide a concrete formalization roadmap for the PROVEIT project.

A Proof and dependency audits

[write] The three sources’ audits are printed as delivered, with two corrections in source 53’s table (marked [write]). The rows of source 51 for monomial automorphisms and diagonal derivations describe second routes to results of other reports ([section 18](#)).

A.1 Source 51

Result	Main input	Status in this report
Conway–Hahn basis, theorem 4.2	Conway normal-form theorem	Classical theorem restated in vector-language.
No set spans, theorem 5.1	Set union of normal-form supports	Complete direct proof.
Hamel class basis, theorem 5.3	NBG+GC; set-sized basis extension	Complete class-recursion proof.
Basis containing monomials, proposition 5.4	NBG+GC; greedy class recursion	Complete proof.
Quotient independent family, theorem 5.7	Disjoint countable support blocks	Complete explicit proof.
Support-window dimension, theorem 5.15	Erdős–Kaplansky theorem	Classical cardinal result applied to $\mathbb{R}^A$; countable lower bound also proved directly.
Support projections, theorem 6.1	Heredity of reverse well ordering	Complete proof.
Associated graded, theorem 6.7	Leading-term multiplication	Complete proof.
Strong matrix classification, theorem 10.3	Flattening of locally finite Hahn families	Complete proof.
Strong real dual, theorem 10.7	Strong matrix theorem with codomain $\mathbb{R}$	Complete proof.
Dilation invariants, theorem 18.3	Strong-dual weights and exponent dilations	Complete proof.
Order dual is zero, theorem 11.2	Infinite surreal bound $\omega x $	Complete proof.
Set-indexed nets eventually constant, theorem 11.8	Every set of surreals has a surreal upper bound	Complete proof.
Monomial automorphisms, theorem 18.1	Hahn support transport and character laws	Complete proof for the stated monomial-line-preserving subgroup.
Diagonal derivations, theorem 18.5	Additive real characters of the exponent group	Complete proof.

Result	Main input	Status in this report
Monomial algebraic independence, theorem 18.7	Uniqueness of finite normal forms	Complete proof.

No computer algebra was needed for these proofs. The main verification risks are foundational rather than computational: distinguishing sets from proper classes, and never replacing strong summability by an unjustified topological limit.

A.2 Source 52

A.2.1 Main dependency chain

The chain for the principal operator result is short:

$$\text{support-preserving division} \implies \text{finite resonance identity} \implies \mathbb{R}(X)\text{-action on } Q_\Gamma;$$

$$\text{im}(\mathfrak{D} - \gamma) = \ker(x \mapsto x_\gamma) \implies \text{coefficientwise commutant} \implies \text{nonsplitting.}$$

The dimension lower bound uses eventual exponential-polynomial independence, while the upper bound uses countable supports in $\mathbb{R}$. The two-layer theorem adds bounded supports approaching zero at a controlled reciprocal rate. No unproved derivative formula at nonreal surreal exponents enters these arguments.

A.2.2 Load-bearing hypotheses

The group $\Gamma = \{0\}$ is excluded from the infinite-dimensional statements: then $H_\Gamma = \text{Fin}_\Gamma = \mathbb{R}$ and every quotient discussed there is zero. The real-exponent condition in the Euler sections is not optional. The polynomial zero set is finite because eigenvalues are distinct real exponents. Hahn sums are always set-indexed and have well-ordered support. Global choice is invoked only for the explicit class-basis discussion; ordinary set-sized choices suffice elsewhere.

The ideal controlling quotient lifts is J_Γ , not finitely supported weights. An infinite reverse-well-ordered weight support can have finite intersection with every individual Hahn support. This distinction was checked separately because it is easy to confuse “finite output support for each input” with “finite global weight support.”

The finite-dimensional canonical basis theorem asserts exact leading-coordinate reduction in the entire subspace, not a numerical finite-prefix algorithm for arbitrary infinite inputs. The no-valuation-basis proof uses nested balls rather than an unjustified sum of basis vectors. The full surreal topology statements concern set-indexed nets; they say nothing automatic about proper-class-indexed notions of convergence.

A.2.3 What has not been supplied

There is no explicit Hamel basis spanning all of H_Γ or **No**, no classification of all invariant subspaces, no formal verification of the new theorems, and no claimed all-rank extension of the Euler quotient. The requested natural-basis question is answered by the Hahn/Hamel distinction; the stated theorems do not require any of these additional constructions.

A.3 Source 53

Statement	Status	Dependency
Conway normal form, $\mathbf{No} = \mathbb{R}((\omega^{\mathbf{No}}))$	Classical	Conway–Gonshor normal form; modern accounts in [11, 9, 17, 7].
Ordered field / natural valuation / three-part split	Classical	Standard Hahn-field consequences of normal form.
Restricted class-product model and coefficient separation	Proved here	Direct reformulation of normal-form existence and uniqueness.
Monomial Hahn basis	Proved here	Conway normal form plus the strong-summability definition.
Monomial Hamel independence and proper finite span	Proved here	Normal-form uniqueness and the geometric-series example.
No set spans $\mathbf{No}$	Proved here	Replacement and the fact that $\mathbf{No}$ is a proper class.
Class Hamel basis under ETR/global choice	Conditional construction here; [write] GBC suffices (proposition 5.4)	Set-sized basis theorem at each stage plus explicit class recursion.
Unbounded support complexity of Hamel bases	Proved here	Finite-union cardinal arithmetic and normal-form supports.
$H(A) \cong \mathbb{R}^A$ and exact dimension	Proved here from standard theorem	Normal form plus Erdős–Kaplansky [15].
Vandermonde geometric family	Proved here	Coefficient comparison and the Vandermonde determinant.
One-dimensional scale layers and $\text{gr } \mathbf{No} \cong \mathbb{R}[\mathbf{No}]$	Proved here	Leading-exponent filtration and coefficient extraction.
Support projection calculus	Proved here	Coefficientwise identities.
Convex-subspace classification	[write] Classical (Hahn, Conrad); direct proof here	Leading-term comparison in an ordered Hahn field.
Diagonal commutant and idempotents	Proved here	One-coordinate projections and coefficient separation.
Positive diagonal classification	Proved here	Lexicographic sign determined by the leading surviving coefficient.
Rank transfer for finite systems	Proved here	Real row/column reduction or coefficientwise equations.
Closedness of $\mathbf{No}_{\text{fin}}$ in the natural valuation	Proved here	Every set of surreal exponents has a strict surreal lower bound.
Algebraic, topological, and summation trichotomy	Proved here	Combination of the span, closure, and normal-form theorems.

B Notation and terminology

Source 51's table:

Symbol	Meaning
No , On	The surreal-number and ordinal classes.
ω^γ	Conway omega-map monomial at exponent γ .
$\text{supp}(x)$	Set of exponents with nonzero coefficient in the Conway normal form of x .
$[\omega^\gamma]x$	Real coefficient of ω^γ in x .
$\sum^H$	Strong or Hahn sum.
$\mathfrak{M}$	The monomial class $\{\omega^\gamma : \gamma \in \mathbf{No}\}$.
$\mathbf{No}_{\text{fin}}$	Finite-support normal forms; the algebraic direct sum of the monomial lines.
$H(A)$	Surreals whose support is contained in the reverse-well-ordered set A .
P_A	Projection retaining exactly the exponents in A .
$v(x)$	Natural valuation – lead(x).
Strong map	A real-linear map preserving every admissible set-indexed Hahn sum and commuting with it.
Summation-finite class	A class meeting every reverse-well-ordered set in a finite set; equivalently, containing no infinite decreasing sequence.

[write] Further symbols of sources 53 and 52 (the renamings are in [section 2.3](#)):

Symbol	Meaning
$c_a(x)$	Real coefficient of ω^a in x (53); $= [\omega^a]x$.
$\text{lead}(x)$, $\text{lc}(x)$	Greatest support exponent and its coefficient.
$F_{<a}$, $F_{\leq a}$	Leading-exponent filtration subspaces.
P_S	Projection retaining exactly the exponents in S (53); $= P_A$.
D_λ	Coefficientwise diagonal operator with multiplier λ (53); $= D_a$.
S_b	Monomial shift $x \mapsto \omega^b x$ (53).
$H(A)$, $H(A)^{\text{fin}}$	Support slice and its finite-support part (53's $\mathcal{H}_A$, $\mathcal{H}_A^{\text{fin}}$).
V_C	Convex subspace of an initial exponent class C (53).
t^γ	$\omega^{-\gamma}$ (52).
H_Γ	$\mathbb{R}((t^\Gamma))$, Γ a set-sized ordered abelian group (52).
Fin_Γ , Lf_Γ	Finite-support part of H_Γ and its valuation closure (52's E_Γ , C_Γ).
$H[A]$, $H_{\geq \gamma}$, $H_{> \gamma}$	Series supported in A ; valuation subspaces (52).
$\mathfrak{D}$, G_p , P_{Z_p}	Euler operator, nonresonant Green operator, resonance projection (52).
Q_Γ	$H_\Gamma/\text{Fin}_\Gamma$, an $\mathbb{R}(X)$ -vector space (52).
M_a , J_Γ	Coefficient multiplier; ideal of weights with reverse-well-ordered support (52).
$\mathfrak{c}$	$2^{\aleph_0}$ (52).

C Provenance of this report

[write] This report was built from three manuscripts, delivered on 1 October 2026 to the repository's docs/incoming/ drop zone (arrival commit 3172be896) as batch 73, cluster X, of that drop zone, and placed in commit 26abf259b. Each records its own pin, and all three pins are the same commit, f7e7d8607 (f7e7d8607d3b1c49c93b844cae20a77ab3689fe7).

Source	Archive (main file)	What it contributed
51 (base)	<code>surreal_numbers_real_vector_class.zip</code> (<code>surreal_numbers_real_vector_class.tex</code> , 2,321 lines; 32-page PDF)	The base text, printed whole: every section, the 20 research questions, its audit and notation appendices.
52	<code>surreal_real_vector_spaces.zip</code> (<code>article.tex</code> , 1,054 lines; 29-page PDF; with <code>code/verify.py</code> , <code>data/verification.json</code> , <code>audit/PROOF_AUDIT.md</code> , <code>audit/sources.json</code> , shipped beside this report as <code>code/52-euler-quotient-verify.py</code> , <code>data/52-euler-quotient-verification.json</code> , <code>52-euler-quotient-PROOF_AUDIT.md</code> and <code>data/52-euler-quotient-sources.json</code>)	Printed whole: its foundations, dimension, valuation and naturality material in Parts I and II , its duals, topology and completion in Part III , its Euler quotient as Part IV , its 10 questions, conclusion and appendices.
53	<code>surreal_vector_space_over_R_article.zip</code> (<code>surreal_vector_space_over_R.tex</code> , 1,819 lines; 28-page PDF; no support files)	Printed whole except the statements and proofs it shares with source 51 (credited where they stand): its answer in one page, coordinate model, support complexity, slices, graded splitting, cut decompositions, convex subspaces, diagonal operators, positivity, covariance, linear systems, closedness and trichotomy, recommendation, Lean map and plan, 31 questions, conclusion and audit.

Where the merge had to choose.

1. *Base.* Source 51: weakest foundations on the shared theorems (NBG with global choice; source 52 uses the same, source 53 assumed more), the whole class including proper-class active weights, and the most complete shared spine.
2. *Shared statements.* Printed once in source 51’s wording, except the window dimension ([theorem 5.15](#)), stated with source 53’s two additional clauses. Source 53’s statements of the same results (normal form, Hahn basis, monomial independence, no set spans, class basis, projection calculus, graded algebra, diagonal calculus, Vandermonde family) are not repeated; their labels sit on the printed statements, and their proofs are printed where they differ (no set spans; class basis by staged Hahn fields). Statements that add a clause are printed as complements (source 53’s split scale layer and countable tail).
3. *Different scope, both printed.* Source 52’s slice versions (monomials not a Hamel basis of H_Γ , the strong dual of H_Γ , positivity in any non-Archimedean field, uniform discreteness) and source 53’s closedness of $\mathbf{No}_{\text{fin}}$ are printed beside source 51’s results with notes on how they relate.
4. *Repository results.* Pointers with second routes: [section 12](#), [section 18](#), [Part IV](#).
5. *Order.* Source 51’s diagonal maps moved from its strong-map section to [section 7](#), beside source 53’s; source 51’s dilation-invariant dual stays with the exponent dilations it uses ([section 18](#)).
6. *Corrections* (each marked [write]): source 53’s foundational hypothesis for the class basis (GBC suffices); its “unique natural coordinate directions” and “merely permutes” (canonical relative to the omega map); classical status of its convex-subspace classification and of source 52’s echelon and no-valuation-basis theorems; source 53’s DOI for Berarducci–Mantova (10.4171/JEMS/762

→ 10.4171/JEMS/769) and source 52’s page range for Kuhlmann (723–735 → 723–736); source 52’s program and record are named by their shipped paths. The arXiv identifiers 2509.22374, 2403.05827 and 2010.01983 were checked against arXiv on 1 October 2026 (titles, authors and versions as cited).

7. *Questions.* Six overlapping questions are printed once with every source’s wording; one of source 53’s is answered by source 51; partial answers from the sibling sources and from other reports are marked.

C.1 Source 52’s provenance, literature scope, and nonclaims

C.1.1 Repository snapshot and reviewed material

The repository branch was resolved on October 1, 2026 to

f7e7d8607d3b1c49c93b844cae20a77ab3689fe7.

The main materials read through the connected GitHub interface were the surreal report guide, the opening 240 source lines of *Three Duals at Surreal Scales*, and the detailed README of the tail-span report. Repository searches for Hamel and valuation-basis terminology supplied navigation clues. Search results were served from an indexed snapshot distinct from the pinned branch, so they are not represented as an exhaustive audit of the pinned tree. The comparison-report reads used the pin, and the report guide’s opening was rechecked at that pin.

The comparison therefore establishes conceptual overlap and avoids claiming the repository’s completion and strongness distinctions as new. It does *not* establish that no similar theorem occurs anywhere among all reports, attachments, or unindexed files. The repository was not modified.

C.1.2 Classical input versus results developed here

Topic	Provenance and status
Conway normal form and monomial arithmetic	Classical input from Conway and Gonshor, with the precise normal-form statement also available in Mantova–Matusinski. Not reproved from the genetic construction here.
Valuation independence and immediate extensions	Established language of valued-vector-space theory. We give direct Hahn proofs of the finite-dimensional and no-global-basis assertions.
Infinite dimension	The Vandermonde method is classical. The full-codimension statement is proved directly with the quotient and coset-support details explicit.
Completion and invisible functionals	Closely related to established Hahn-space methods and explicit repository precedents. Included as necessary structure, not a claim of first discovery.
Euler localization, commutant, and nonsplitting	Deductions developed and proved in this manuscript. The unified formulation is proposed for further research; historical priority is not certified.
Two rational-calculus layers and lift ideal	Proved here with explicit supports and the correct reverse-well-ordered kernel ideal. Broader literature comparison remains desirable.

Topic	Provenance and status
Class Hamel basis	Conditional on the stated class theory and global-choice framework. No class quotient is treated as a set of proper-class cosets.

The targeted primary-literature search included normal forms and surreal derivations, valuation bases, infinite vector-space dimension, formalization, and Euler/diagonal-operator quotient terminology. It was not an exhaustive MathSciNet, zbMATH, or citation-network search. No named historical open problem is claimed solved. Mathematical correctness and historical novelty are separate questions.

C.2 The sources' abstracts

Source 51.

The surreal numbers form a proper-class vector space over their canonical copy of the real field. The immediate question—what is its natural basis?—has two different answers, depending on what “basis” means. The canonical answer is the monomial class $\mathfrak{M} = \{\omega^\gamma : \gamma \in \mathbf{No}\}$. Every surreal has a unique Conway normal form, a set-supported, reverse-well-ordered real linear combination of these monomials. Thus $\mathfrak{M}$ is a genuine *Hahn basis*, a basis for strong transfinite summation, and the canonical basis of the associated graded vector class. It is not a Hamel basis, because Hamel combinations are finite and most surreals have infinite normal-form support.

Working in NBG set theory with global choice, we prove that a Hamel class basis does exist, but no set-sized family spans $\mathbf{No}$, every such basis is a proper class, and every Hamel basis contains a proper class of infinite-support vectors. The quotient by the finite-support skeleton already contains an explicit proper-class linearly independent family. On each infinite reverse-well-ordered support window A , the set-sized space of all series supported in A has Hamel dimension $|\mathbb{R}|^{|A|}$; in particular, a countable window has continuum dimension, although its monomial Hahn basis is countable.

The normal form also yields a canonical Boolean algebra of support projections and identifies the associated graded algebra with the real group algebra of the additive surreal class. We classify strongly real linear maps by Hahn-compatible column families, and specialize this to an exact classification of strongly summation-preserving real-valued functionals. Such a functional is a weighted coefficient sum whose active exponents meet every reverse-well-ordered set only finitely; this permits proper-class active sets, including the ordinal exponents. In sharp contrast, every order-bounded real linear functional is zero, hence there is no nonzero positive real linear state and no nonzero order-compatible real-valued seminorm on all of $\mathbf{No}$.

Finally, the full natural valuation topology admits no non-eventually-constant set-indexed convergent net, so Conway normal forms are strong formal sums rather than Schauder expansions in that topology. We describe monomial-preserving ordered-field automorphisms, diagonal strong derivations, and an explicit proper-class algebraically independent family. A detailed research program closes the article. Classical inputs are Conway normal form, the omega map, Hahn–Neumann support arithmetic, and the Erdős–Kaplansky dimension theorem; the structural package assembled here is offered as an unrefereed research draft, not as a claim of priority.

Source 52.

The surreal numbers form a proper-class vector space over the reals. Their natural coordinates are Conway’s monomials ω^a , but these are a Hahn basis, not a Hamel basis: arbitrary vectors require set-sized, reverse-well-ordered supports. We make this distinction precise and develop its consequences.

For every nontrivial set-sized ordered abelian group Γ , the finite-monomial span Fin_Γ has the largest possible algebraic defect in $H_\Gamma = \mathbb{R}((t^\Gamma))$: both H_Γ and $H_\Gamma/\text{Fin}_\Gamma$ have real dimension $|H_\Gamma|$.

Finite-dimensional subspaces have canonical reduced valuation bases, whereas the full Hahn space has no valuation-independent Hamel basis. We prove an automorphism obstruction to a natural Hamel basis, classify the valuation completion of the monomial span, and separate strong from merely continuous real-valued coordinate functionals.

The main further development concerns nonzero additive subgroups $\Gamma \subseteq \mathbb{R}$. The Euler derivation $\mathfrak{D}(t^\gamma) = \gamma t^\gamma$ makes $H_\Gamma/\text{Fin}_\Gamma$ canonically a vector space over $\mathbb{R}(X)$ of dimension $2^{\aleph_0}$. Every real-linear endomorphism of H_Γ commuting with $\mathfrak{D}$ is automatically a strong coefficient multiplier. Consequently Fin_Γ has no $\mathfrak{D}$ -invariant complement, despite having algebraic complements. For dense Γ , the completion defect itself splits into two continuum-dimensional $\mathbb{R}(X)$ -layers. Complete proofs, explicit witnesses, finite exact-arithmetic checks, a provenance audit, and further research questions are included.

Source 53.

The surreal numbers form an ordered vector space over their canonical copy of $\mathbb{R}$, but the usual phrase “choose a basis” conceals three separate issues: **No** is a proper class; a Hamel basis permits only finite combinations; and Conway normal forms are transfinite Hahn sums rather than ordinary convergent series. This report separates those issues and gives a precise answer. The canonical monomials ω^a form a Hahn or summation basis, not a Hamel basis. The finite monomial span is the class group algebra $\mathbb{R}[\mathbf{No}]$, whereas all of **No** is the corresponding proper-class Hahn field $\mathbb{R}((\omega^{\mathbf{No}}))$.

From this viewpoint we prove a collection of linear-algebraic consequences. No set can span **No**; under an explicit class-choice hypothesis a class Hamel basis exists and may be chosen to contain all monomials, but every such basis has vectors of arbitrarily large support. For a set-sized reverse-well-ordered exponent set A , the support slice $H(A)$ is canonically $\mathbb{R}^A$ and has Hamel dimension $|\mathbb{R}|^{|A|}$ when A is infinite. A concrete Vandermonde family of continuum many independent geometric-series surreals already occurs in a countable slice. The leading-exponent filtration has one-dimensional successive layers and associated graded algebra $\mathbb{R}[\mathbf{No}]$. Support cuts give canonical linear projections; their commutant consists exactly of coefficientwise diagonal operators. We classify the positive diagonal operators and all convex real subspaces. Finite real linear systems over **No** reduce coefficientwise to the same systems over $\mathbb{R}$. Finally, we prove a topology/summation trichotomy: the algebraic span and natural-valuation closure of the monomials are both the finite-support subspace, while closure under admissible Hahn sums is all of **No**. The final sections map these results to existing Lean developments in PROVEIT and propose a program of further research and formalization.

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